

AKTU



B. Tech I-Year

Gateway Classes



Semester -I & II Common to All Branches

BEE101/201: FUNDAMENTALS OF ELECTRICAL ENGG.

UNIT-2 : Steady State Analysis of Single Phase AC Circuits



Gateway Series for Engineering

- ✓ Topic Wise Entire Syllabus
- ✓ Long - Short Questions Covered
- ✓ AKTU PYQs Covered
- ✓ DPP
- ✓ Result Oriented Content



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AKTU



B. Tech I-Year

Gateway Classes



BEE101 / BEE201: FUNDAMENTALS OF ELECTRICAL ENGINEERING

Unit-2: : Steady State Analysis of Single Phase AC Circuits

Syllabus

Representation of Sinusoidal waveforms – Average and effective values, Form and peak factors. Analysis of single phase AC Circuits consisting R-L-C combination (Series and Parallel) Apparent, active & reactive power, Power factor. Concept of Resonance in series & parallel circuits, bandwidth and quality factor. Three phase balanced circuits, voltage and current relations in star and delta connections.



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FUNDAMENTALS OF ELECTRICAL ENGINEERING

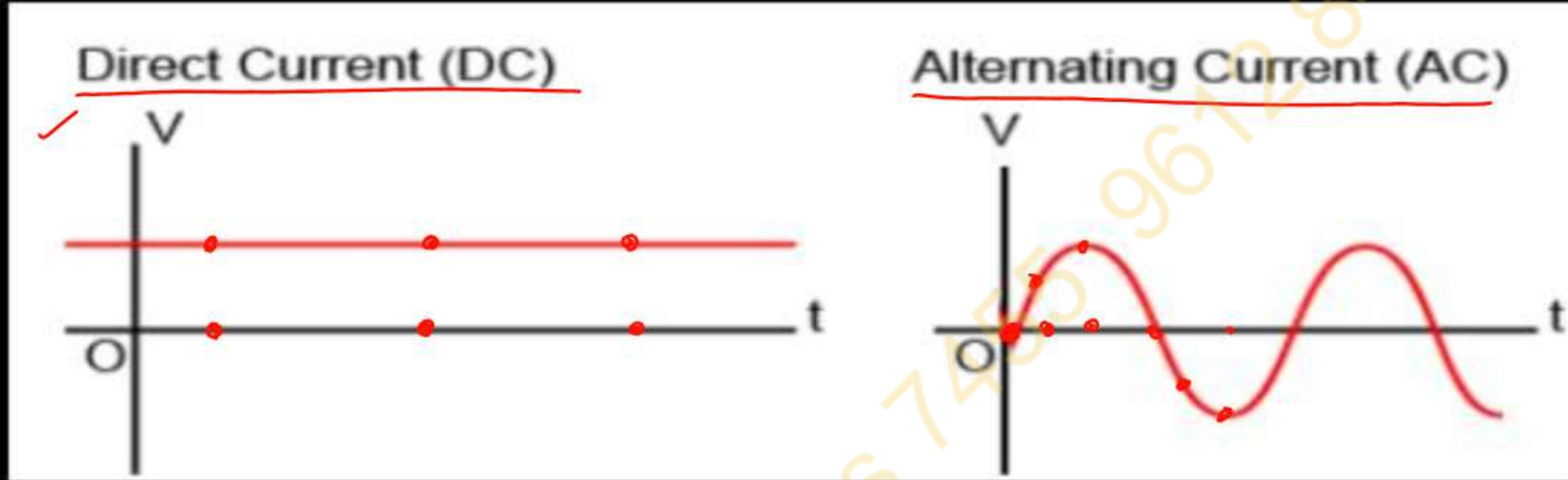
UNIT -2 : Steady State Analysis of 1- Φ Circuits

Lec-1

Today's Target

- ✓ Basic Terminology ✓
- ✓ Average Value ✓
- ✓ RMS Value ✓

Direct Current (D.C.) Supply:- A DC power supply is one that provides a constant magnitude and direction with respect to time.



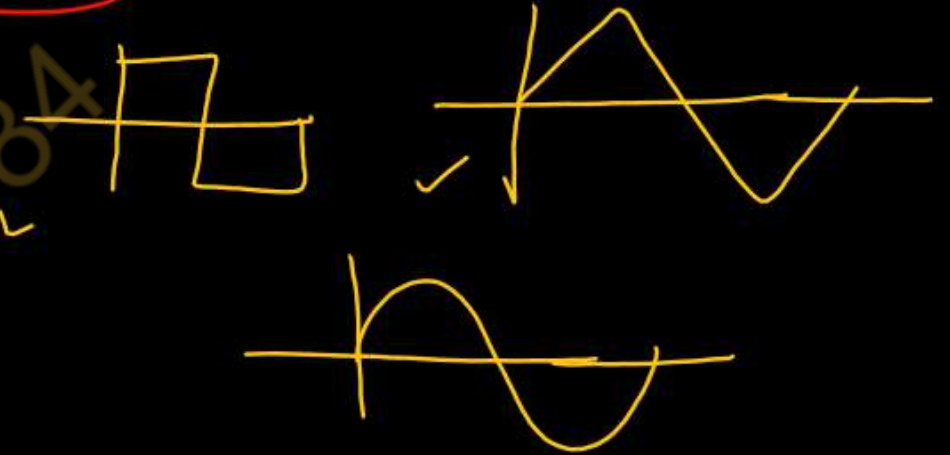
Alternating Current (A.C.) Supply:- Alternating supply is one whose magnitude and direction changes periodically with time.

Advantage of AC over DC

1. Varying voltage level of AC by using transformer.
2. More Economical than DC ✓
3. Generation of high voltage AC is much easier & Cheaper than DC ✓
4. AC can be easily converted to DC. ✓
5. AC machine are more simple in construction and requires less maintenance

Why We should use Pure Sine Wave ?

- ✓ Easy to write equation for pure sine wave.
- Any other wave form can be obtained by a series of sine or cosine wave form.
- Sine or cosine wave can pass through linear circuit without any distortion.
- ✓ Analysis of electrical networks with sine inputs is very easy.



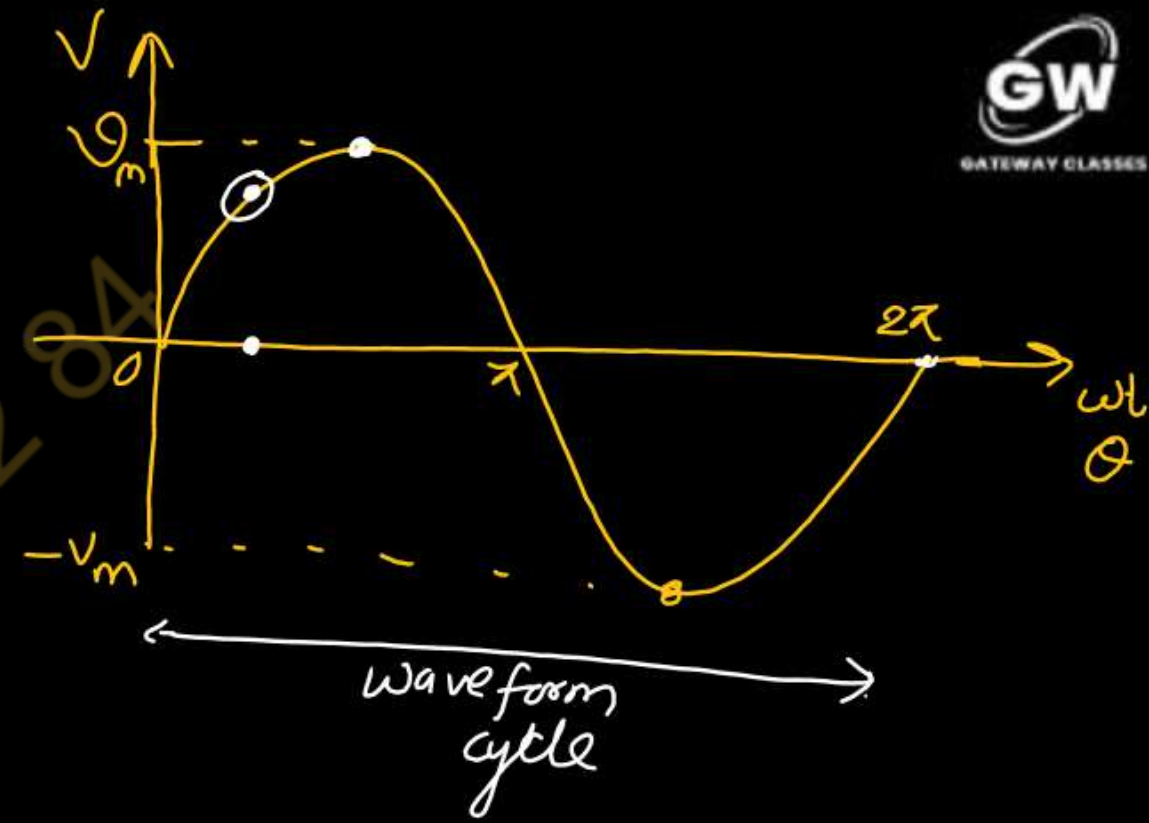
Equation of Alternating Quantity (AKTU-2002-03):-

For a pure sine wave

$$v = V_m \sin \omega t$$

$$v = V_m \sin \theta$$

This equation is also known as instantaneous equation

**Important Terms related to Alternating Quantity -**

Instantaneous Value:- The value of any alternating quantity at any instant of time

Waveform Cycle:- Each repetition of positive and negative cycle of A.C quantity

Time period:- Time taken to complete one cycle (T)

Frequency:- No of cycle completed in one second $(F) = \frac{1}{T}$

Alternating Quantity

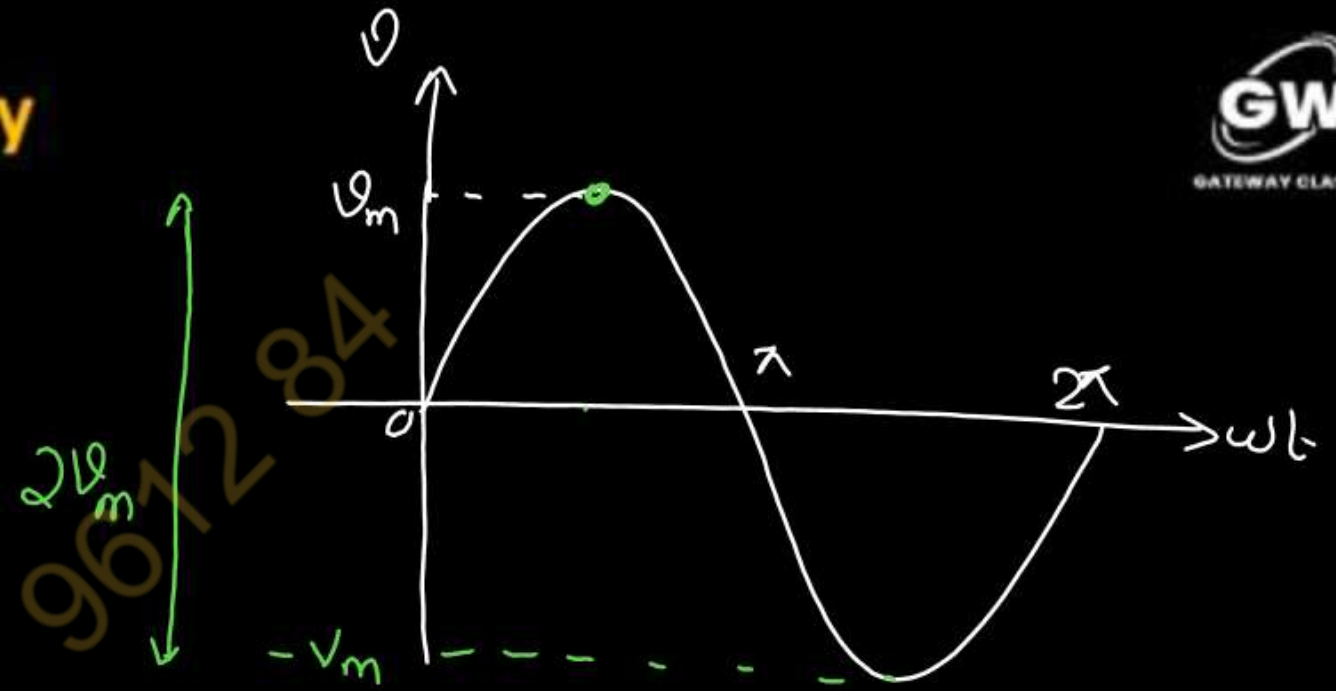
- **Amplitude:-** The maximum value of A.C. quantity is known as amplitude (V_m)

- **Angular Frequency:-** It is expressed in rad/sec.

$$\omega = 2\pi f$$

- **Peak to Peak Value:-** Value of A.C. quantity from positive maximum to negative maximum

$$V_{p.p} = 2V_m$$



Examples of Alternating Quantity

Ex-1 A sinusoidally varying alternating current of frequency 60 Hz has a maximum value of 15 amp.

(i) Write equation for instantaneous value, Find the value of current after 1/200 second

(ii) Find the time to reach 10 amperes for the first time?

(AKTU-2002-03)

Solⁿ Given $f = 60 \text{ Hz}$, $I_m = 15 \text{ amp}$

$$\omega = 2\pi f = 2\pi \times 60 = 120\pi$$

(i) Instantaneous Equation

$$i^o = I_m \sin \omega t$$

$$\boxed{i^o = 15 \sin 120\pi t}$$

find $i^o = ?$ at $t = \frac{1}{200} \text{ Sec.}$

$$i^o = 15 \sin 120\pi \times \frac{1}{200}$$

$$i^o = 14.26 \text{ amp}$$

(ii) $i^o = 10$, $t = ?$

$$i^o = I_m \sin \omega t$$

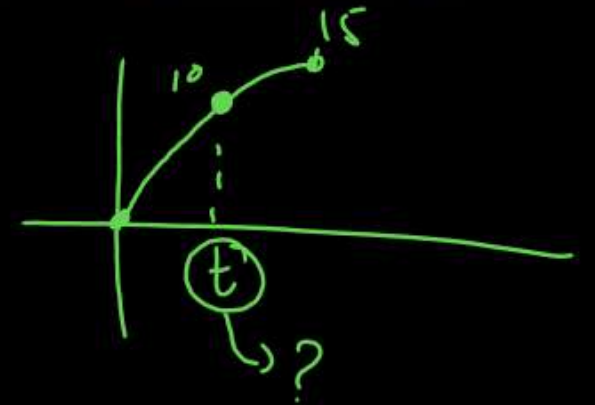
$$10 = 15 \sin 120\pi t$$

$$\sin 120\pi t = \frac{10}{15} = \frac{2}{3}$$

$$120\pi t = \sin^{-1}\left(\frac{2}{3}\right)$$

$$t = \frac{\sin^{-1}\left(\frac{2}{3}\right)}{120\pi}$$

$$\boxed{t = 1.93 \text{ ms}}$$



→ radian

Examples of Alternating Quantity

Ex-2 The equation of an alternating current is $i = 42.42 \sin 628t$ Determine Maximum Value, Frequency.
(AKTU-2017-18, similar question in 2002-03, 2003-4, 2005-06, 2007-08)

Sol.-

$$i = 42.42 \sin 628t$$

$$i = I_m \sin \omega t$$

$$I_m = 42.42 \text{ ap}$$

$$\omega = 628$$

$$2\pi f = 628$$

$$f = \frac{628}{2\pi} = \frac{628}{2 \times 3.14}$$

$$f = 50 \text{ Hz}$$

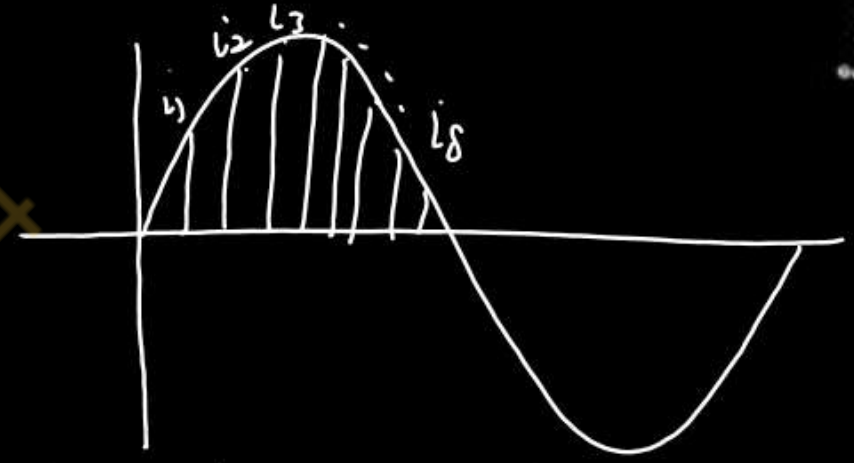
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The average of all the instantaneous values of an alternating voltage and currents over one complete cycle is called **Average Value**.

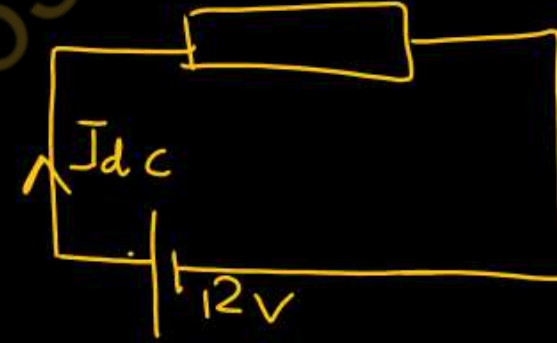
Or

The Average Value (also known as Mean Value) of an Alternating Current (AC) is expressed by that Direct Current (DC) which transfers across any circuit the same amount of charge as is transferred by that Alternating Current (AC) during the same time

Average Value by Graphical Method:-

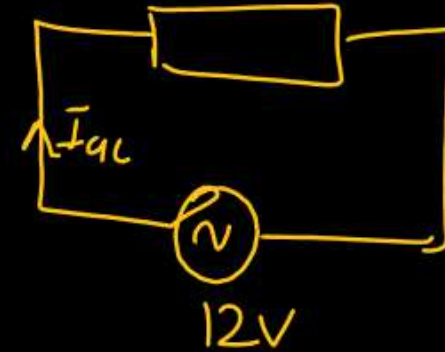


$$I_{avg} = \frac{i_1 + i_2 + i_3 + \dots + i_n}{n}$$



Charge transferred by d.c
Current

||



Charge transferred by A.c
Current. I_{ac}

Average Value (Mean Value)

Average Value by Analytical Method:-

$$I_{avg} = \frac{\text{area under the curve}}{\text{Duration of the curve}}$$

Ex for a Sine wave

$$v = V_m \sin \theta$$

$$V_{avg} = \frac{\int_0^\pi v \cdot d\theta}{\pi} \longrightarrow \text{area}$$

$$= \frac{1}{\pi} \int_0^\pi V_m \sin \theta \cdot d\theta$$

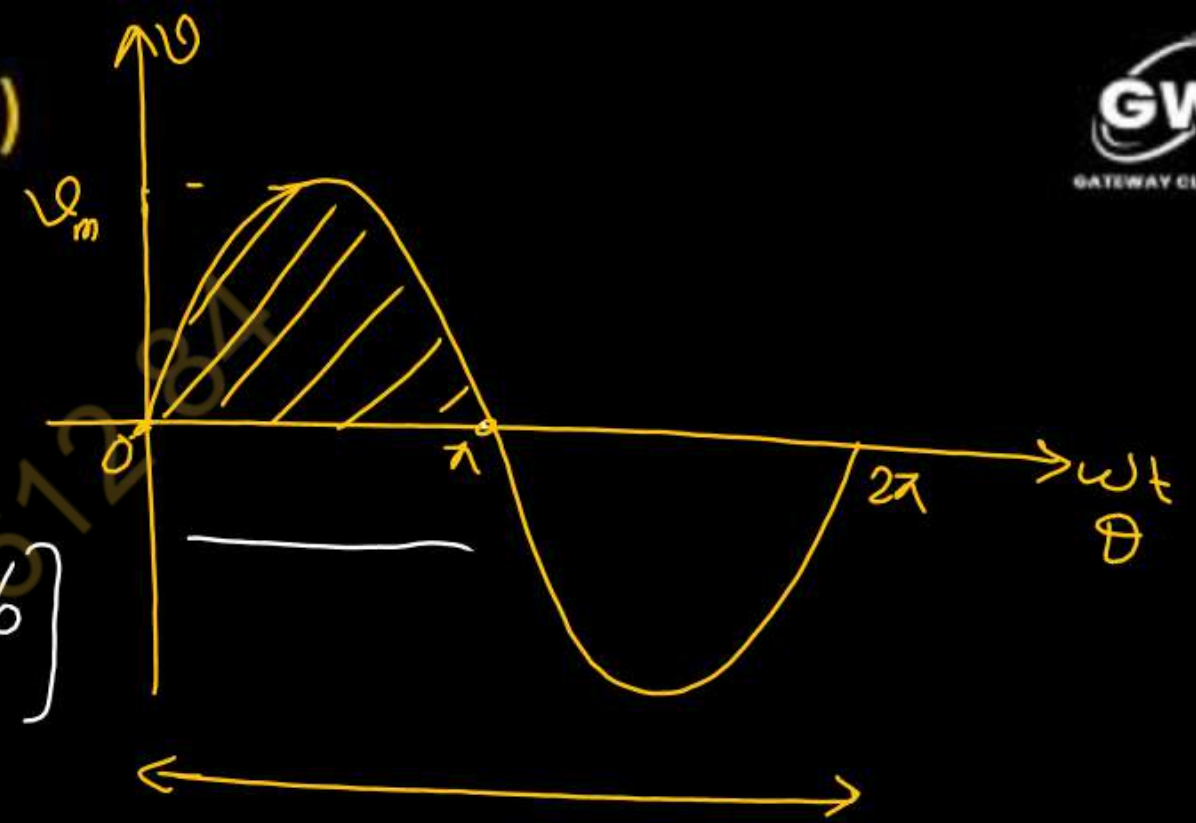
$$= \frac{V_m}{\pi} \left[-\cos \theta \right]_0^\pi$$

$$V_{avg} = \frac{V_m}{\pi} \left[-\cos \theta \Big|_0^\pi + \cos \theta \Big|_0^\pi \right]$$

$$V_{avg} = \frac{V_m}{\pi} \left[+1 + 1 \right]$$

$$V_{avg} = \frac{2V_m}{\pi}$$

for an Sine wave.

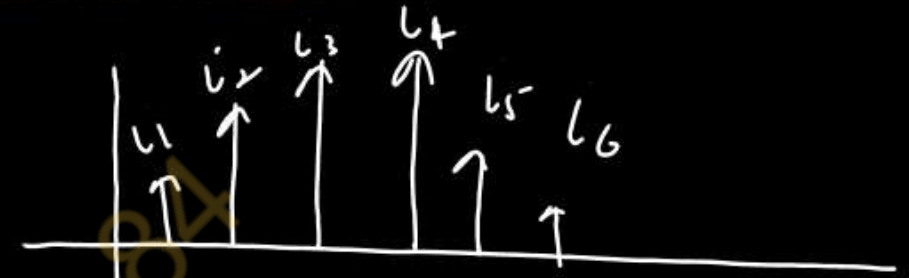


Root Mean Square (R.M.S.) Value/ Effective Value



Definition: The RMS value of AC current is equal to that amount of DC current which produces the same heating effect flowing through the same resistance for the same time.

For example, if 3A (RMS) AC current is flowing through a circuit, And it will produce the same amount of heat (energy) as will be produced by 3A DC current.



$$I_{rms} = \sqrt{\frac{i_1^2 + i_2^2 + i_3^2 + \dots + i_n^2}{n}}$$

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Analytical Method:-

$$I_{rms} = \sqrt{\frac{\text{Area of half cycle squared wave}}{\text{Duration of wave}}}$$

Ex For a Sin wave

$$i^{\circ} = I_m \sin \theta$$

first square the wave

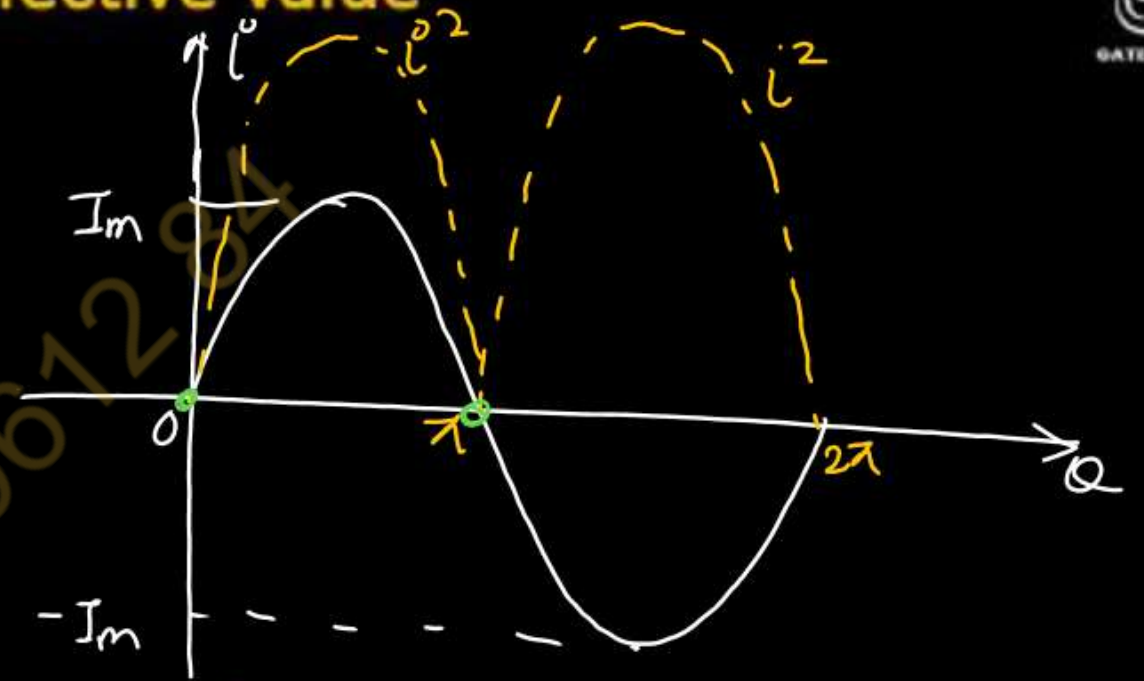
$$i^{\circ 2} = I_m^2 \sin^2 \theta$$

$$I_{rms} = \sqrt{\frac{\int_0^{\pi} i^{\circ 2} d\theta}{\pi}}$$

$$I_{rms}^2 = \frac{1}{\pi} \int_0^{\pi} I_m^2 \sin^2 \theta d\theta$$

$$\begin{aligned} I_{rms}^2 &= \frac{I_m^2}{\pi} \int_0^{\pi} \sin^2 \theta d\theta \\ &= \frac{I_m^2}{\pi} \int_0^{\pi} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta \\ &= \frac{I_m^2}{2\pi} \int_0^{\pi} (1 - \cos 2\theta) d\theta \\ &= \frac{I_m^2}{2\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi} \end{aligned}$$

$$I_{rms}^2 = \frac{I_m^2}{2\pi} \left[\left(\pi - \frac{\sin 2\pi}{2} \right) - \left(0 - \frac{\sin 0}{2} \right) \right]$$



$$I_{rms}^2 = \frac{I_m^2}{2\pi} (\cancel{\pi})$$

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

FUNDAMENTALS OF ELECTRICAL ENGINEERING

UNIT -2 : Steady State Analysis of 1- Φ Circuits

Lec-2

Today's Target

- ✓ Form Factor & Peak Factor
- ✓ PYQ & Examples
- ✓ DPPs

Form Factor & Peak Factor

- **Form Factor:-** It is the ratio of r.m.s value to the average value of an alternating quantity.

$$\text{Form factor} = \frac{\text{R.m.s Value}}{\text{Average Value}}$$

For Sine wave

$$= \frac{E_{rms}}{E_{avg}} = \frac{V_m/\sqrt{2}}{\frac{2V_m}{\pi}} = 1.11 =$$

- **Peak Factor/ Crest Factor :-** It is the ratio of Peak Value to the rms value of an alternating quantity

$$\text{Peak factor} = \frac{\text{Peak Value}}{\text{rms Value}}$$

for Sine wave

$$= \frac{V_m}{\frac{V_m}{\sqrt{2}}} = \sqrt{2} = 1.414 \quad \underline{\text{Ans}}$$

Ex-1 Determine Form Factor and Peak Factor for Full wave rectified Signal ($V_m = 10\text{volt}$) ? (AKTU-2002-03)

from 0 to π

$$v = V_m \sin \theta = 10 \sin \theta$$

$$V_{avg} = \frac{1}{\pi} \int_0^{\pi} 10 \sin \theta d\theta$$

$$= \frac{10}{\pi} [-\cos \theta]_0^{\pi}$$

$$= \frac{10}{\pi} [-\cos \pi + \cos 0]$$

$$V_{avg} = \frac{2 \times 10}{\pi} = \frac{20}{\pi}$$

$$v = 10 \sin \theta$$

$$v^2 = 100 \sin^2 \theta$$

$$V_{rms}^2 = \frac{\int_0^{\pi} 100 \sin^2 \theta d\theta}{\pi}$$

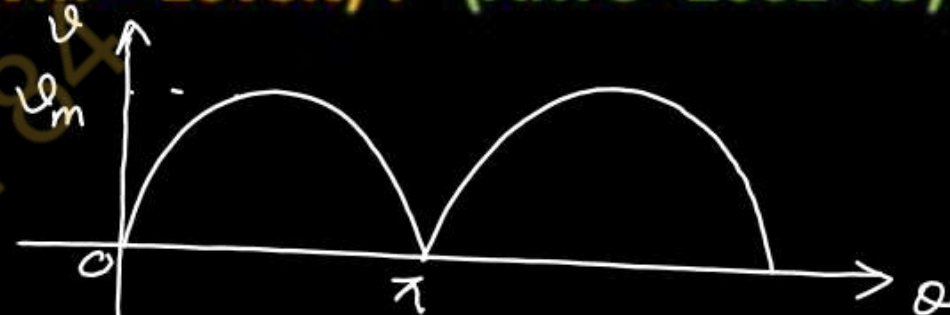
$$= \frac{100}{\pi} \int_0^{\pi} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta$$

$$= \frac{50}{\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi}$$

$$= \frac{50}{\pi} \left[\pi - \frac{\sin 2\pi}{2} - 0 + \frac{\sin 0}{2} \right]$$

$$V_{rms}^2 = 50$$

$$V_{rms} = \sqrt{50}$$



$$\text{Form factor} = \frac{R_{ms}}{avg} = \frac{\sqrt{50}}{\left(\frac{20}{\pi}\right)}$$

$$= 1.11$$

$$\text{Peak factor} = \frac{Peak}{R_{ms}} = \frac{10}{\sqrt{50}}$$

$$= 1.414$$

Ans

Ex-2 Determine Form Factor and Peak Factor for Half wave rectified Signal ($V_m = 10\text{volt}$) ? (AKTU- 2003-04, 2020-21)

Sol

$$v = V_m \sin \theta \quad \begin{matrix} 0 \text{ to } \pi \\ \pi \text{ to } 2\pi \end{matrix}$$

$$v_{avg} = \frac{1}{2\pi} \int_0^{2\pi} v \cdot d\theta$$

$$= \frac{1}{2\pi} \left[\int_0^{\pi} V_m \sin \theta d\theta + \int_{\pi}^{2\pi} 0 d\theta \right]$$

$$= \frac{V_m}{2\pi} \left[-\cos \theta \right]_0^{\pi}$$

$V_{avg} = \frac{V_m}{\pi} = \frac{10}{\pi}$

Imp

For rms Value

$$v^2 = V_m^2 \sin^2 \theta \quad \begin{matrix} 0 \text{ to } \pi \\ \pi \text{ to } 2\pi \end{matrix}$$

$$v_{rms}^2 = \frac{1}{2\pi} \int_0^{2\pi} v^2 d\theta$$

$$= \frac{1}{2\pi} \left[\int_0^{\pi} V_m^2 \sin^2 \theta d\theta + \int_{\pi}^{2\pi} 0 d\theta \right]$$

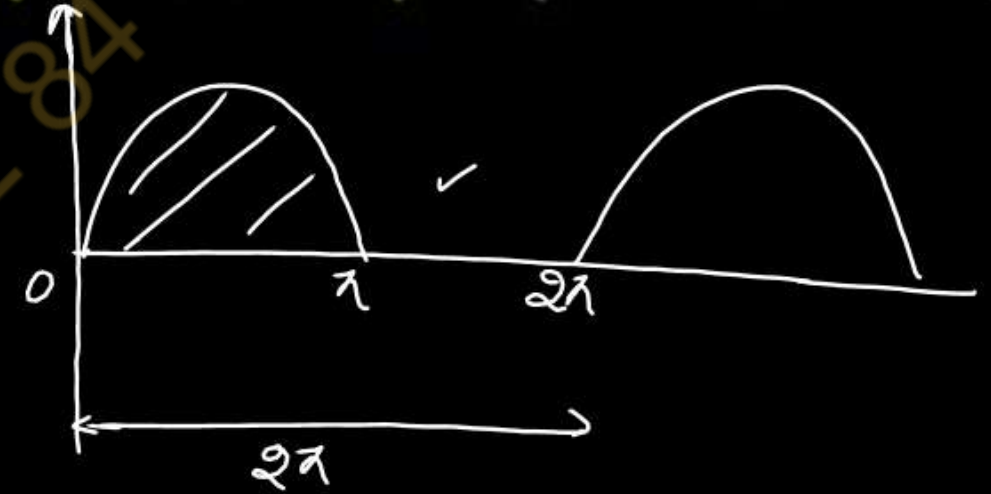
$$= \frac{V_m^2}{2\pi} \left[\int_0^{\pi} \left(\frac{1 - \cos 2\theta}{2} \right) d\theta \right]$$

$$= \frac{V_m^2}{4\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{\pi}$$

$$= \frac{V_m^2}{4\pi} \left[\pi - \frac{\sin 2\pi}{2} - 0 + \frac{\sin 0}{2} \right]$$

$$v_{rms}^2 = \frac{V_m^2}{4}$$

$V_{rms} = \frac{V_m}{2} = \frac{10}{2} = 5 \text{ volt}$



$$\text{Form factor} = \frac{5}{\left(\frac{10}{\pi}\right)} = 1.57$$

$$\text{Peak factor} = \frac{10}{5} = 2$$

$$y = mx + c$$

$$p=10$$

$$B=1$$

$$m=10$$

Numerical

Ex-3 Determine Form Factor and Peak Factor for the given Signal ? (AKTU- 2004-05, 2017-18)

$$V = 10t \quad \longrightarrow \quad 0 \text{ to } 1$$

$$1 \text{ to } 2$$

$$V = -10t + c$$

$$\text{at } t=2, V=0$$

$$0 = -10 \times 2 + c$$

$$\boxed{c = 20}$$

$$V = -10t + 20 \quad 1 \text{ to } 2$$

$$\begin{cases} V = 10t & 0 < t < 1 \\ V = -10t + 20 & 1 < t < 2 \end{cases}$$

$$V_{avg} = \frac{1}{2} \int_0^2 V \cdot dt$$

$$= \frac{1}{2} \left[\int_0^1 10t \, dt + \int_1^2 (-10t + 20) \, dt \right]$$

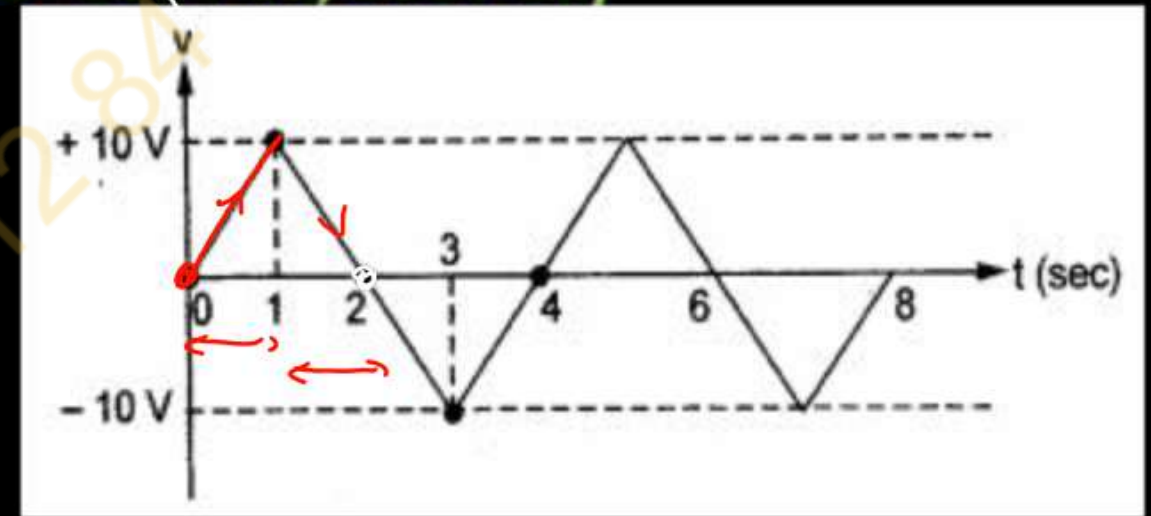
$$= \frac{1}{2} \left[\left[\frac{10t^2}{2} \right]_0^1 + \left[-\frac{10t^2}{2} + 20t \right]_1^2 \right]$$

$$= \frac{1}{2} \left[\left(\frac{10 \times 1}{2} \right) + \left[\left(-\frac{10 \times 4}{2} + 20 \times 2 \right) - \left(-\frac{10}{2} + 20 \right) \right] \right]$$

$$= \frac{1}{2} \left[5 + [+20 - 15] \right]$$

$$= \frac{1}{2} [5 + 5] = \frac{10}{2} = 5 \text{ volt}$$

$$\boxed{V_{avg} = 5 \text{ volt}}$$



$$V = 10t \quad 0 < t < 1$$

$$= -10t + 20 \quad 1 < t < 2$$

$$V^2 = 100t^2 \quad 0 < t < 1 \quad \text{--- (1)}$$

$$= (-10t + 20)^2$$

$$= (100t^2 + 400 - 400t) \quad 1 < t < 2 \quad \text{--- (2)}$$

$$V_{rms}^2 = \frac{1}{2} \int_0^2 V^2 dt$$

$$= \frac{1}{2} \left[\int_0^1 100t^2 dt + \int_1^2 (100t^2 + 400 - 400t) dt \right]$$

$$= \frac{1}{2} \left[100 \left[\frac{t^3}{3} \right]_0^1 + \left[100 \left[\frac{t^3}{3} \right]_1^2 + 400[t]_1^2 - 400 \left[\frac{t^2}{2} \right]_1^2 \right] \right]$$

$$= \frac{1}{2} \left[\left(100 \cdot \frac{1}{3} \right) + \left[100 \left(\frac{8}{3} - \frac{1}{3} \right) + 400[2-1] - 400 \left(\frac{4}{2} - \frac{1}{2} \right) \right] \right]$$

$$= \frac{1}{2} \left[\frac{100}{3} + \frac{700}{3} + 400 - \frac{400 \times 3}{2} \right]$$

$$= \frac{1}{2} \left[\frac{800}{3} - 200 \right]$$

$$V_{rms}^2 = \frac{1}{2} \left[\frac{800 - 600}{3} \right] = \frac{1}{2} \frac{200}{3} = \frac{100}{3}$$

$$V_{rms} = \sqrt{\frac{100}{3}} = \frac{10}{\sqrt{3}} = 5.77 \text{ Volt}$$

$$\text{Form factor} = \frac{5.77}{5} = 1.15$$

$$\text{Peak factor} = \frac{10}{5.77} = 1.73$$

Ans

Ex-4 Determine Form Factor and Peak Factor for the given Signal ? (AKTU- 2012-13)

$$v = A \quad 0 \text{ to } \pi$$

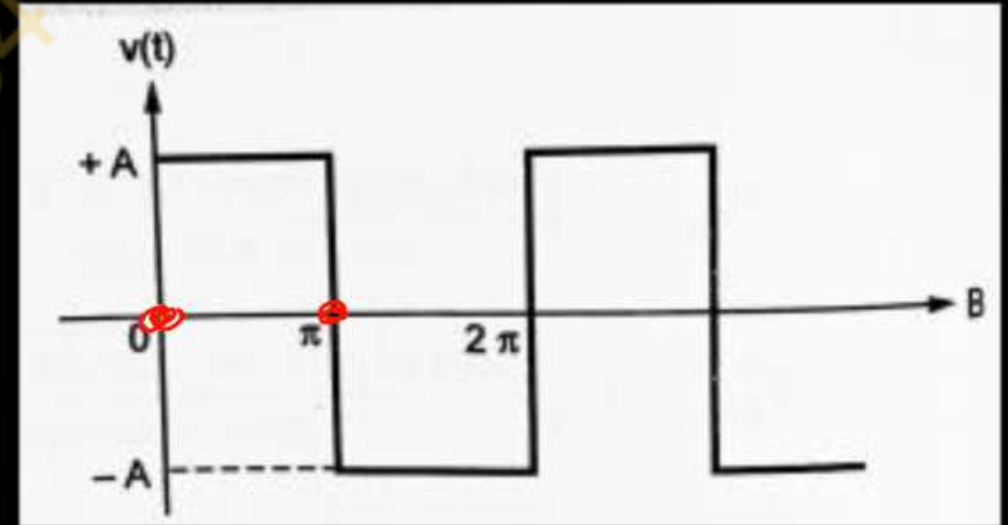
$$V_{avg} = \frac{1}{\pi} \int_0^{\pi} A \cdot dt$$
$$= \frac{A}{\pi} [t]_0^{\pi}$$

$$V_{avg} = \frac{A}{\pi} \cdot \pi = A$$

$$V_{rms}^2 = \frac{1}{\pi} \int_0^{\pi} A^2 dt$$
$$= \frac{A^2}{\pi} [t]_0^{\pi}$$

$$V_{rms}^2 = \frac{A^2}{\pi} \cdot \pi = A^2$$

$$V_{rms} = A$$



$$m = \frac{200}{2} = 100$$

Numerical

Ex-5 Determine Form Factor and Peak Factor for the given Signal ? (AKTU- 2004-05, 2017-18)

$$v = 100t \quad \text{---} \quad 0 < t < 2$$

$$V_{avg} = \frac{1}{2} \int_0^2 v \cdot dt$$

$$= \frac{1}{2} \int_0^2 100t \, dt$$

$$= \frac{100}{2} \left[\frac{t^2}{2} \right]_0^2$$

$$= \frac{100}{4} [2^2 - 0]$$

$$\boxed{V_{avg} = 100 \text{ volt}}$$

$$v^2 = 100^2 t^2$$

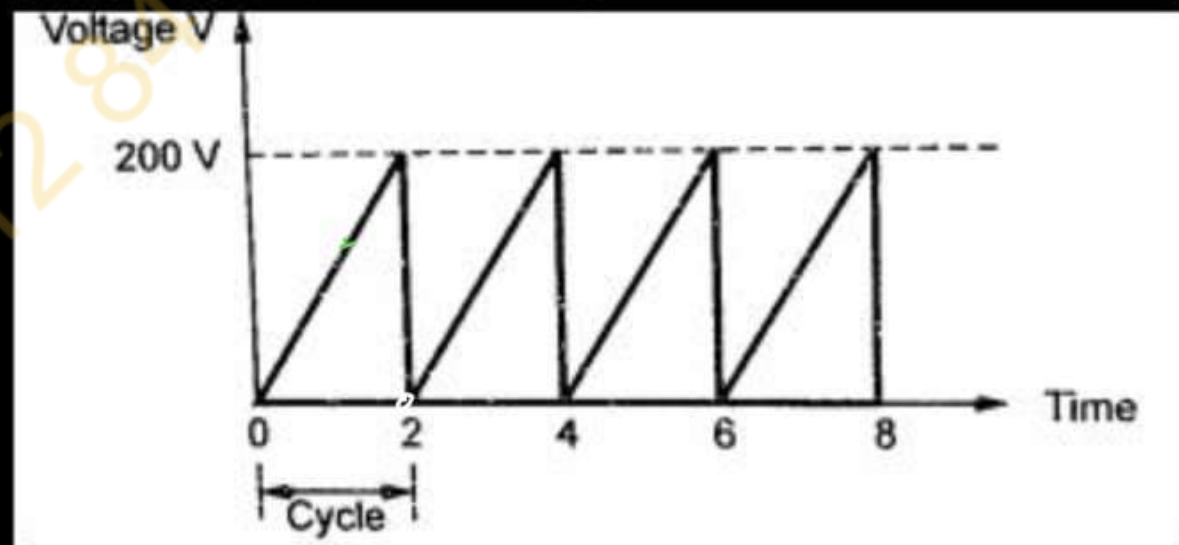
$$V_{rms}^2 = \frac{1}{2} \int_0^2 100^2 t^2 \, dt$$

$$= \frac{100^2}{2} \left[\frac{t^3}{3} \right]_0^2$$

$$= \frac{100^2}{2 \times 3} [8 - 0]$$

$$= \frac{100^2}{6} \times 8 = \frac{40000}{3}$$

$$V_{rms} = \sqrt{\frac{40000}{3}} = 115.4 \text{ Volt}$$

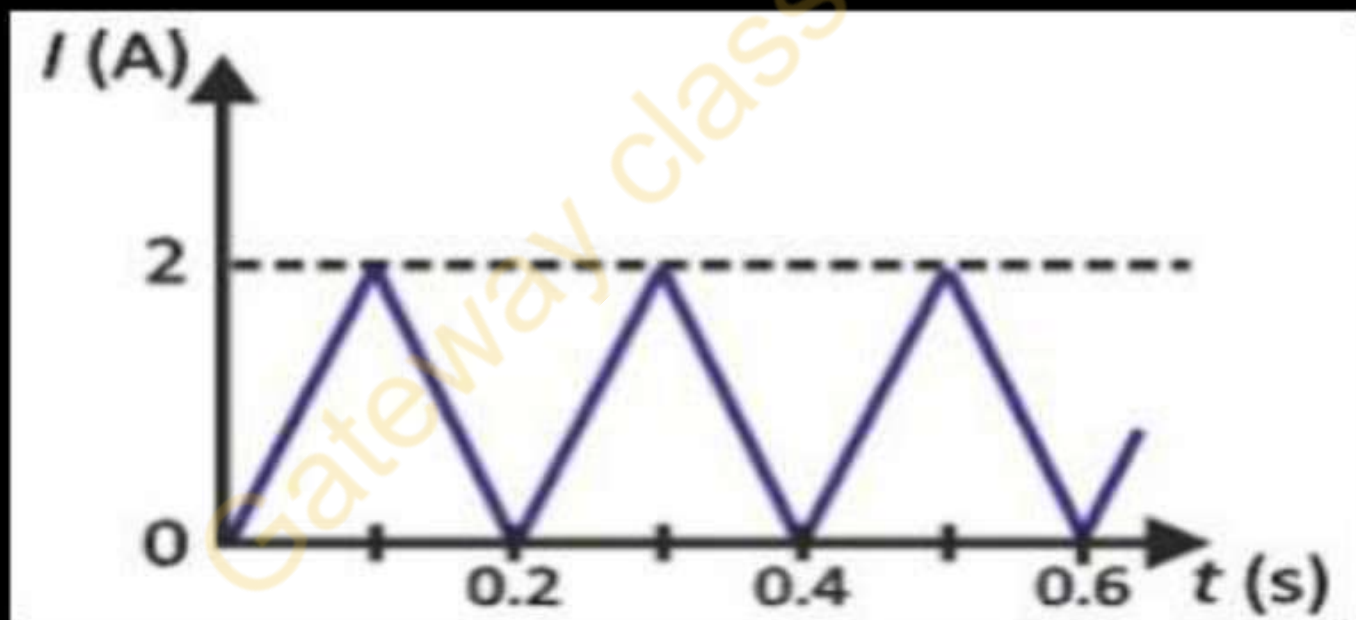


Q.1 A sinusoidally varying alternating current of frequency 50 Hz has a maximum value of 10 amperes: (i) Write down the equation for instantaneous value. (ii) Find the value of current after $1/100$ second. (iii) Find the time to reach 7 amp for the first time and (iv) Find its average value.

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Q2 Find the Average value and RMS value of the following wave.

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FUNDAMENTALS OF ELECTRICAL ENGINEERING

UNIT -2 : Steady State Analysis of 1- Φ Circuits

Lec-3

Today's Target-

- ✓ Concept of Leading and Lagging
- ✓ Use of Scientific Calculator
- ✓ Polar and Rectangular system
- ✓ Analysis of AC circuits
- ✓ Pure R,L,C & Series R-L , R-C Circuits
- ✓ DPPs

Concept of Phase and Phase Difference

- **Phase** :- The Phase of an AC quantity is the angle travelled by phasor representing that alternating quantity up to the instant of consideration, measured from reference.

$$i = I_m \sin(\omega t + \phi)$$

- **when $\phi = 0$:** At $t=0$ AC quantity has Zero instantaneous Value.
- **When ϕ is positive :** At $t=0$ AC quantity has some positive instantaneous Value.
- **When ϕ is negative :** At $t=0$ AC quantity has some negative instantaneous Value.



Phase difference :- The difference between the phase of two Alternating Quantities of same frequency is known as phase difference.

Zero Phase difference :- Same phase, In phase
Phase diff = 0

$$V = V_m \sin \omega t, \quad I = I_m \sin \omega t$$

Leading Phase Difference :-

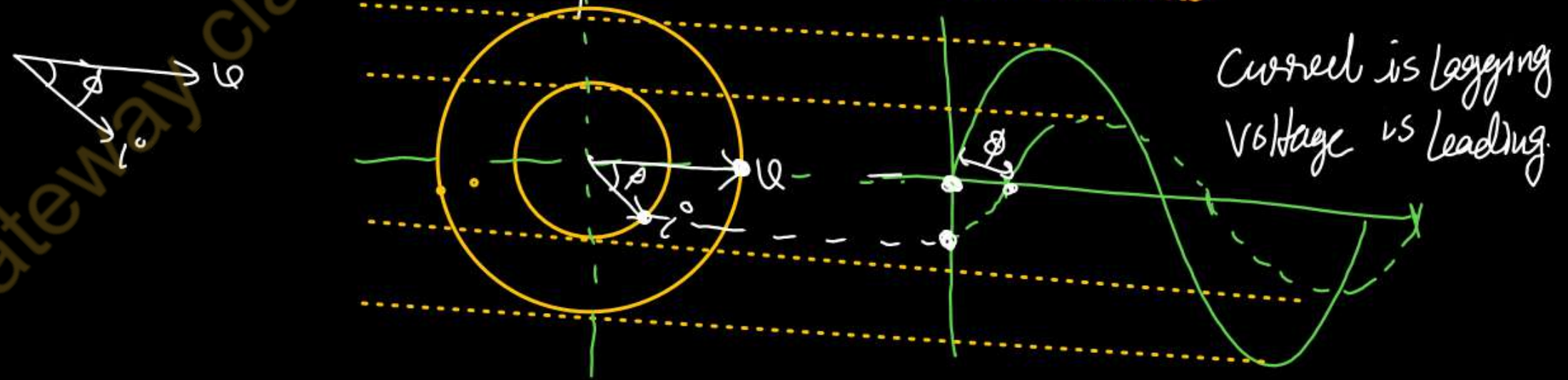
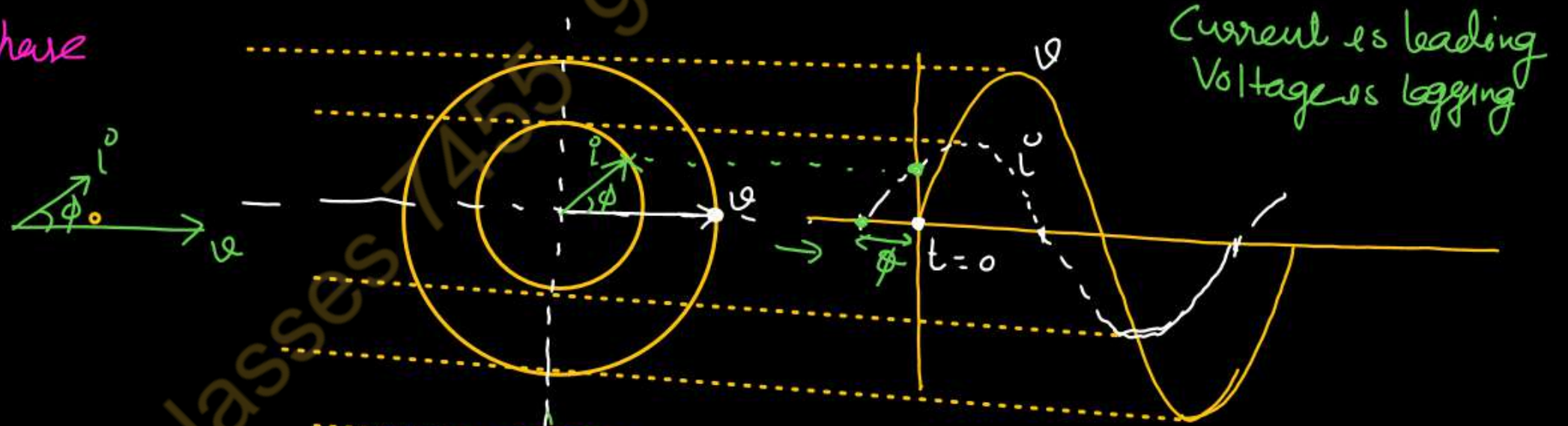
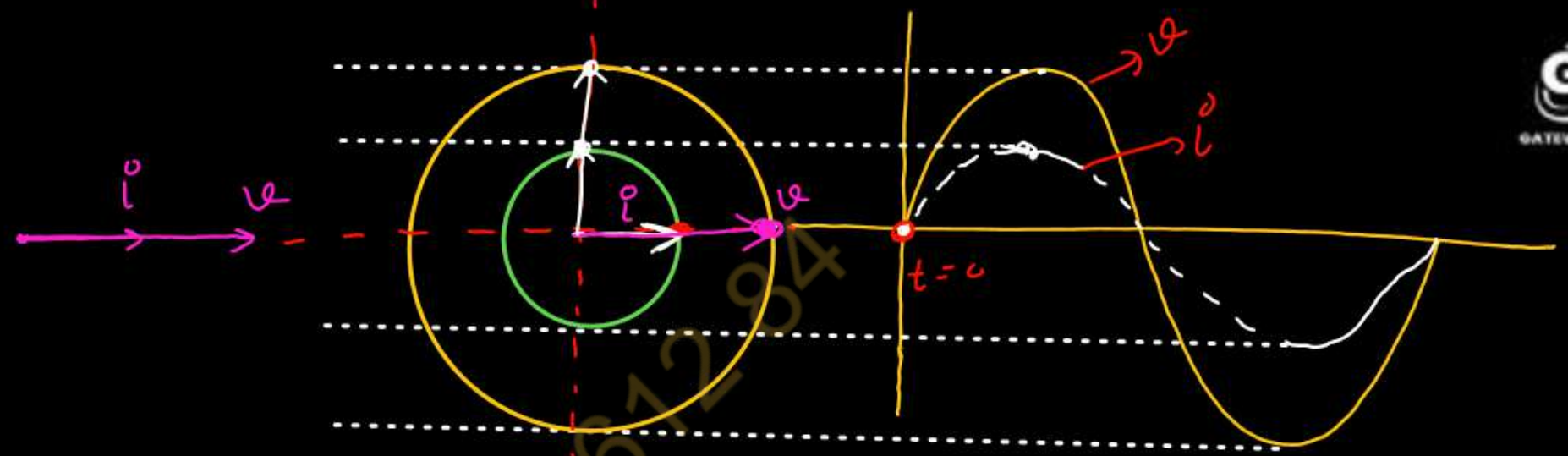
$$V = V_m \sin \omega t$$

$$I = I_m \sin(\omega t + \phi)$$

Lagging Phase Difference :-

$$V = V_m \sin \omega t$$

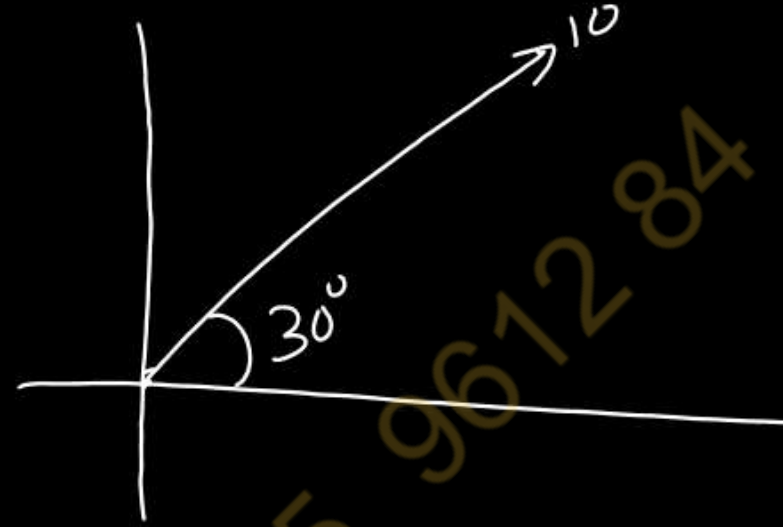
$$I = I_m \sin(\omega t - \phi)$$



Concept of Polar System:-

Handwritten diagram illustrating the polar representation of a complex number. It shows a magnitude r and an angle θ relative to the positive horizontal axis. The angle is labeled "angle".

$$z = 10 \angle 30^\circ$$



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Concept of Rectangular System:-

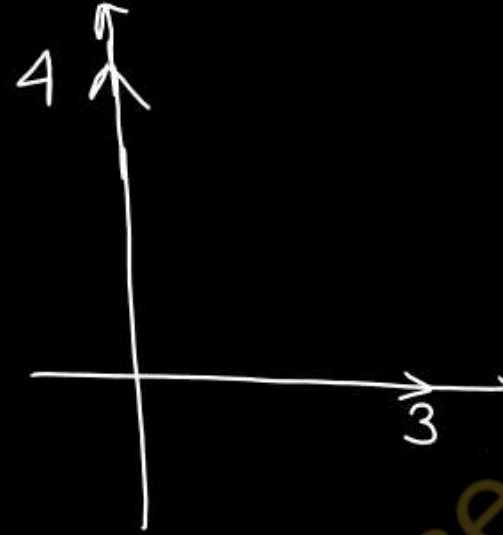
$$\text{Real} \leftarrow A + jB \rightarrow \text{Imag}$$

$$V = 3 + 4j$$

$$z = r \cos \theta + j r \sin \theta$$

$$r = \sqrt{A^2 + B^2}$$

$$\theta = \tan^{-1} \frac{B}{A}$$



$$-1 = j^2$$

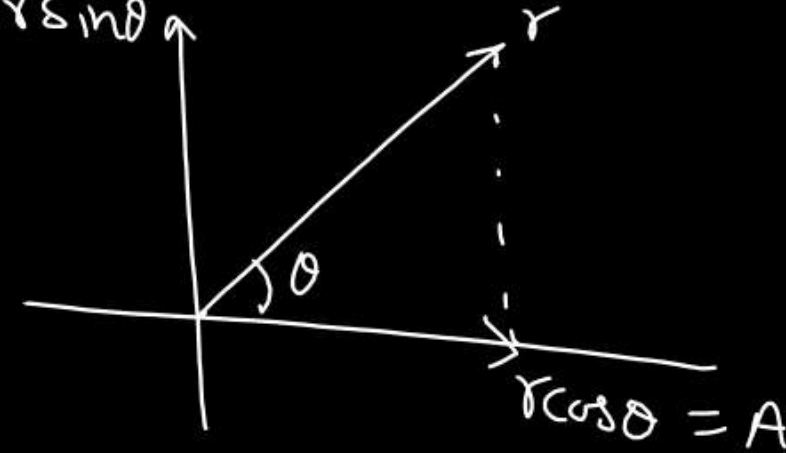


$$j = \sqrt{-1}$$

$$j^2 = -1$$

$$j \cdot j^2 = -j$$

$$B = r \sin \theta$$



Polar to Rectangular conversion & Vice versa

Complex mode ON

Polar to Rectangular conversion

$$r \angle \theta \longrightarrow A + iB$$

$$10 \angle 30^\circ \longrightarrow 10 \boxed{[]} \boxed{[]} \boxed{[]} 30 \boxed{[]} = 8.66 + 5j$$

Add.

Sub

Div

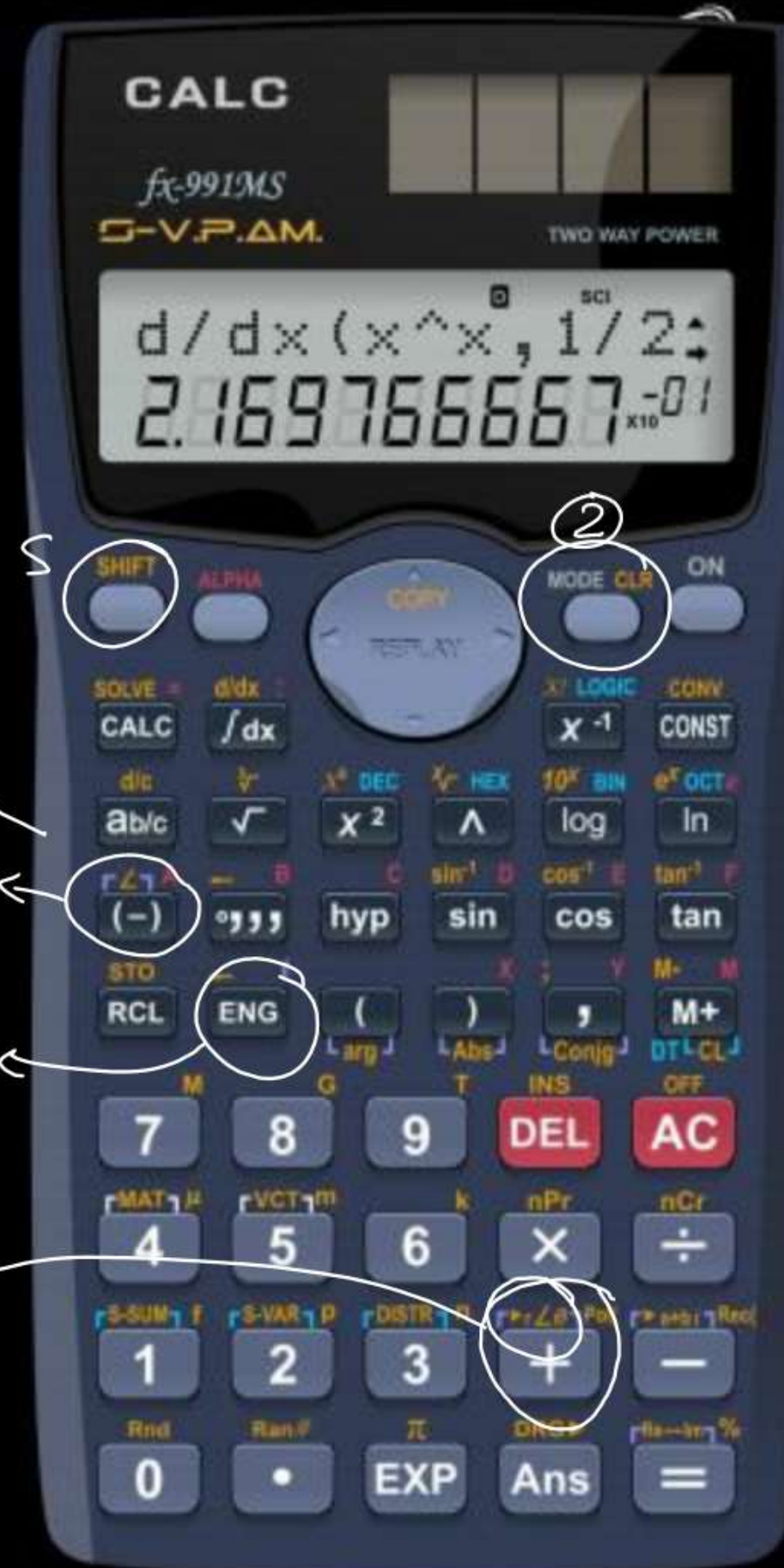
Multi

Always kept Complex quantity in bracket

Rectangular to Polar conversion

$$(3 + 5j) \rightarrow r \angle \theta$$

$$(3 + 5 \boxed{[]} \boxed{[]}) \boxed{[]} \boxed{[]} = 5.8 \angle 59.03$$



$\boxed{[]} \boxed{[]}$

Current & Voltage Equations

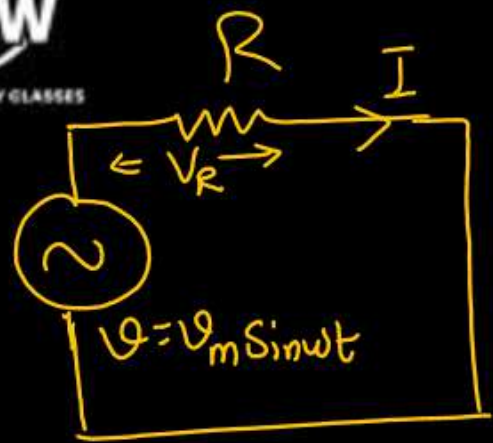
Reactance and Impedance

Phase Relationship

Power in AC Circuits

Waveform Analysis

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In Circuit

$$V = V_m \sin \omega t \quad \text{--- (1)}$$

for Current

$$I = \frac{V}{R} = \frac{V_m}{R} \sin \omega t$$

$$I = I_m \sin \omega t \quad \text{--- (2)}$$

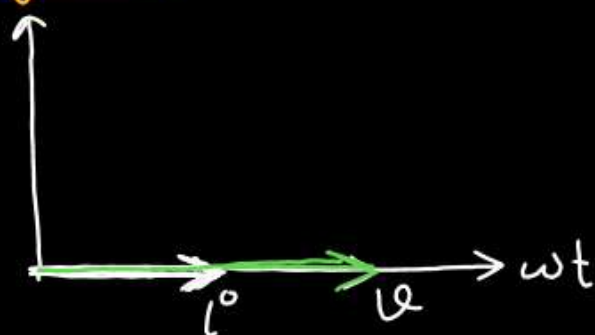
Where $I_m = \frac{V_m}{R}$

Phase diff. = 0

Voltage and Current are in Same phase

A.C. Through Pure Resistive Circuit

Phasor diagram



Power $\rightarrow$

Instantaneous power

$$\begin{aligned} P &= V \cdot I \\ &= V_m \sin \omega t \cdot I_m \sin \omega t \\ &= V_m I_m \sin^2 \omega t \\ &= V_m I_m \left(\frac{1 - \cos 2\omega t}{2} \right) \end{aligned}$$

$$P = \frac{V_m I_m}{2} - \frac{V_m I_m}{2} \cos 2\omega t$$

$\rightarrow \text{avg.} = 0$

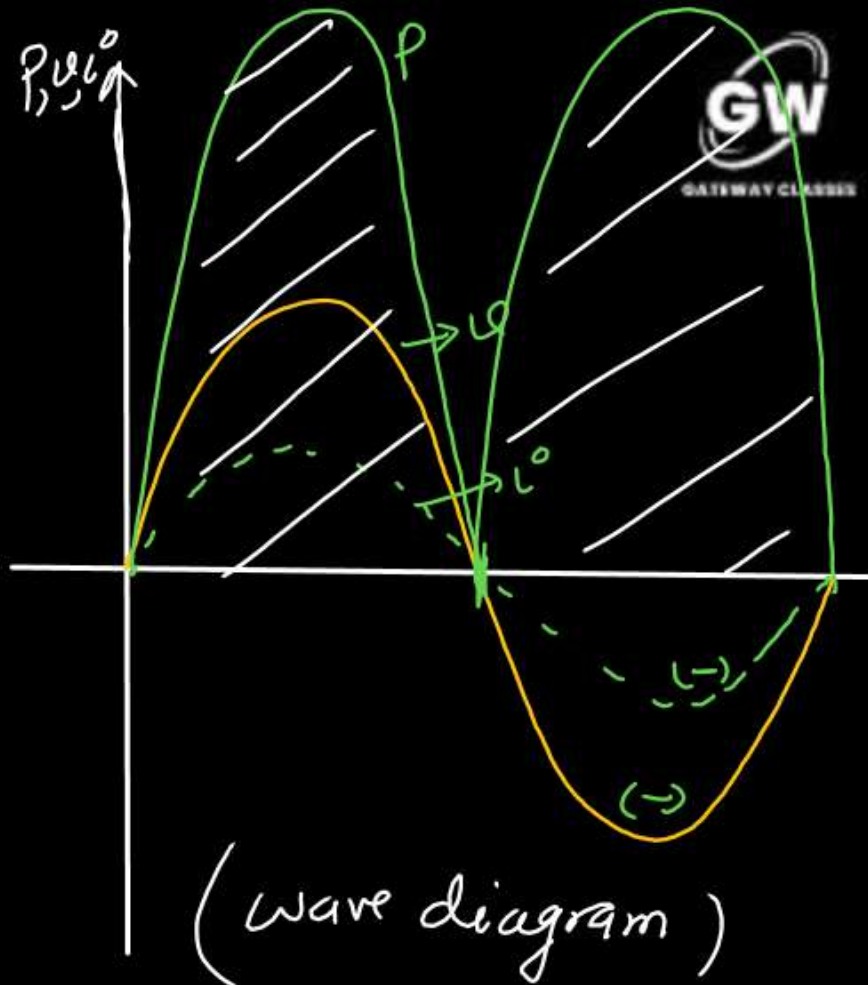
Average power

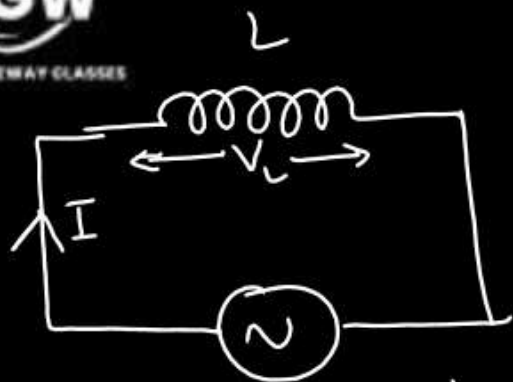
$$P_{\text{avg}} =$$

Over a complete cycle average value of any alt. quantity is always zero.

$$\begin{aligned} P_{\text{avg}} &= \frac{V_m I_m}{2} \\ &= \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \end{aligned}$$

$$P_{\text{avg}} = V \cdot I$$





$$V = V_m \sin \omega t$$

$$V = V_m \sin \omega t \quad \text{--- (1)}$$

$$e = -L \frac{di}{dt}$$

$$V_L = -e = L \frac{di}{dt}$$

$$\frac{di}{dt} = \frac{V}{L} = \frac{V_m \sin \omega t}{L}$$

Integrate both side

$$i = \frac{V_m}{L} \int \sin \omega t \, dt$$

$$i = \frac{V_m}{\omega L} (-\cos \omega t)$$

$$i = I_m \sin(\omega t - \frac{\pi}{2}) \quad \text{--- (2)}$$

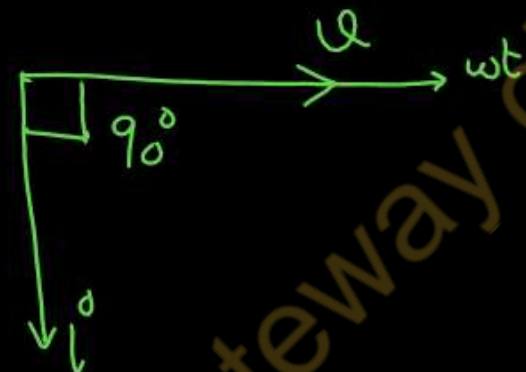
$$I_m = \frac{V_m}{\omega L}$$

$$\omega L = \frac{V_m}{I_m} = X_L = \text{Inductive Reactance, } (\Omega)$$

$$X_L = \omega L$$

from Equ (1) and (2)
we can see that current is lagging behind the voltage by 90°

Phasor diagram



A.C. Through Pure Inductive Circuit

Power (Instantaneous)

$$P = VI$$

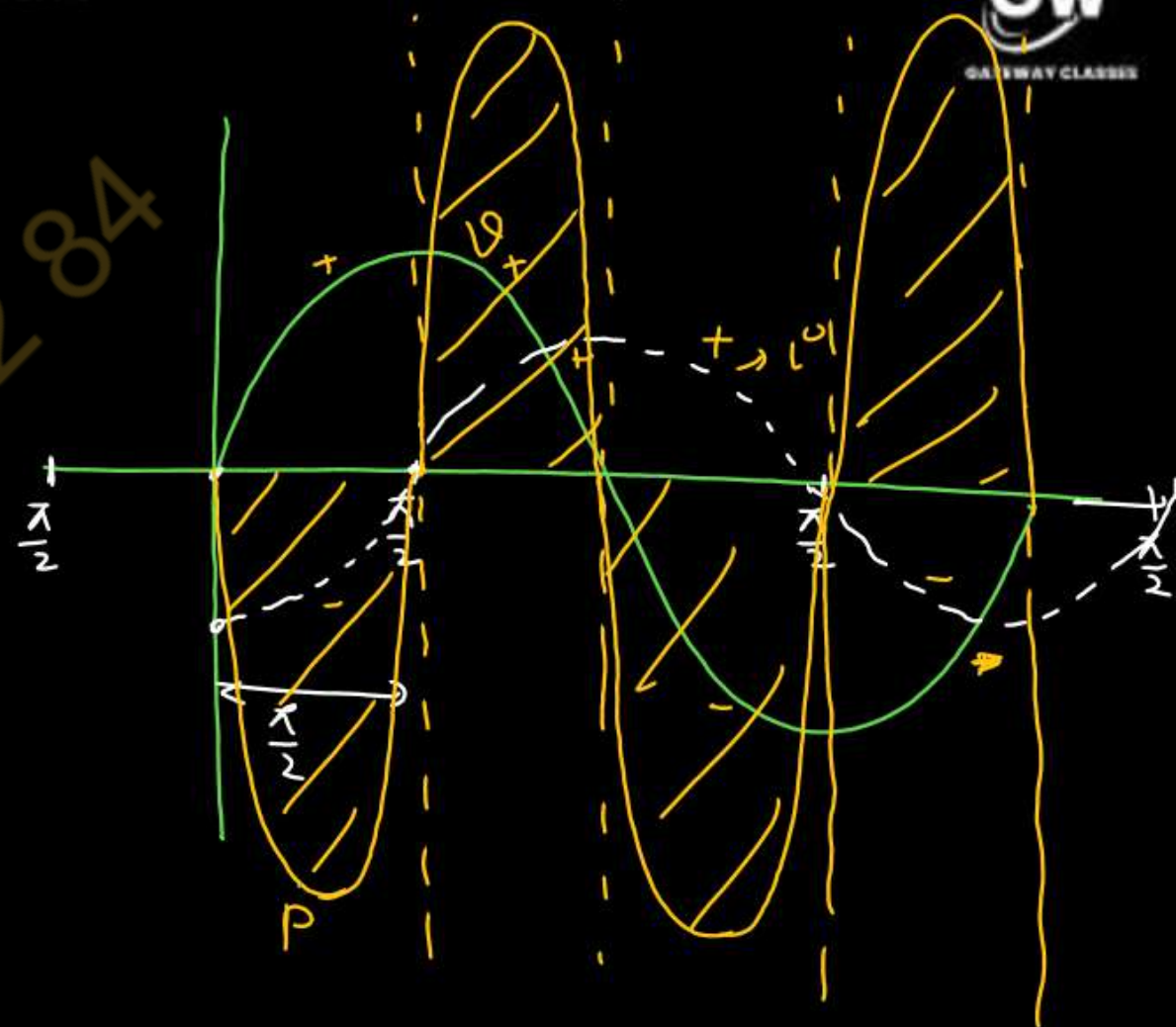
$$P = V_m \sin \omega t (-I_m \cos \omega t)$$

$$P = -\frac{V_m I_m}{2} \sin \omega t \cos \omega t$$

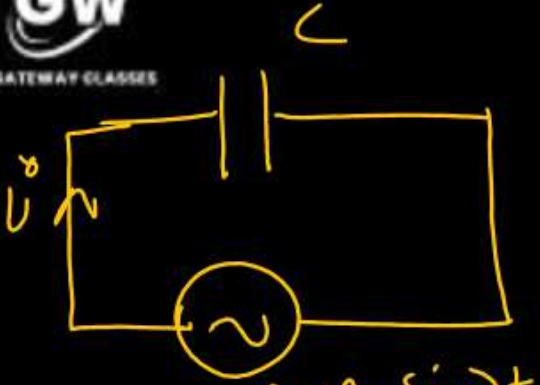
$$P = -\frac{V_m I_m}{2} \sin 2\omega t$$

Average Power $\rightarrow$ At $90^\circ = 0$

$$P_{avg} = 0$$



A.C. Through Pure Capacitive Circuit



$$V = V_m \sin \omega t$$

$$V = V_m \sin \omega t \quad \text{--- (1)}$$

$$i^o = \frac{dq}{dt} = \frac{d}{dt} C V$$

$$i^o = \frac{d}{dt} (C V_m \sin \omega t)$$

$$i^o = C V_m \cos \omega t \cdot \omega$$

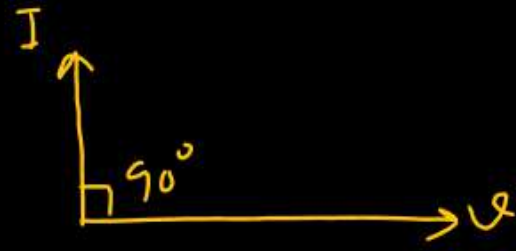
$$i^o = \omega C V_m \cos \omega t$$

$$i^o = I_m \sin(\omega t + \frac{\pi}{2})$$

So

--- (2)

from Equ (1) & (2)
we can see that current
is leading by $\frac{\pi}{2}$ by voltage



also
we have

$$I_m = \omega C V_m$$

$$\frac{V_m}{I_m} = \frac{1}{\omega C} = X_C$$

= Capacitive
Reactance

$$X_C = \frac{1}{\omega C} \Omega$$

Instan. Power

$$P = V i^o$$

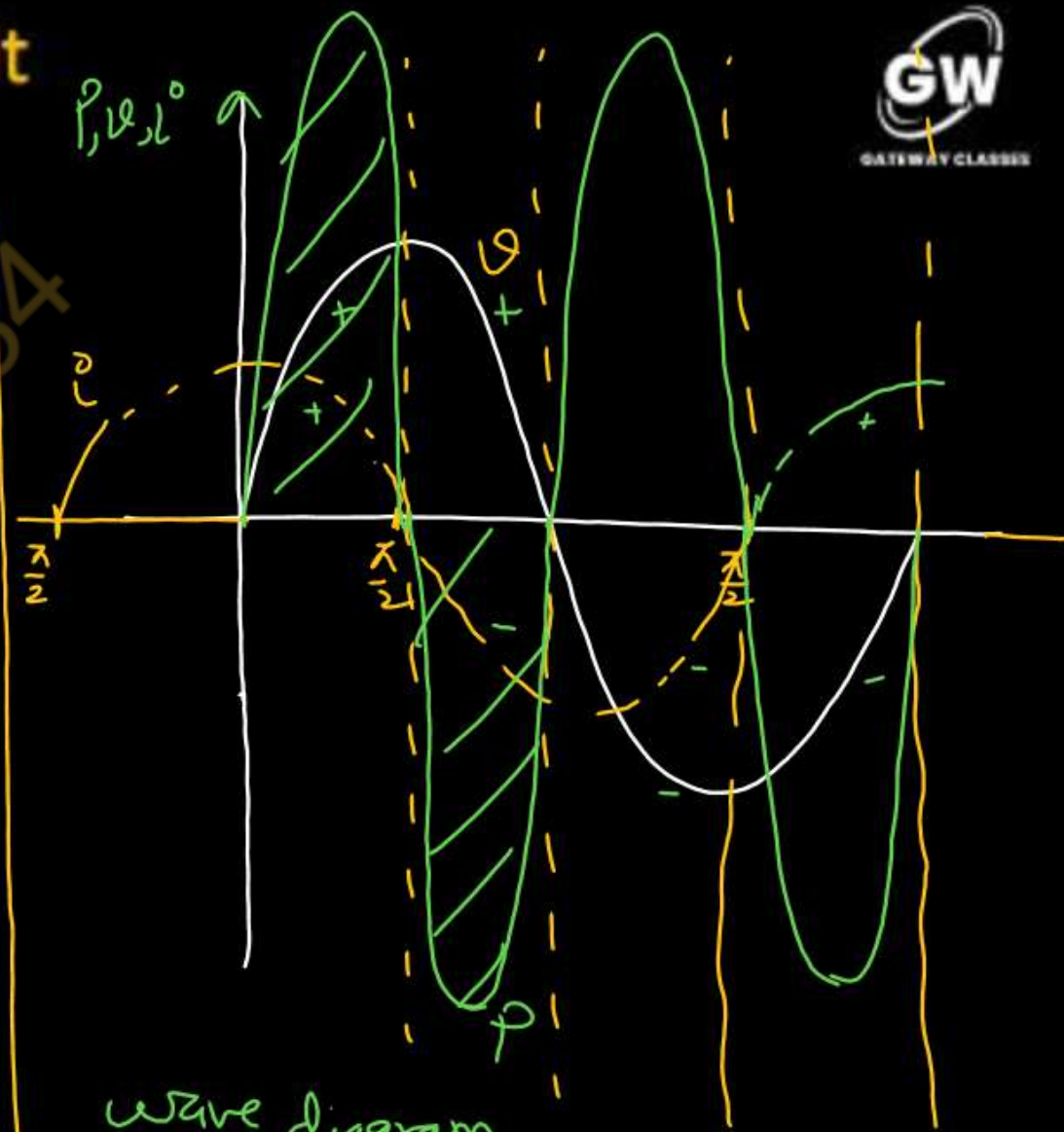
$$= V_m \sin \omega t \cdot I_m \cos \omega t$$

$$P = \frac{V_m I_m}{2} 2 \sin \omega t \cos \omega t$$

$$P = \frac{V_m I_m}{2} \sin 2 \omega t$$

Average Power $\rightarrow$ Alt. quan = 0

$$P_{avg} = 0$$



Wave diagram

FUNDAMENTALS OF ELECTRICAL ENGINEERING

UNIT -2 : Steady State Analysis of 1- Φ Circuits

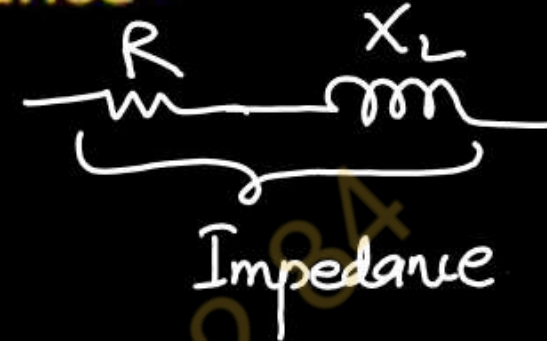
Lec-4

Today's Target-

- ✓ Concept of Impedance
- ✓ Analysis of series R-L , R-C , R-L-C Circuits
- ✓ Concept of Power
- ✓ Power Factor
- ✓ DPPs

Concept of Impedance

Resistance $\rightarrow R$



Reactance $\rightarrow$ due to Inductor $(X_L) = 2\pi fL = \omega L$

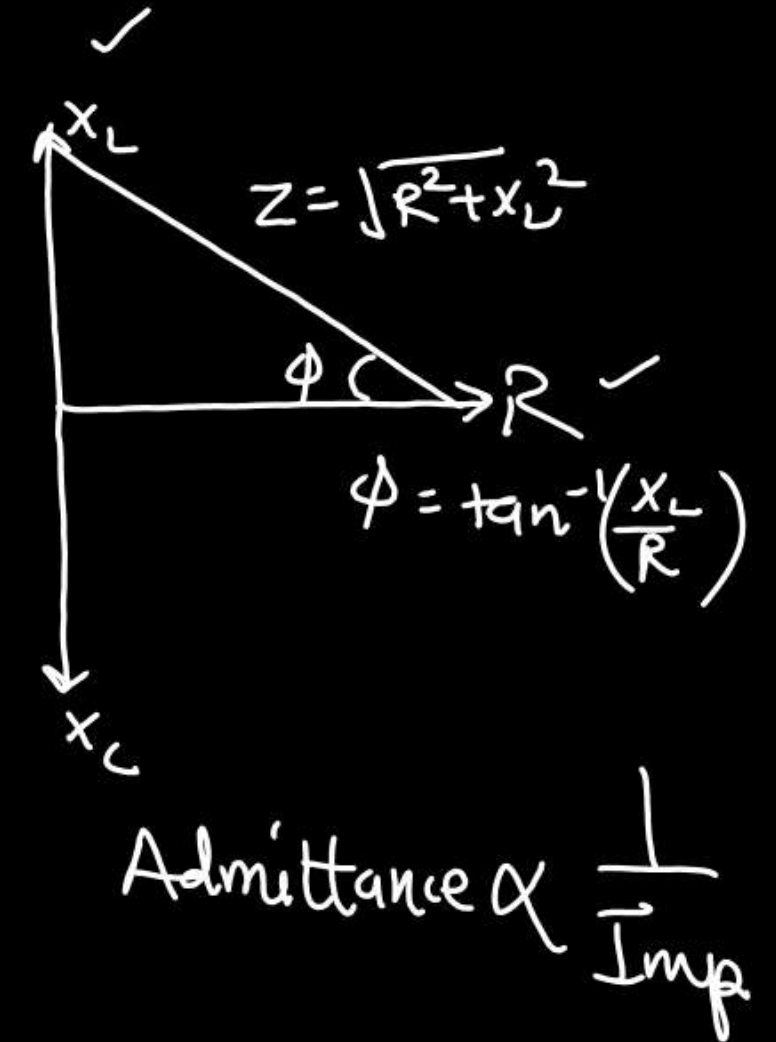
due to Capacitor $(X_C) = \frac{1}{2\pi fC} = \frac{1}{\omega C}$

Impedance $\rightarrow$ Combination of Resistance & Reactance

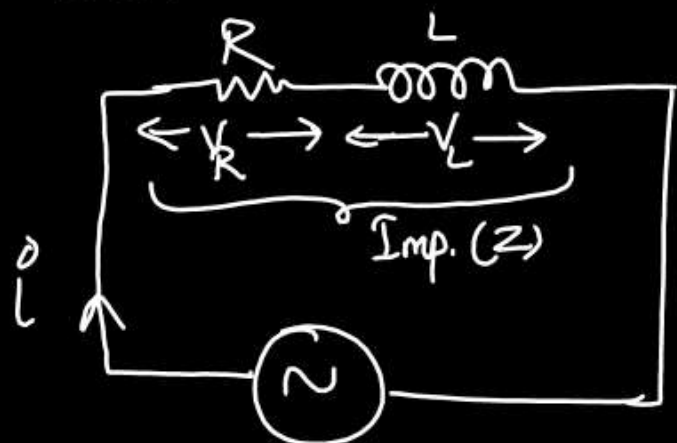
$Z = R + jX_L \rightarrow$ Series R-L Ckt (+ sign)

$Z = R - jX_C \rightarrow$ Series R-C Ckt. (- sign)

$Z = R + j(X_L - X_C) \rightarrow$ Series R-L-C Circuit



Series R-L Circuit



$$V = V_m \sin \omega t \quad \text{--- ①}$$

$$\vec{V} = \vec{V}_R + \vec{V}_L$$

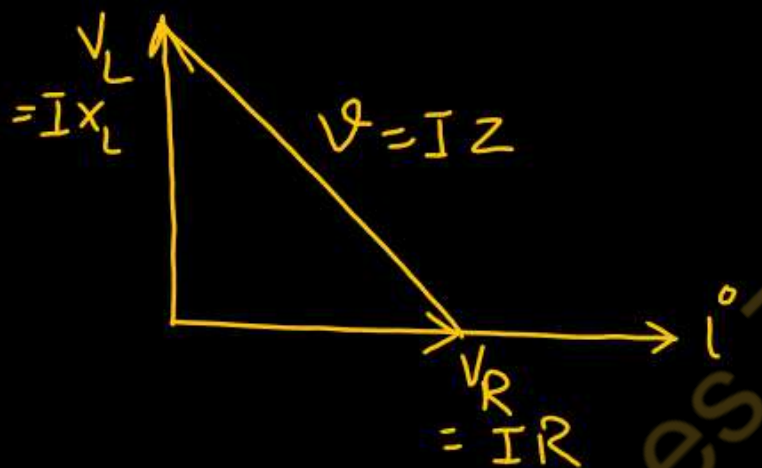
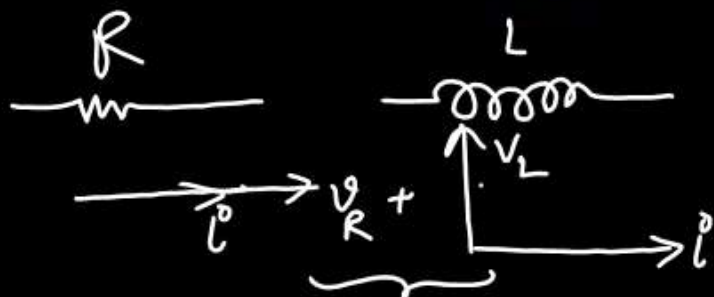
$$\vec{i}Z = \vec{i}R + i\vec{X}_L$$

$$\vec{Z} = \vec{R} + \vec{X}_L$$

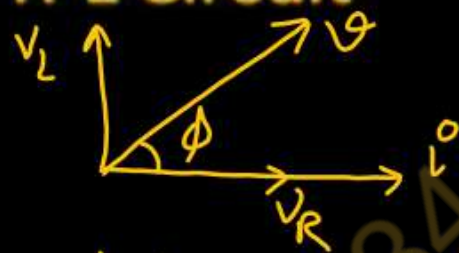
$$|\vec{Z}| = \sqrt{R^2 + X_L^2}$$

$$\angle \phi = \tan^{-1} \left(\frac{X_L}{R} \right)$$

A.C. Through Series R-L Circuit



Phasor



Current is lagging behind the voltage by ϕ .

$$i = I_m \sin(\omega t - \phi) \quad \text{--- ②}$$

Instantaneous power

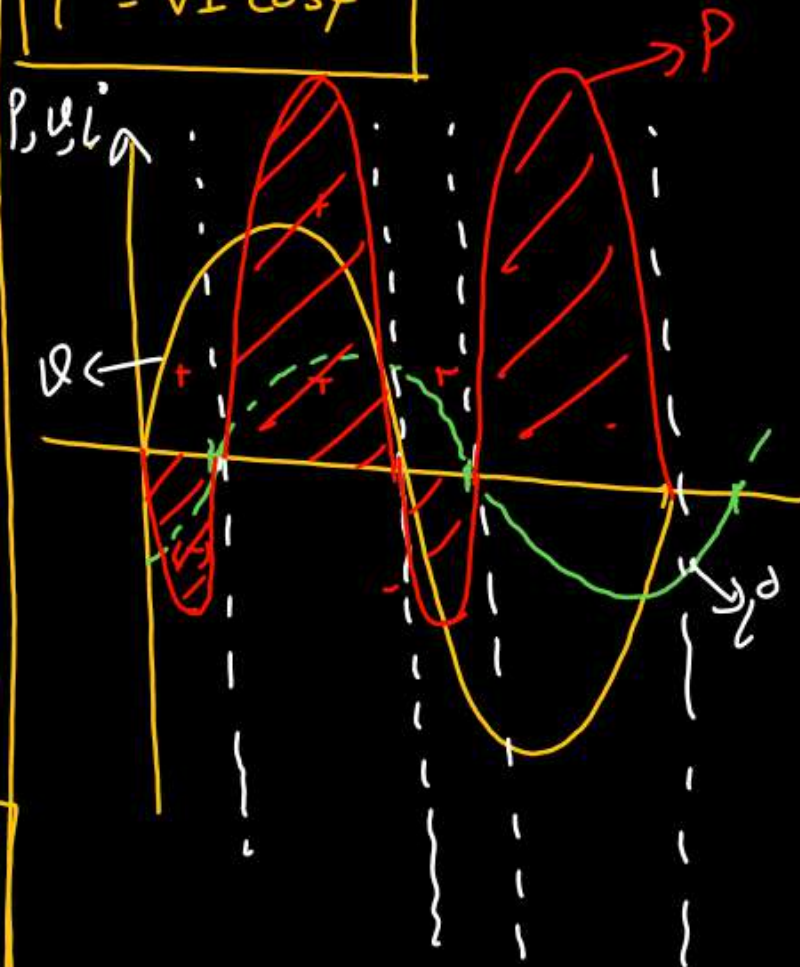
$$\begin{aligned} P &= VI \\ &= V_m \sin \omega t \cdot I_m \sin(\omega t - \phi) \\ &= \frac{V_m I_m}{2} \left[2 \sin \omega t \sin(\omega t - \phi) \right] \end{aligned}$$

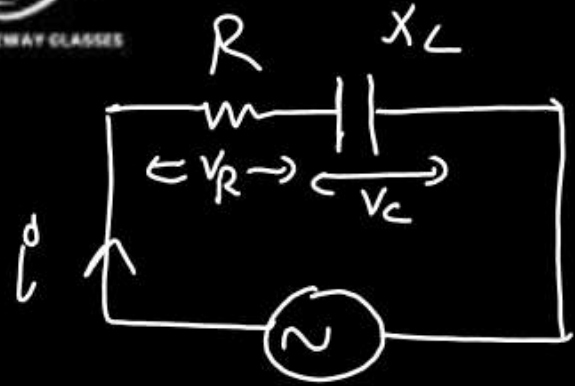
$$P = \frac{V_m I_m}{2} \left[\cos(\phi) - \cos(2\omega t - \phi) \right]$$

Average power

$$\begin{aligned} P_{avg} &= \frac{V_m I_m}{2} \cos \phi \\ &= \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \cdot \cos \phi \end{aligned}$$

$$P = VI \cos \phi$$





$$V = V_m \sin \omega t \quad \text{--- (1)}$$

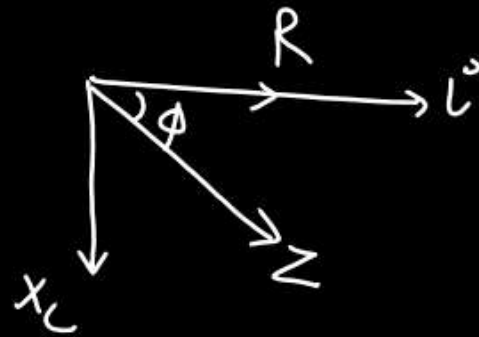
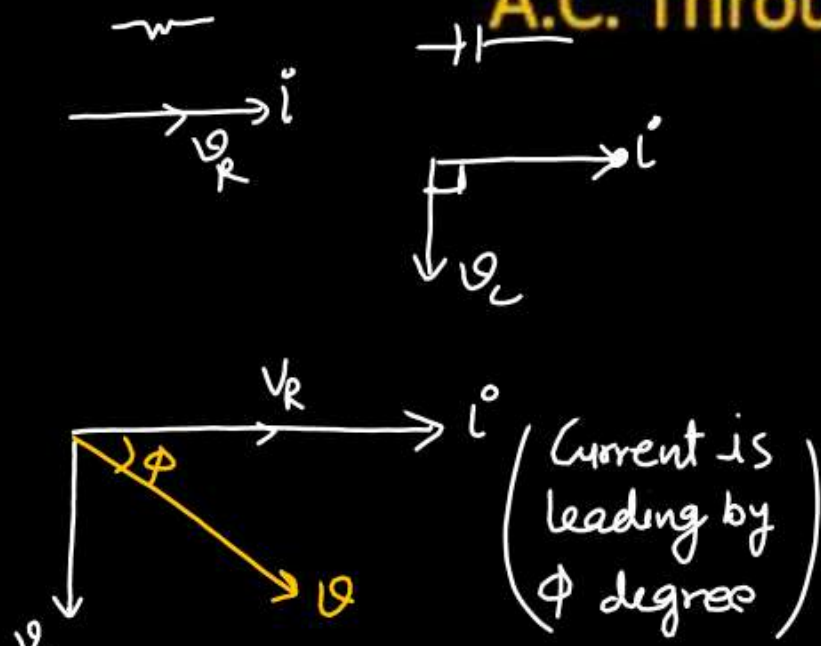
$$\vec{V} = \vec{V}_R + \vec{V}_C$$

$$\vec{iZ} = \vec{iR} + \vec{iX}_C$$

$$\vec{Z} = \vec{R} + \vec{X}_C$$

$$|\vec{Z}| = \sqrt{R^2 + X_C^2}$$

$$\angle \phi = \tan^{-1} \left(-\frac{X_C}{R} \right)$$



$$i = I_m \sin(\omega t + \phi) \quad \text{--- (2)}$$

A.C. Through Series R-C Circuit

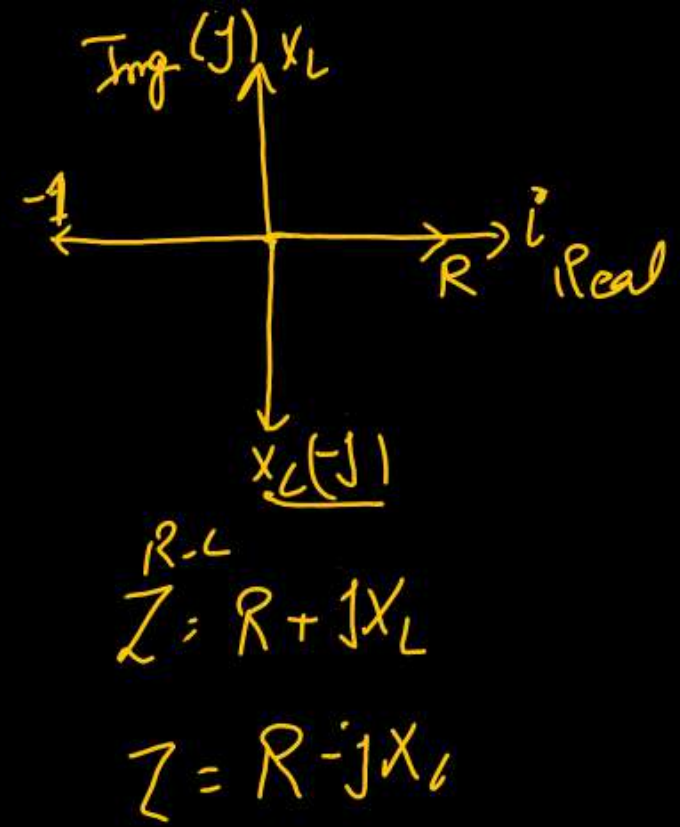
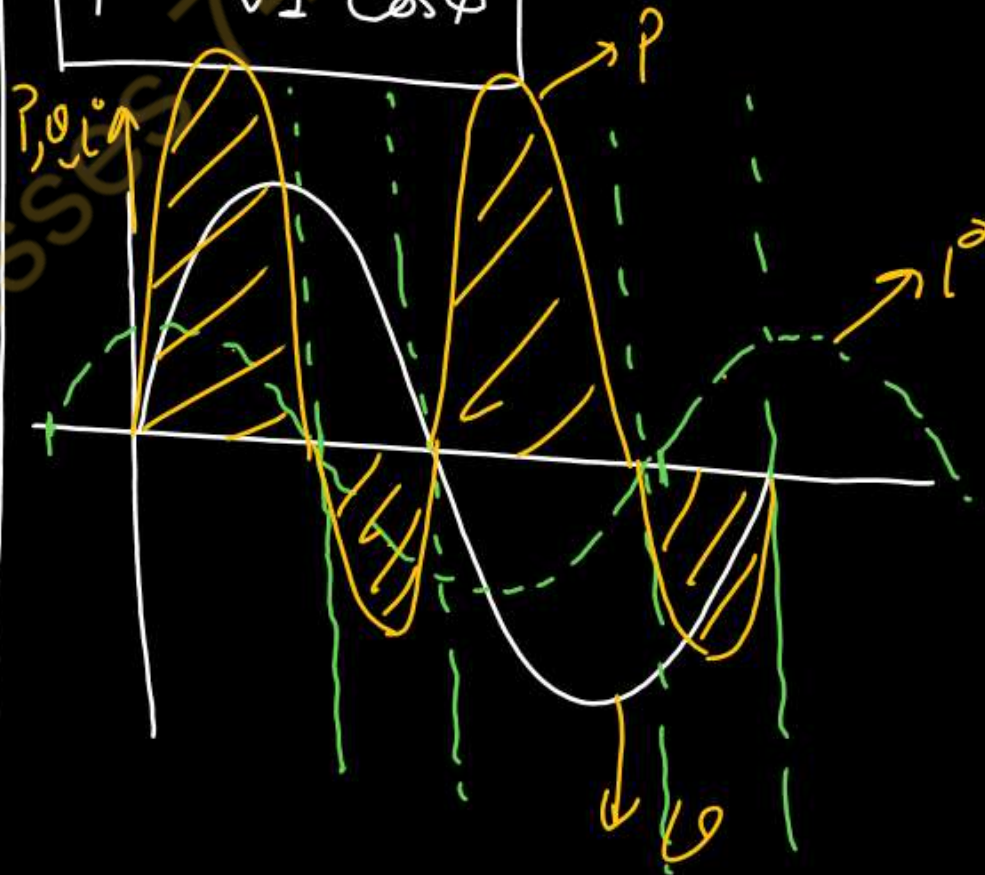
Instantaneous power

$$\begin{aligned} P &= VI \\ &= V_m \sin \omega t \cdot I_m \sin(\omega t + \phi) \\ &= \frac{V_m I_m}{2} [\cos(\phi) - \cos(2\omega t + \phi)] \end{aligned}$$

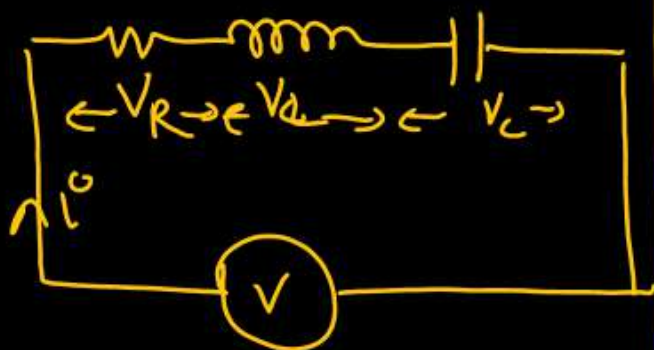
Average Power

$$P_{avg} = \left(\frac{V_m}{\sqrt{2}} \right) \left(\frac{I_m}{\sqrt{2}} \right) \cos \phi$$

$$P = VI \cos \phi$$



A.C. Through Series R-L-C Circuit



$$\vec{V} = \vec{V}_R + \vec{V}_L + \vec{V}_C$$

$$\vec{Z} = \vec{R} + \vec{X}_L + \vec{X}_C$$

$$|\vec{Z}| = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

Case (i)

when $X_L > X_C$

Then Circuit will be Inductive

$$|Z| = R + j(X_L - X_C)$$

$$= \sqrt{R^2 + (X_L - X_C)^2}$$

$$\phi = \tan^{-1} \left(\frac{X_L - X_C}{R} \right)$$

Case (ii) $X_C > X_L$ (Capacitive)

$$Z = R + jX_L - jX_C$$

$$= R - j(X_C - X_L)$$

$$|Z| = \sqrt{R^2 + (X_C - X_L)^2}$$

$$\phi = \tan^{-1} \left(\frac{X_C - X_L}{R} \right)$$

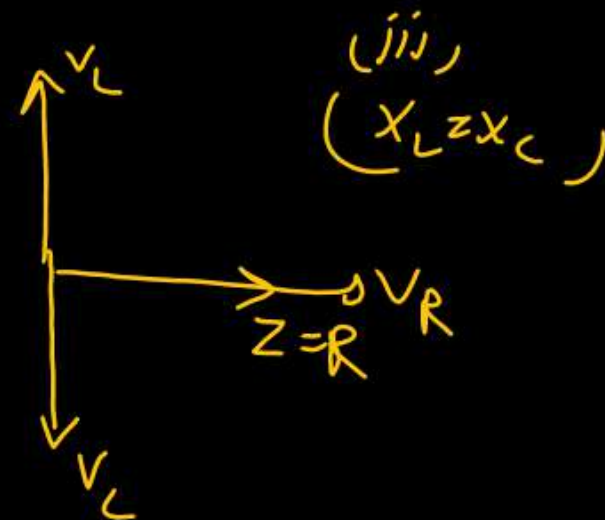
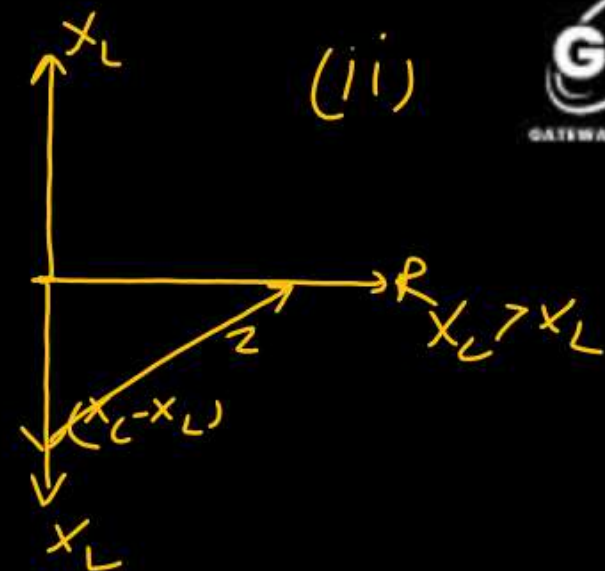
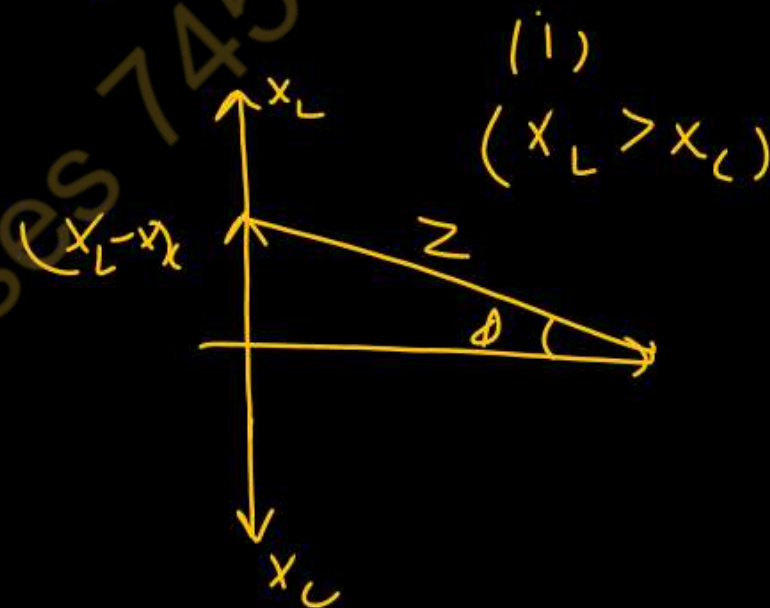
Case (iii)

when $X_L = X_C$

$$\boxed{Z = R}$$

Circuit will be Resistive

Special Case



Concept of Power:-

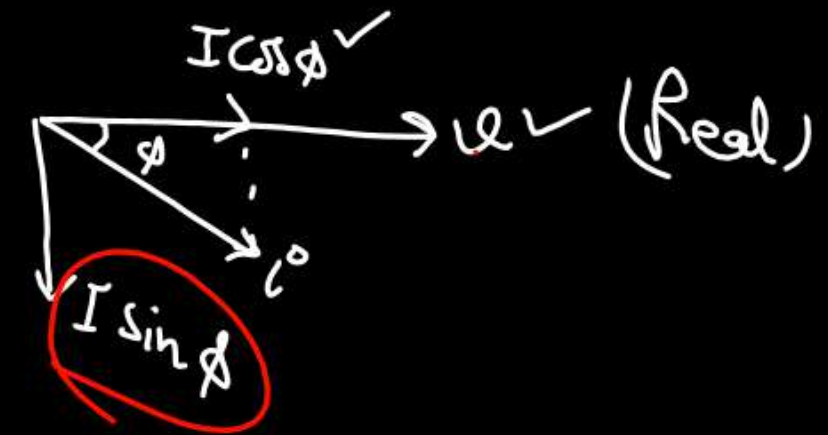
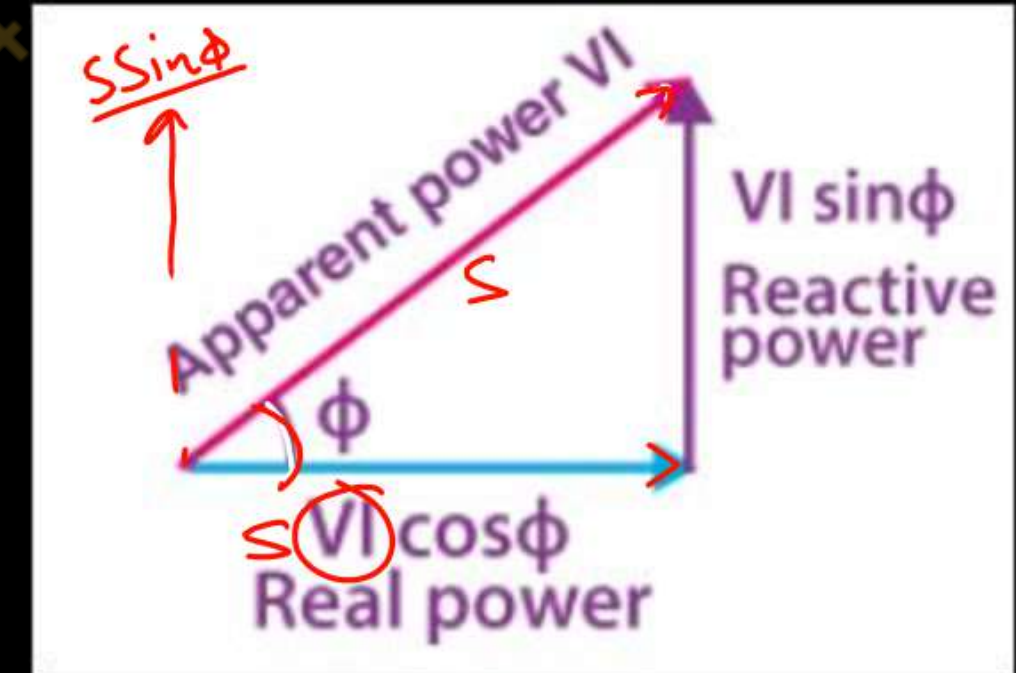
- ✓ **Apparent Power:-** Apparent Power (S) is Product of rms value of Voltage (V) and current (I).
$$S = VI$$

Unit- KW
- ✓ **Active/True Power :-** True Power (P) is Product of applied Voltage (V) and active component of current (I).
Unit- KVA

$$P = VI \cos \phi = S \cos \phi$$

- Reactive Power :-** Reactive power (Q) is Product of applied Voltage (V) and reactive component of current (I).
Unit- KVAR

$$Q = VI \sin \phi = S \sin \phi$$



Power factor is the

- Ratio between true power and apparent power is called the for this circuit.
- Ratio of resistance to impedance. $\cos \phi = \frac{R}{Z}$
- Cosine of phase angle between voltage and current. $Pf = \cos \phi$

$$\cos \phi = \frac{VI \cos \phi}{VI} = P.f$$

Cause of Low Power Factor.

- All AC motor transformers has low power factor. ✓
- Most of the loads has low power factors/
- Industrial heating furnace, induction furnace has low power factor.

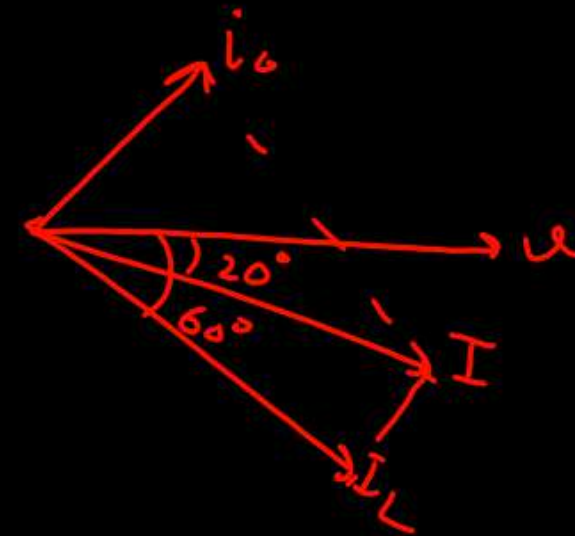
Improvement of Power Factor.

- Using Phase advancers ✓
- Using Capacitor Bank, Synchronous Condenser ✓

Effects of low power factor

- a) Large Copper Losses b) Large kVA rating c) Poor Voltage Regulation.

Power factor always follows the nature of current.



- Ex.-1** An A.C. circuit having a pure resistance of 10 ohm and is connected to an a.c. supply of 230 volt and 50Hz. Calculate (a) Current (b) Power Consume (c) Equation of voltage and current.

Sol: $R = 10\Omega$, $V_r = 230V$, $f = 50Hz$, $\omega = 2\pi \times 50 = 100\pi$

$$I^0 = \frac{V}{R} = \frac{230}{10} = 23 \text{ Amp} , \quad V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$P = I^2 R = 23^2 \times 10$$

$$V_m = V_{rms} \sqrt{2} = 230\sqrt{2}$$

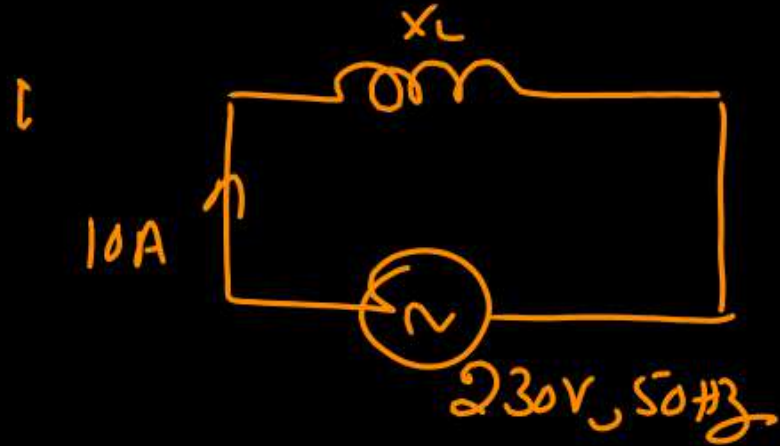
$$V = V_m \sin \omega t = 325.26 \sin 100\pi t$$

$$I = I_m \sin \omega t = 23\sqrt{2} \sin 100\pi t$$

Ans

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- **Ex.-2.** A Pure inductive coil allow a current of 10A to flow from an a.c. supply of 230 volt and 50Hz. Calculate
(a) Inductive reactance (b) Inductance of coil (c) Power Consumed (c) Equation of voltage and current.



$$(a) X_L = \frac{V}{I} = \frac{230}{10} = 23 \Omega$$

$$(b) 2\pi fL = X_L = 23$$

$$L = \frac{23}{2\pi f} = \frac{23}{100\pi} = 0.073 \text{ H}$$

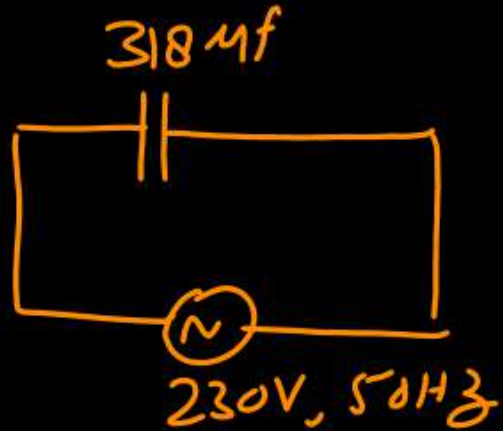
$$P = 0$$

$$(d) V = 230\sqrt{2} \sin 100\pi t$$

$$I = 23\sqrt{2} \sin(100\pi t - \frac{\pi}{2})$$

Ad

Ex.-3. A 318 microfarad capacitor is connected across a supply of 230 volt and 50Hz. Calculate (a) Capacitive reactance (b) RMS value of Current (c) Equation of voltage and current.



$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi \times 50 \times 318 \times 10^{-6}}$$

$$(a) \boxed{X_C = 10 \Omega}$$

$$(b) I^0 = \frac{V}{X_C} = \frac{230}{10} = \underline{23 A}$$

$$\left. \begin{aligned} (c) \quad v &= 230\sqrt{2} \sin 100\pi t \\ i^0 &= 23\sqrt{2} \sin \left(100\pi t + \frac{\pi}{2} \right) \end{aligned} \right\}$$

Ex.-4. An Electrical circuit connected to $v=100 \sin(628t+60)$ it draw current $i=15 \sin(628t+30)$, identify the circuit, also find the value of Resistance, Reactance, Power, Power Factor. **AKTU-2020-21**

$$V = 100 \sin(628t + 60)$$

$\downarrow$
 $V_m = 100$

$$I_m = 15$$

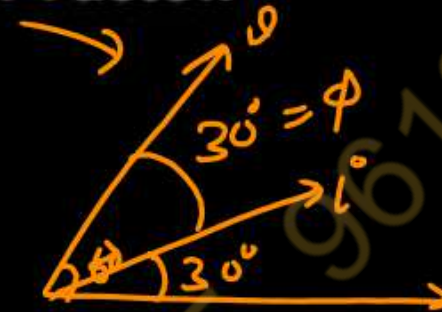
$$\omega = 628$$

$$Z = \frac{V_m}{I_m} = \frac{100}{15} = 6.67$$

$$\phi = 30^\circ$$

$$Z = 6.67 \angle +30^\circ = 5.77 + 3.33j$$

$\downarrow$ R $\rightarrow$ X_L



as Voltage is leading
& Current is lagging

Circuit will be inductive

$$P = I^2 R$$

$$= \left(\frac{15}{\sqrt{2}}\right)^2 \cdot 5.77$$

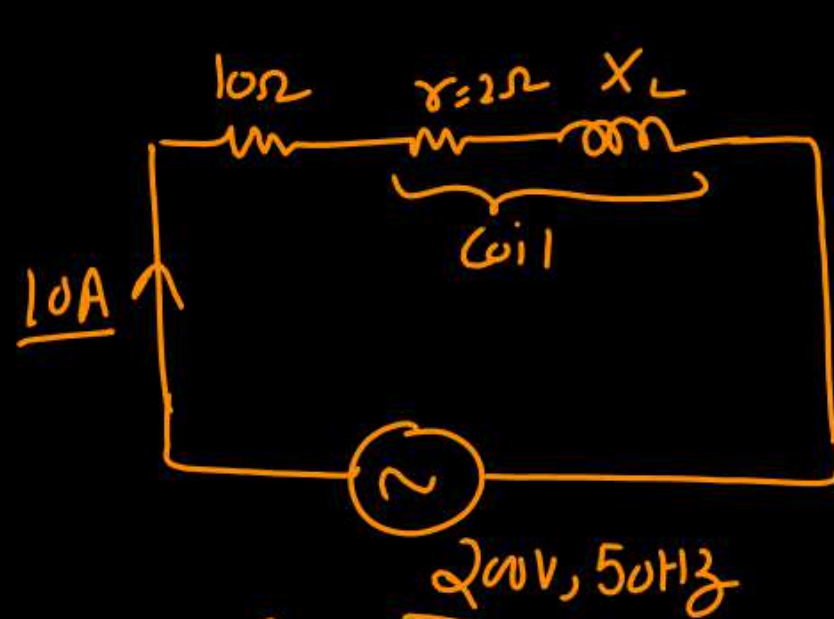
$$P = 649.12 \text{ W}$$

$$P.f = \cos \phi$$

$$= \cos 30^\circ$$

$$= 0.86 \text{ lag}$$

- Ex.-5** A non inductive resistance of 10 ohms is connected in series with an inductive coil across 200V, 50 Hz a.c. supply. The current drawn by the series combination is 10 A. The resistance of the coil is 2 ohms. Determine i) Inductance of the coil, (ii) Power factor (iii) Voltage across the coil. **AKTU-2004-05**



$$Z = \frac{V}{I} = \frac{200}{10} = 20 \Omega$$

$$Z = \sqrt{(R+r)^2 + X_L^2}$$

$$20 = \sqrt{12^2 + X_L^2}$$

$$20^2 = 12^2 + X_L^2$$

$$X_L^2 = 256$$

$$X_L = 16 \Omega$$

$$L = \frac{X_L}{2\pi f} = 0.05 H$$

$$\text{Power factor} = \frac{R}{Z} = \frac{12}{20} = 0.6$$

(iii)

$$V = I(R + jX_L)$$

$$V_{\text{coil}} = 10(2 + j16) = 161.24 \angle 82.87^\circ$$

Q 1. The voltage applied to a circuit is $v = 100 \sin (\omega t + 30^\circ)$ and the current flowing in the circuit is $i = 15 \sin (\omega t + 60^\circ)$. Determine the impedance, resistance, reactance, power and power factor of the circuit.

AKTU 2003-04

Q.2 Given $v = 200 \sin 377t$ volt and $i = 8 \sin (377t - 30^\circ)$ amp for an a.c. circuit. Determine:- a.) The power factor b.) True Power, c.) Apparent Power d.) Reactive Power.

AKTU- 2008-09

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FUNDAMENTALS OF ELECTRICAL ENGINEERING

UNIT -2 : Steady State Analysis of 1- Φ Circuits

Lecture- 5

Today's Target-

- ✓ Series Resonance ✓ *M. Imp*
- ✓ Parallel Resonance ✓ *↓*
- ✓ Bandwidth in series resonance ✓
- ✓ PYQ ✓
- ✓ DPPs ✓

Condition of Resonance in Series RLC circuit:-

- Power Factor is unity. ✓
- Voltage and current are in same phase. ✓
- A series resonant circuit has the capability to draw heavy current and power from the mains; it is also called **acceptor circuit**.

Parameters at Resonance Frequency:

(i) Resonance Frequency

At Resonance

$$X_L = X_C$$

$$2\pi f_r L = \frac{1}{2\pi f_r C}$$

$$f_r^2 = \frac{1}{4\pi^2 LC}$$

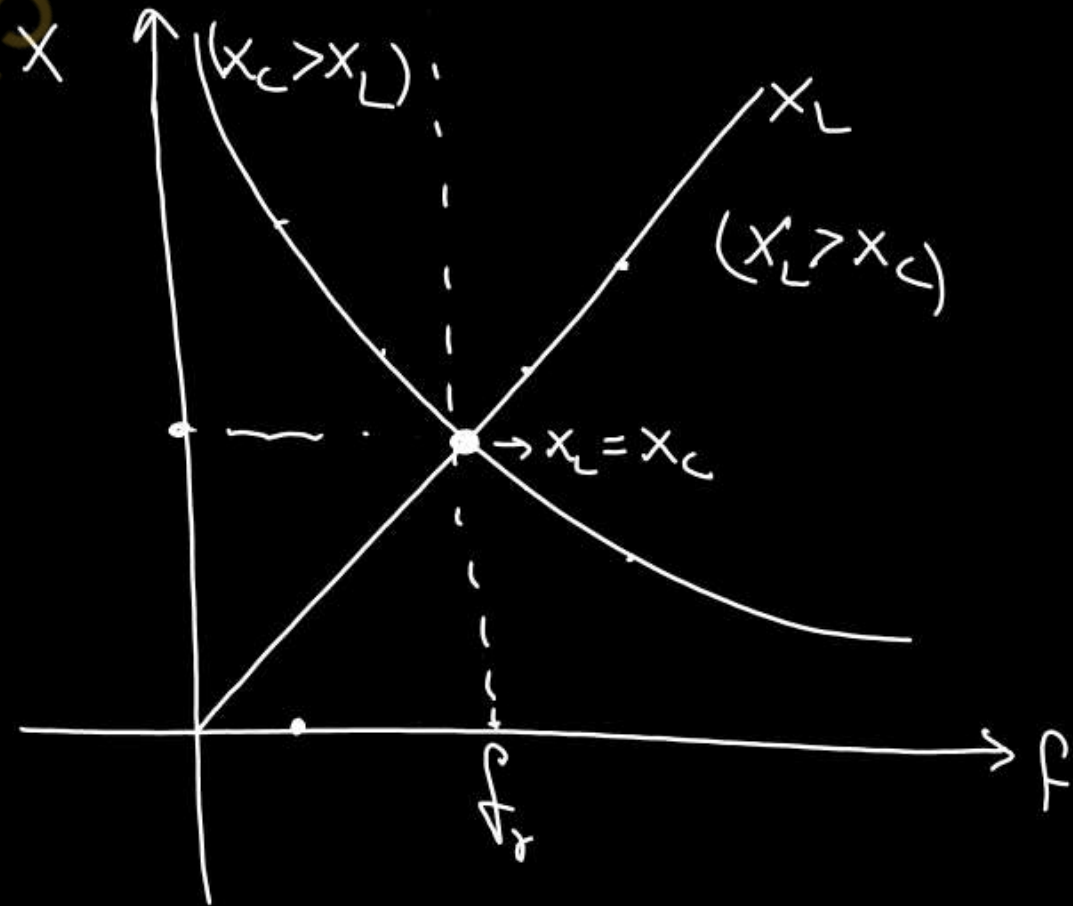
$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$X_L = 2\pi f L$$

$$X_L \propto f$$

$$X_C = \frac{1}{2\pi f C}$$

$$X_C \propto \frac{1}{f}$$



Resonance in Series RLC circuit

Parameters at Resonance Frequency:

(ii) **Impedance at Resonance:** At resonance, the total impedance of series RLC circuit is equal to resistance i.e. $Z = R$, impedance has only real part but no imaginary part and this impedance at resonant frequency is called dynamic impedance.

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \Rightarrow \boxed{Z = R} \rightarrow \text{minimum}$$

(iii) Current at Resonance:

$$\uparrow I = \frac{V}{Z_{\min}} \downarrow, \text{ Current will be maximum}$$

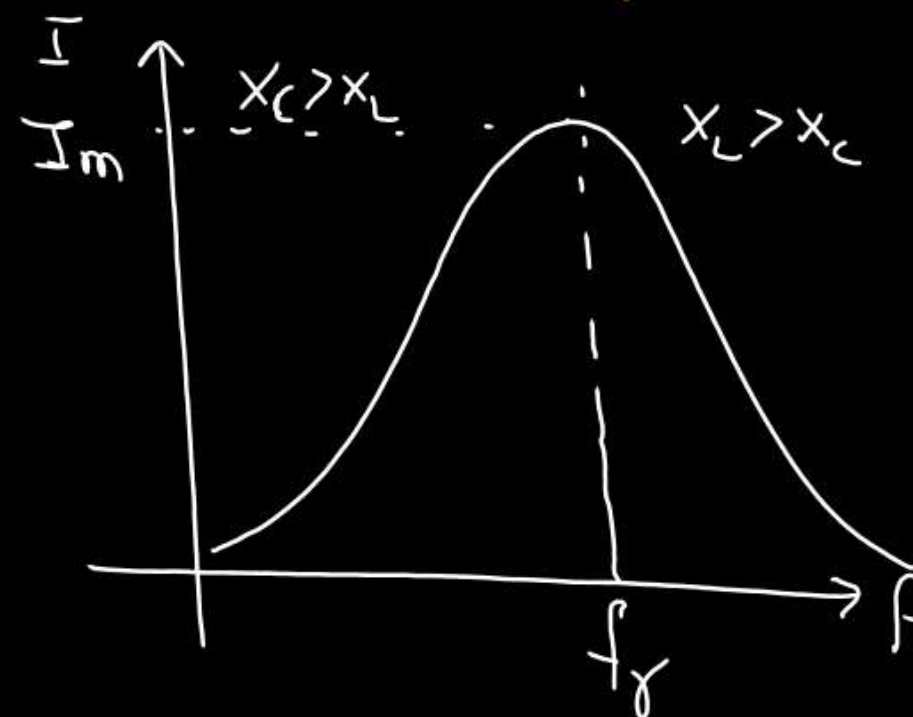
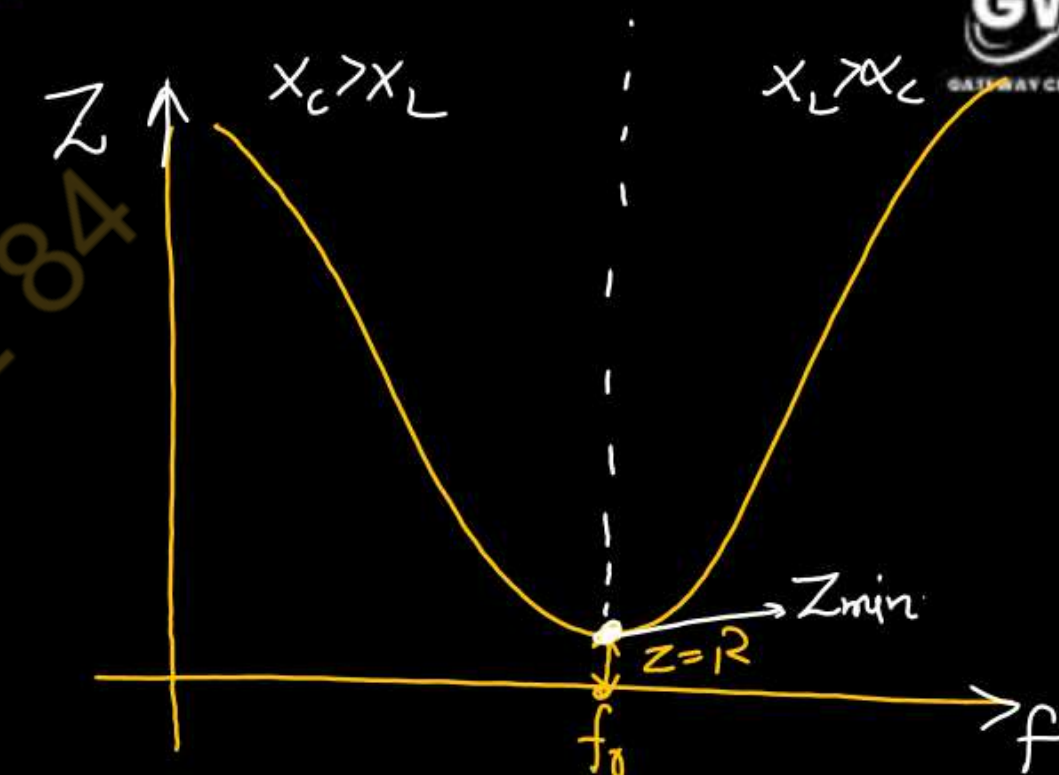
(iv) **Quality Factor:-** The ratio of a voltage developed across the inductance or Capacitance at resonance to the applied voltage

$$Q \text{ factor} = \frac{V_L}{V} = \frac{IX_L}{IR} \quad \text{or} \quad \frac{V_C}{V} = \frac{IX_C}{IR}$$

$$\boxed{Q = \frac{\omega L}{R}} \quad \text{or} \quad \boxed{Q = \frac{1}{\omega CR}}$$

$$Q = \frac{\omega L}{R} = \frac{1}{\sqrt{LC}} \cdot \frac{L}{R} = \sqrt{\frac{L}{C}} \cdot \frac{1}{R}$$

$$\boxed{Q = \frac{1}{R} \sqrt{\frac{L}{C}}}$$



Resonance in Series RLC circuit

- Series resonance RLC circuit is called as voltage magnification circuit, because the magnitude of voltage across the inductor and the capacitor is equal to Q times the input sinusoidal voltage V .

Application of Series RLC Resonant Circuit

- Since resonance in series RLC circuit occurs at particular frequency, so it is used for filtering and tuning purpose as it does not allow unwanted oscillations that would otherwise cause signal distortion, noise and damage to circuit to pass through it.

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Bandwidth of Series RLC Circuit at resonance

The bandwidth of any system is the range of frequencies for which the current or output voltage is equal to 70.7% of its value at the resonant frequency, and it is denoted by BW.

Frequency f_1 is the frequency at which the current is 0.707 times the current at resonant value, and it is called the **lower cut-off frequency**. The frequency f_2 is the frequency at which the current is 0.707 times the current at resonant value (i.e. maximum value), and is called the **upper cut-off frequency**. The bandwidth, or BW, is defined as the frequency difference between f_2 and f_1 .

Current in Circuit is given by

$$I = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}} \quad \text{--- (1)}$$

At resonance. Current will be max

$$I_m = \frac{V}{R}$$

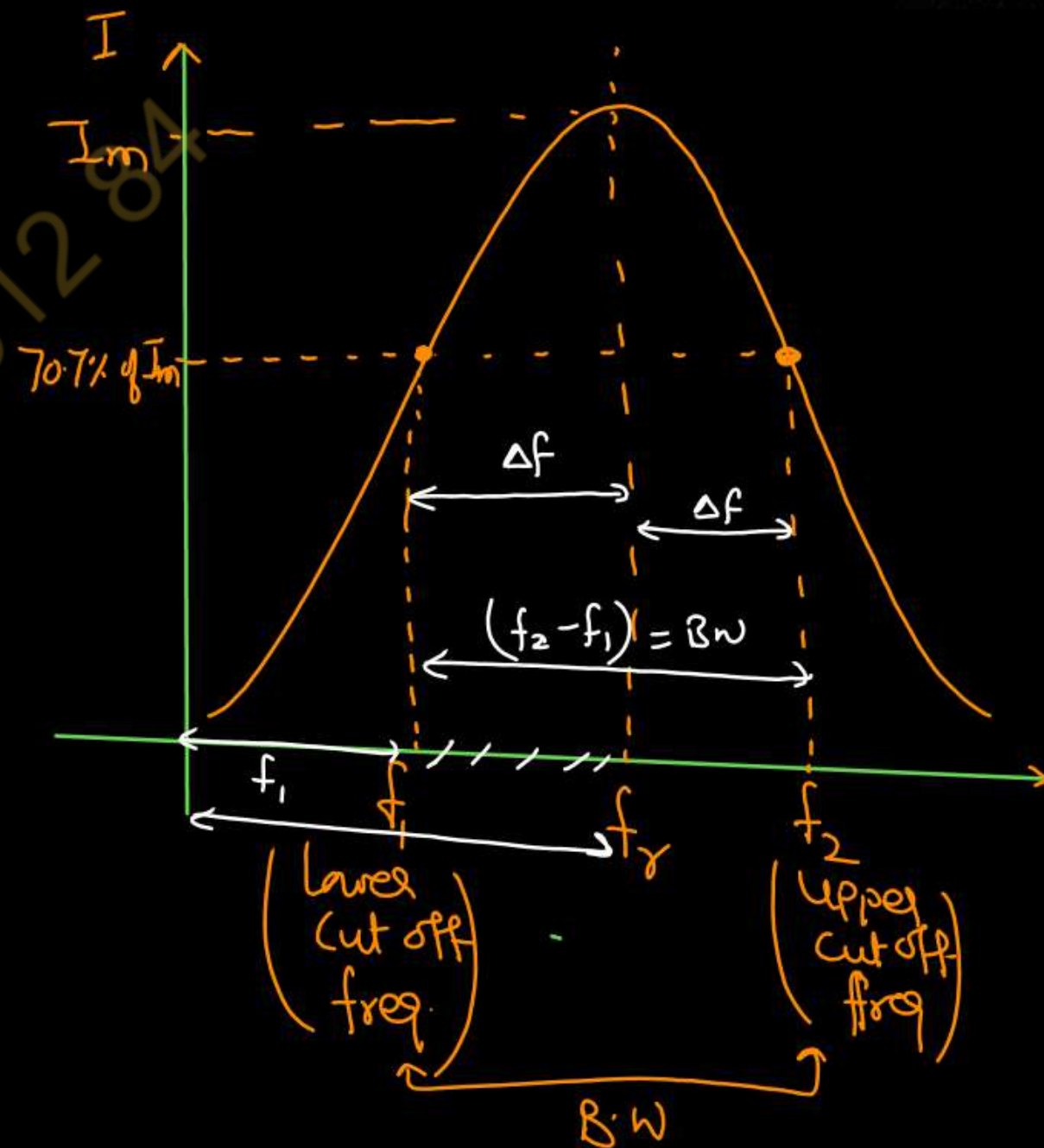
and $I = \frac{I_m}{\sqrt{2}} = \frac{V}{R\sqrt{2}} \quad \text{--- (2)}$



$$P_m = I_m^2 R$$

$$\frac{P_m}{2} = \frac{I_m^2}{2} R = \left(\frac{I_m}{\sqrt{2}}\right)^2 \cdot R = I^2 R$$

R.m.s values



$$\frac{V}{R\sqrt{2}} = \frac{V}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$2R^2 = R^2 + (X_L - X_C)^2$$

$$(X_L - X_C)^2 = R^2$$

$$\text{or } X_L - X_C = \pm R$$

$$\omega_1 L - \frac{1}{\omega_1 C} = -R \quad \text{--- (3)}$$

$$\& \omega_2 L - \frac{1}{\omega_2 C} = +R \quad \text{--- (4)}$$

Add Equⁿ (3) & (4)

$$L(\omega_1 + \omega_2) - \frac{1}{C} \left(\frac{1}{\omega_1} + \frac{1}{\omega_2} \right) = 0$$

$$L(\omega_1 + \omega_2) - \frac{1}{C} \left(\frac{\omega_1 + \omega_2}{\omega_1 \omega_2} \right) = 0$$

Bandwidth of Series RLC Circuit at resonance

$$L - \frac{1}{C\omega_1\omega_2} = 0$$

$$\frac{1}{C\omega_1\omega_2} = L$$

$$\omega_1\omega_2 = \frac{1}{LC} \quad \text{--- (5)}$$

$$\omega_r^2 = \frac{1}{LC}$$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$f_r = \sqrt{f_1 f_2}$$

Now Subtract Equ 3 from 4

$$L(\omega_2 - \omega_1) - \frac{1}{C} \left(\frac{1}{\omega_2} - \frac{1}{\omega_1} \right) = 2R$$

$$L(\omega_2 - \omega_1) - \frac{1}{C} \left(\frac{\omega_1 - \omega_2}{\omega_1 \omega_2} \right) = 2R$$

$$L(\omega_2 - \omega_1) + \frac{1}{C} \left(\frac{\omega_2 - \omega_1}{\omega_1 \omega_2} \right) = 2R$$

$$(\omega_2 - \omega_1) \left[L + \frac{1}{C\omega_1\omega_2} \right] = 2R$$

$$(\omega_2 - \omega_1) \left[L + \frac{1}{L} \right] = 2R$$

$$2L(\omega_2 - \omega_1) = 2R$$

$$\omega_2 - \omega_1 = \frac{R}{L}$$

$$f_2 - f_1 = \frac{R}{2\pi L}$$

$$BW = \frac{R}{2\pi L}$$

and

$$\Delta f = \frac{R\omega}{2} = \frac{R}{4\pi L}$$

Now cutoff freq. are

$$f_1 = f_r - \Delta f$$

$$f_1 = f_r - \frac{R}{4\pi L} \quad \text{L.C.f}$$

$$f_2 = f_r + \Delta f$$

$$f_2 = f_r + \frac{R}{4\pi L} \quad \text{U.C.f}$$

Condition of Resonance in Parallel RLC circuit:-

- **Parallel Resonance** means when the circuit current is in phase with the applied voltage of an AC circuit containing an inductor and a capacitor connected together in parallel.

Parameters at Resonance Frequency:

➤ (i) Resonance Frequency

At Resonance

$$I_c = I_L \sin \phi$$

$$\frac{1}{X_c} = \frac{1}{Z} \cdot \frac{X_L}{Z_L}$$

$$\frac{1}{X_c \cdot X_L} = \frac{1}{Z^2}$$

$$Z^2 = X_L \cdot X_c$$

$$\frac{1}{\omega_c} = (\sqrt{R^2 + X_L^2})^2$$

$$R^2 + X_L^2 = \left(\frac{L}{C}\right)$$

$$R^2 + (2\pi f_r L)^2 = \frac{L}{C}$$

$$(2\pi f_r L)^2 = \frac{L}{C} - R^2$$

$$2\pi f_r L = \sqrt{\frac{L}{C} - R^2}$$

where $Z_L = R_L + jX_L$

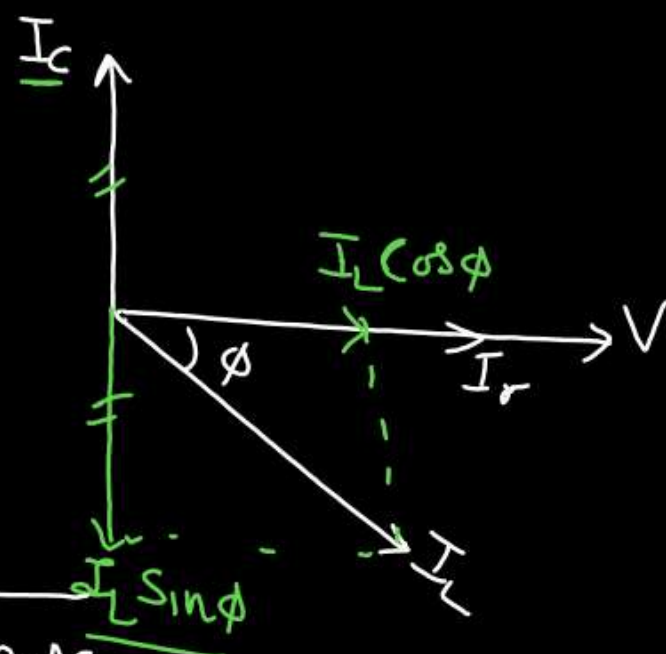
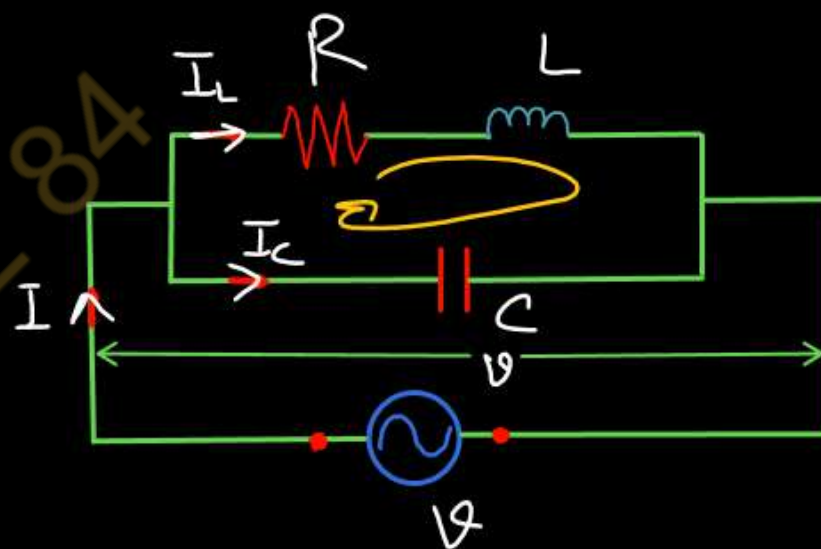
$$f_0 = \frac{1}{2\pi L} \sqrt{\frac{L}{C} - R^2}$$

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

very small

if R is very small as compare to L then

$$f_r = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$$



(ii) Impedance at Parallel Resonance:

- At parallel resonance line current $I_r = I_L \cos \phi$ or

$$\frac{V}{Z_r} = \frac{V}{Z_L} \cdot \frac{R}{Z_L}$$

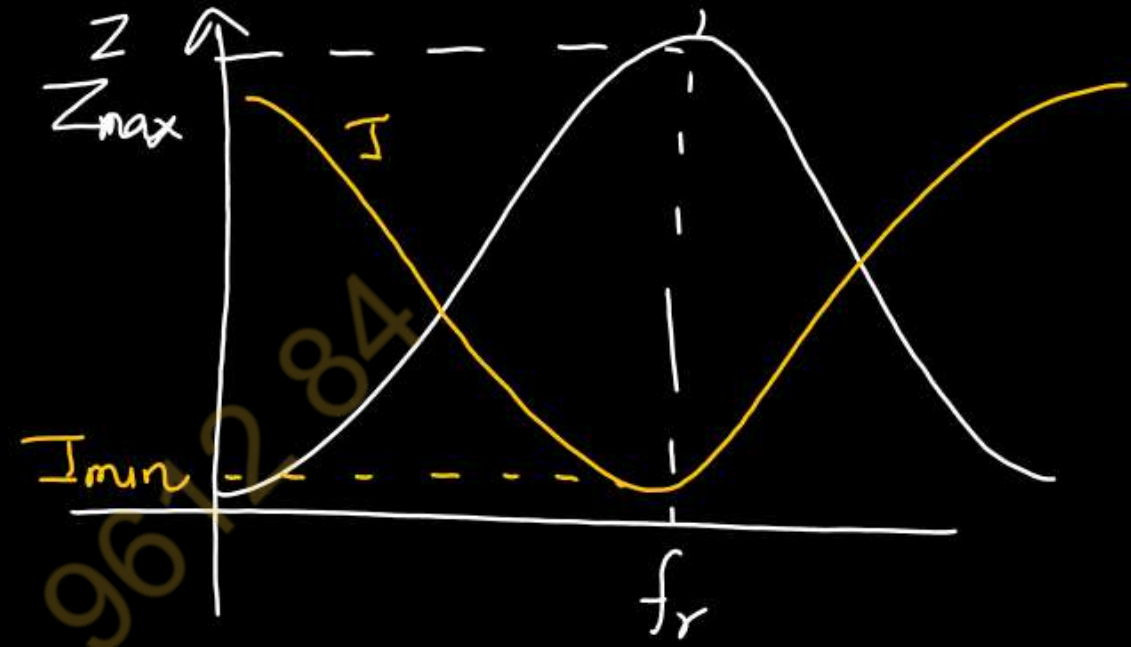
$$\frac{1}{Z_r} = \frac{R}{Z_L^2}$$

$$Z_r = \frac{Z_L^2}{R} = \frac{L}{CR}$$

$$\frac{L}{CR}$$

➤ $Z_r = \frac{L}{CR}$

- The circuit impedance is purely resistive because there is no frequency term present in it. Circuit impedance Z_r will be in Ohms.
- The value of Z_r will be very high because the ratio L/C is very large at parallel resonance.
- The value of circuit current, $I_r = V/Z_r$ is very small because the value of Z_r is very high.
- parallel resonant circuit can draw a very small current and power from the mains, therefore, it is also called as Rejector Circuit.



(iii) Current at Resonance: I_{min} current

(iv) Quality Factor:- The ratio of a ~~voltage~~ developed across the inductance or Capacitance at resonance to the ~~applied voltage~~ Supply current

$$Q = \frac{I_{L-C}}{I} = \frac{\frac{V}{X_C}}{\frac{V}{Z_r}} = \frac{Z_r}{X_C}$$

$$Q = \frac{\frac{L}{R}}{\frac{1}{\omega C}} = \frac{\omega L}{R}$$

- The ratio of a Current across the inductance or Capacitance at resonance to the Current at resonance.
- Parallel resonance RLC circuit is called as **Current magnification** circuit.

Ex.-1. A 120 V, 60 Watt Lamp is to be operated on 220 V, 50 Hz supply mains, In order that lamp should operate on correct voltage Calculate value of (i) Non inductive resistance (ii) Pure Inductance (iii) Pure Capacitance

AKTU-2005-06

(i) $V_R = IR$
 $R = \frac{V_R}{I} = \frac{120}{0.5} = 200\Omega$

(ii) $\vec{V}^2 = \vec{V}_R^2 + \vec{V}_L^2$
 $\vec{V}_L^2 = \vec{V}^2 - \vec{V}_R^2$
 $V_L = \sqrt{200^2 - 120^2}$

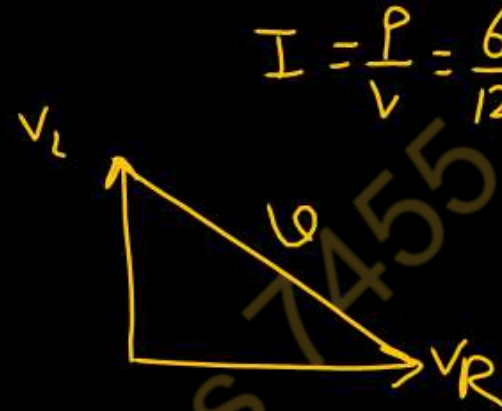
$V_L = 160V$

$IX_L = V_L$
 $X_L = \frac{V_L}{I} = \frac{160}{0.5} = 320\Omega$

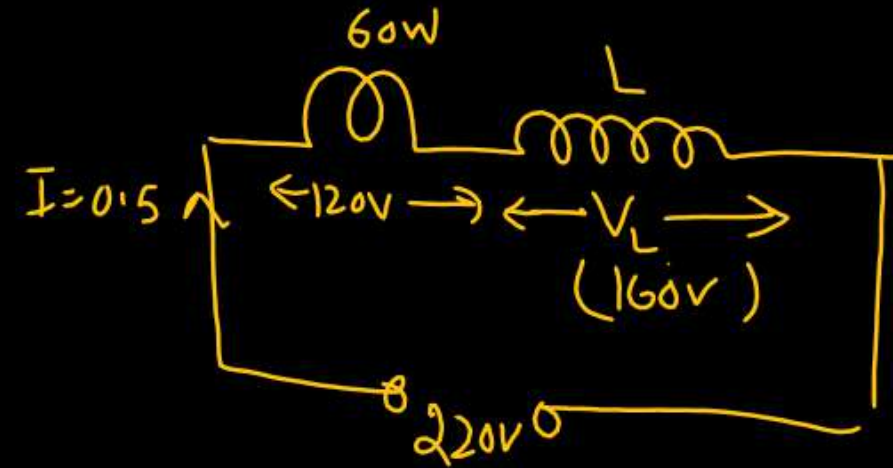
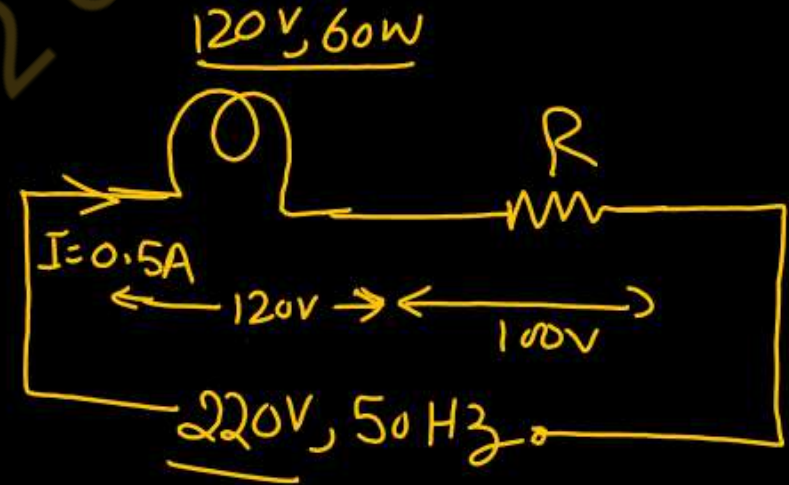
$L = \frac{320}{2\pi f}$
 $= \frac{320}{2\pi \times 50}$

$L = 1.01 H$

(iii) Capacitor = ?
H.W



$I = \frac{P}{V} = \frac{60}{120} = 0.5A$



Ex.2. A resistance of 120 ohms and a capacitive reactance of 250 ohms are connected in series across a A.C. voltage source. If a current of 0.9 A is flowing in the circuit find out (i) power factor (ii) supply voltage (iii) voltage across resistance and capacitance (iv) Active and reactive power.

Given

$$Z = R - jX_c$$

$$Z = 120 - 250j$$

$$Z = (120 - 250j) \angle -64.38^\circ$$

$$= 277.3 \angle -64.38^\circ$$

$$\phi = 64.38^\circ$$

(i) $P.f = \cos \phi = \cos 64.38^\circ = 0.43$

(ii) $V = IZ$

$$= 0.9(120 - 250j)$$

$$V = 108 - 225j$$

$\downarrow \quad \quad \downarrow$
 $V_R \quad \quad V_C$

$$= 249.57 \angle 64.35^\circ$$

$$I = \frac{V}{Z \angle \theta} = L - \theta$$

$$V_R = 108 \text{ volt}$$

$$V_C = 225 \text{ volt}$$

(iv) $P = VI \cos \phi$

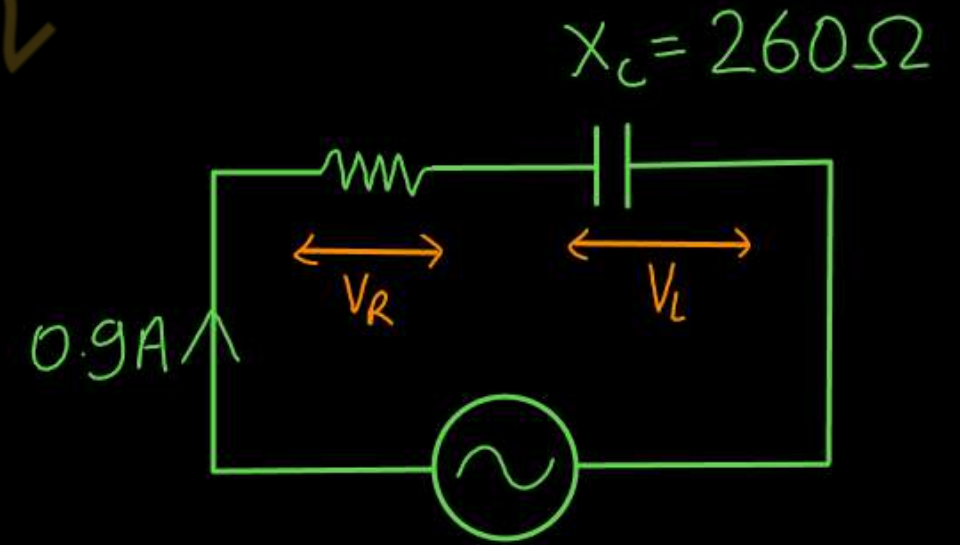
$$= 249.57 \times 0.9 \times 0.43$$

$$P = 96.58 \text{ watt}$$

$$Q = VI \sin \phi$$

$$= 249.57 \times 0.9 \times \sin 64.38^\circ$$

$$Q = 202.52 \text{ VAR}$$



Ex.-3. A coil having a resistance of 6 ohm and an inductance of 0.0255 H is connected across a 230v, 50 Hz a.c. supply. Calculate (i) Current (ii) power factor (iii) Active power (iv) Reactive Power (v) Apparent Power (vi) It is desired to improve power factor to 0.8 .What value of capacitance to be connected in series R and what is reduction in reactive power. **AKTU- 2009-10**

$$L = 0.0255 \text{ H}$$

$$X_L = 2\pi 50 \times 0.0255$$

$$X_L = 8.01 \Omega$$

$$Z = 6 + 8j$$

$$(i) I = \frac{V}{Z} = \frac{230}{(6+8j)} = 23 \angle -53.13$$

$$(ii) p.f = \cos \phi = \cos 53.13 = 0.6 (\text{lag})$$

$$P = VI \cos \phi = 230 \times 23 \times 0.6 \\ = 3174 \text{ watts}$$

$$Q = VI \sin \phi \\ = 230 \times 23 \times \sin 53.13$$

$$Q_1 = 4231.99 \text{ VAR}$$

$$S = (3.1 + 4.2j) \text{ KVA}$$

$$\cos \phi = 0.8 = \frac{R}{Z}$$

$$Z = \frac{R}{0.8} = \frac{6}{0.8} = 7.5 \Omega$$

$$\text{So } Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$(7.5)^2 = 6^2 + (8 - X_C)^2$$

$$20.25 = (8 - X_C)^2$$

$$8 - X_C = 4.5$$

$$\boxed{X_C = 3.5 \Omega}$$

$$X_C = \frac{1}{2\pi f C}$$

$$C = \frac{1}{2\pi \times 50 \times 3.5} = 9.09 \times 10^{-4} \text{ f}$$

$$Q_2 = 230 \times \left(\frac{230}{7.5}\right) \sin(36.86) \\ = 4231 \text{ VAR}$$

Reduction

Ex.-4 A Series RLC circuit consisting a resistance of 20 ohm , inductance of 0.2 H and Capacitance of 150 μ F is connected across 230V, 50 Hz source. Calculate (i) Impedance (ii) Current (iii) Power Factor (iv) Frequency Required to Make power factor Unity (v) Quality Factor **AKTU- 2012-13**

$$(i) L = 0.2H, X_L = 2\pi fL = 100\pi \times 0.2 = 62.8\Omega$$

$$C = 150\mu f, X_C = \frac{1}{2\pi fC} = \frac{1}{100\pi \times 150 \times 10^{-6}} = 21.22\Omega$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{20^2 + (62.8 - 21.22)^2}$$

$$\boxed{Z = 46.13\Omega}$$

$$I = \frac{V}{Z} = \frac{230}{46.13} = 4.98A$$

$$P.f = \frac{R}{Z} = \frac{20}{46.13} = 0.43$$

$$(iv) f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$= \frac{1}{2\pi\sqrt{0.2 \times 150 \times 10^{-6}}}$$

$$f_r = 29.05 \text{ Hz}$$

$$(v) Q = \frac{\omega L}{R} = \frac{2\pi \times 29.05 \times 0.2}{20}$$

$$\boxed{Q = 1.8}$$

Ex.-5. A 20 ohm resistor is connected in series with an inductor and a capacitor across a variable frequency 25 V supply. When the Frequency is 400 Hz, the current is at the maximum value of 0.5 A and the potential difference across the capacitor is 150 V Calculate the resistance and inductance of the inductor. **AKTU- 2001-02**

$$f_r = 400 \text{ Hz}$$

$$I_m = 0.5 \text{ A}$$

$$V_c = 150 \text{ V}$$

$$V = 25 \text{ V}$$

At resonance

$$I_m = \frac{V}{(20+r)}$$

$$0.5 = \frac{25}{20+r}$$

$$20+r = 50$$

$$\boxed{r = 30}$$

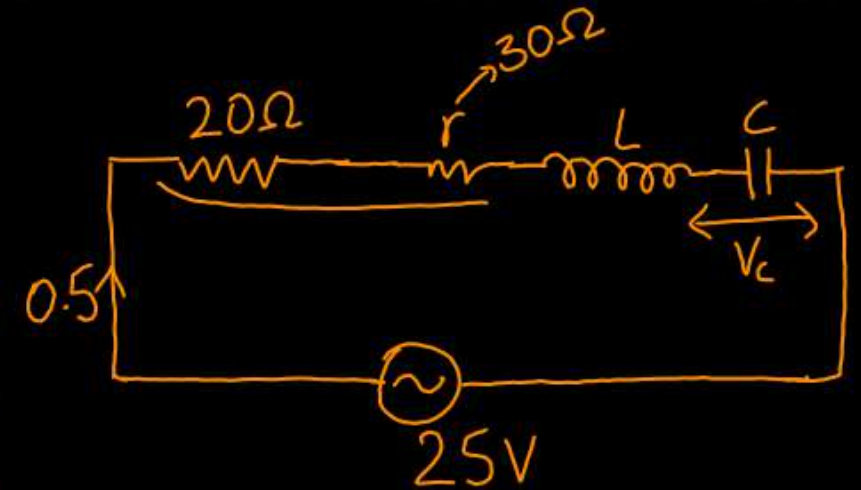
$$X_c = \frac{V_c}{I} = \frac{150}{0.5} = 300 \Omega$$

At resonance

$$X_L = X_c$$

$$X_L = 300 \Omega$$

$$\boxed{L = \frac{X_L}{2\pi f} = \frac{300}{100\pi} = 0.95 \text{ H}}$$



Ex.-6. A series RLC circuit consist of $R = 1000\Omega$, $L = 100\text{mH}$ and $C = 10\mu\text{F}$. The applied voltage across the circuit is 100V .

- (i) Find the resonant frequency of the circuit.
- (ii) Find the quality factor of the circuit at the resonance frequency.
- (iii) At what angular frequencies do the half power points occur?
- (iv) Calculate the bandwidth of the circuit.

AKTU 2021-22

Sol. (i) $f_r = \frac{1}{2\pi\sqrt{LC}}$

$Q = \frac{\omega L}{R}$

(iii) $f_1 = f_r = \frac{R}{4\pi L}$

$f_2 = f_r + \frac{R}{4\pi L}$

$B.W = f_2 - f_1$

Gateway classes 7455 951284

Ex.-8. A Series circuit consists of a resistance of 10 ohm and inductance of 50mH and a variable capacitance in series across a 100V , 50 Hz supply. Calculate (i) Value of capacitance to produce resonance (ii) Voltage across the capacitance (iii) Q- Factor. **AKTU 2018-19**

$$X_L = 2\pi fL = 100\pi \times 50 \times 10^{-3} \\ = 15.7 \Omega$$

for resonance

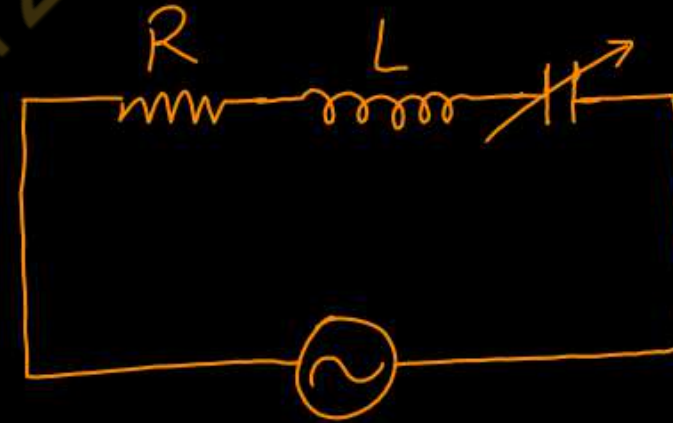
$$X_C = X_L = 15.7 \Omega$$

$$C = \frac{1}{2\pi \times 50 \times 15.7} = 2.02 \times 10^{-4} \text{ f}$$

$$I = \frac{V}{R} = \frac{100}{10} = 10 \text{ amp}$$

$$V_C = I \times X_C = 10 \times 15.7 = 157 \text{ volt}$$

$$Q = \frac{\omega L}{R}$$



Ex.-7. A Series a.c. circuit has resistance of 15 ohm and inductive reactance of 10ohm. Calculate the value of a capacitor which is connected across this series combination so that system has unity power factor. The frequency of a.c. supply of a.c. supply is 50 Hz. **AKTU- 2005-06**

$$f_r = 50 \text{ Hz}$$

$$R = 15 \Omega$$

$$X_L = 10 \Omega$$

$$X_C = X_L = 10 \Omega$$

$$C = \frac{1}{2\pi f X_C}$$

$$= \frac{1}{100\pi \times 10} = 3.18 \times 10^{-4} \text{ f}$$

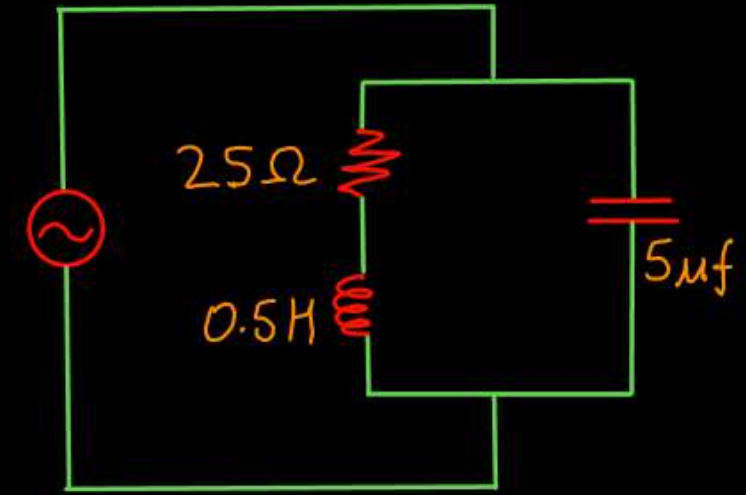
Gateway classes 7455 9612 84

Ex.-8. For the circuit shown below, determine : (i) Resonant frequency, (ii) Total impedance of the circuit at resonance (iii) Quality Factor **AKTU- 2014-15**

$$f_r = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}} = ?$$

$$Z = \frac{L}{RC} = ?$$

$$Q = \frac{\omega L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$$



Gateway classes 7455 961284

Q 1. A series RLC circuit is composed of 10 ohm resistance , 0.1H inductance and 50 microfarad capacitance. A voltage $v(t) = 141.4 \cos(100\pi t)$ V is impressed upon the circuit . Determine (i) The expression for instantaneous current, (ii) The voltage drops V_R, V_L, V_C across resistor capacitor and inductor (iii) Draw the phasor diagram using all the voltage relation. **AKTU- 2014-15**

Ex.-2. A Series RLC circuit has $R= 10 \Omega$, $L= 0.1 \text{ H}$ and $C= 8 \mu\text{f}$ Determine : (i) Resonant frequency, (ii) Half power Frequencies (iii) Bandwidth (iv) Quality Factor **AKTU- 2005-06**

Gateway classes 7455 961292

FUNDAMENTALS OF ELECTRICAL ENGINEERING

UNIT -2 : Steady State Analysis of 1- Φ Circuits

Lecture- 6

Today's Target-

- ✓ PYQ Parallel Circuit, Series Parallel Circuit
- ✓ DPPs

Concept of Admittance

• Admittance:-

$$\frac{R + jX}{Z \text{ (Impedance)}}$$

$$Z = R + jX_L$$

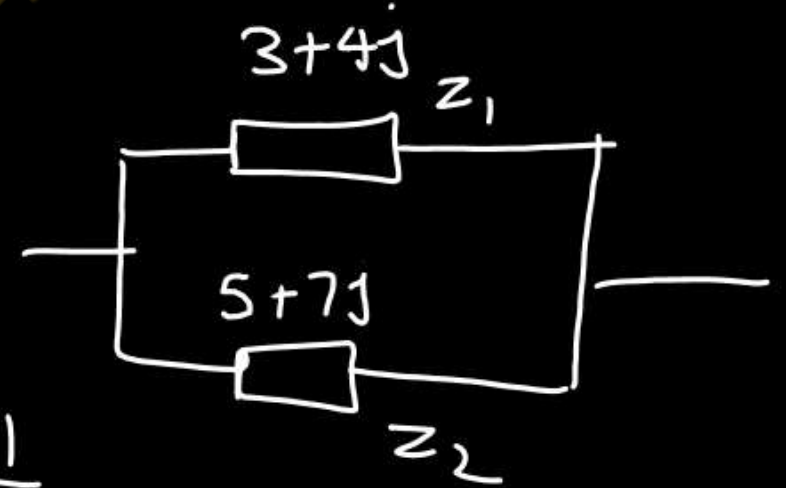
$$Y = \frac{1}{Z} = \frac{1}{R + jX_L} \times \frac{R - jX_L}{R - jX_L}$$

$$= \frac{R - jX_L}{R^2 + X_L^2}$$

$$= \underbrace{\frac{R}{R^2 + X_L^2}}_{\text{Conductance } G} - j \underbrace{\frac{X_L}{R^2 + X_L^2}}_{\text{Susceptance } B}$$

$$Z = R + jX_L$$

$$Y = G + jB$$



$$\frac{1}{Z} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

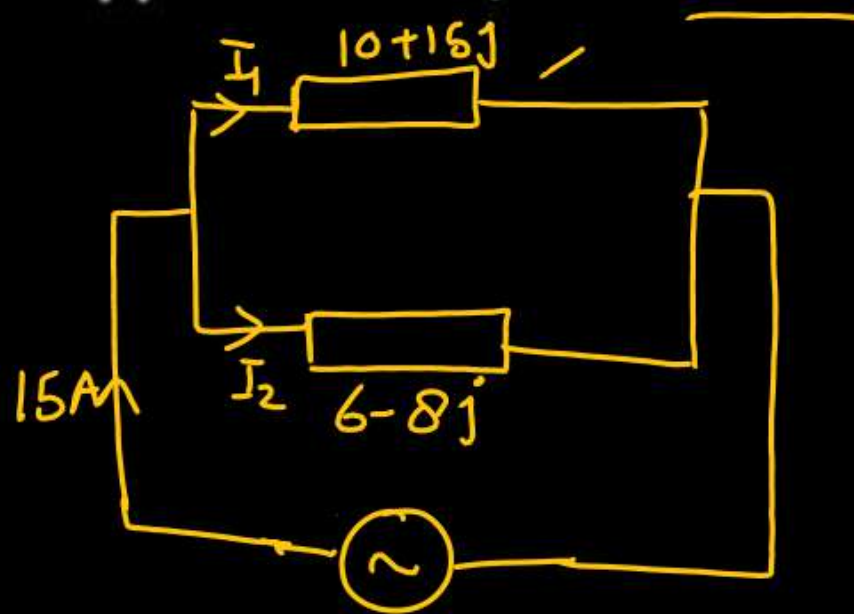
$$Z^{-1} = Z_1^{-1} + Z_2^{-1}$$

$$Z^{-1} = (3+4j)^{-1} + (5+7j)^{-1}$$

$$Z = [ANS]^{-1}$$

$$= 1.81 + 2.54j$$

- Ex.-1** Two impedances given by $Z_1 = (10+j15) \Omega$ and $Z_2 = (6-j8) \Omega$ are connected in parallel. If the total current supplied is 15 A, calculate the current and power absorbed by each branch **AKTU-2021-22**



Current divider.

$$I_1 = \frac{15 \times (6-j8)}{(10+j15) + (6-j8)}$$

$$= \frac{15 \times (6-j8)}{(16+j7)}$$

$$I_1 = 1.96 - 8.36j = 8.58 \angle -76.75^\circ \text{ A}$$

$$I_2 = \frac{15 \times (10+j15)}{(16+j7)}$$

$$I_2 = 13 + 8.36j = 15.48 \angle 32.68^\circ \text{ A}$$

$$P_1 = I_1^2 R_1$$

$$= (8.58)^2 \times 10$$

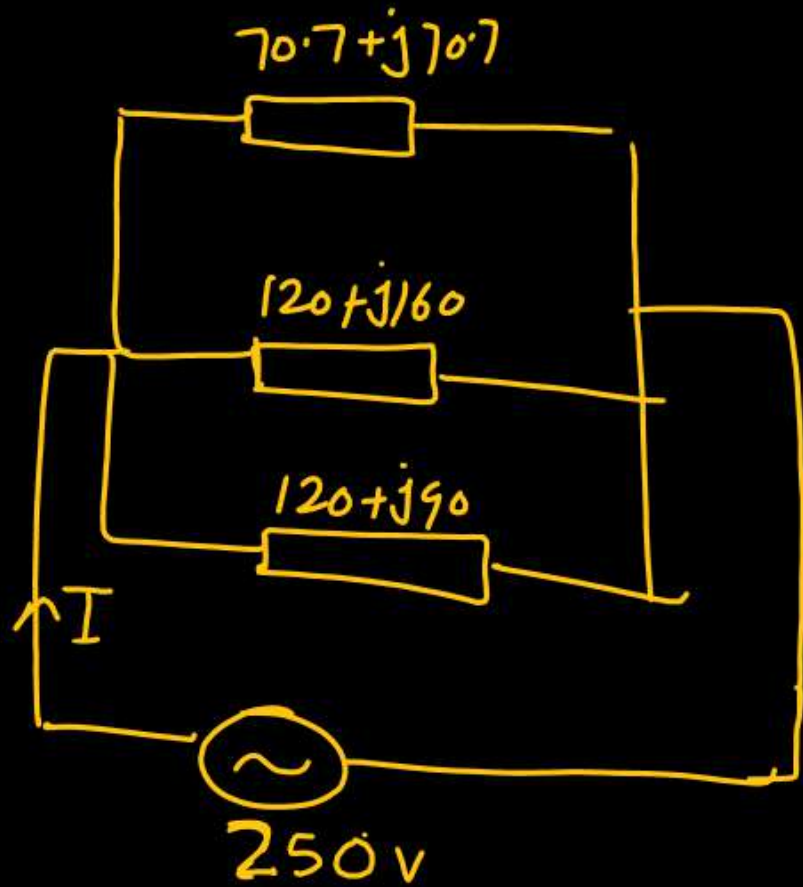
$$P_1 = 736.16 \text{ watt}$$

$$P_2 = I_2^2 R_2$$

$$= (15.48)^2 \times 6$$

$$= 1437.78 \text{ watt}$$

- **Ex.-2.** Three impedances of $(70.7+j70.7)\text{ohm}$, $(120+j160)\text{ohm}$ and $(120+j90)\text{ohm}$ are connected in parallel across a 250 V supply. Determine (i) admittance of the circuit (ii) supply Current and (iii) power factor **AKTU-2021-22**



$$Z = \left[(70.7 + j70.7)^{-1} + (120 + j160)^{-1} + (120 + j90)^{-1} \right]^{-1} \quad \text{(ii)}$$

$$Z = 33.16 + 32.448j$$

$$Z = 46.39 \angle 44.37^\circ$$

$$Y = \frac{1}{Z} = 0.015 - 0.015j$$

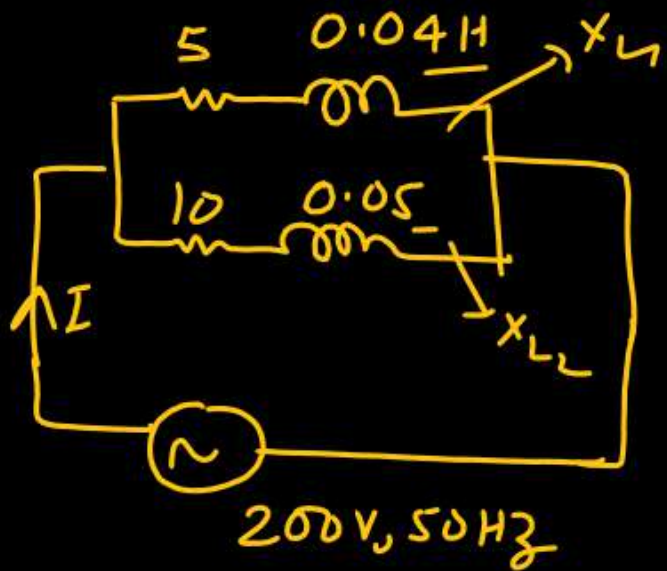
$$\begin{aligned} \text{(i)} \quad i &= \frac{V}{Z} = \frac{250}{Z} \\ &= 5.38 \angle -44.37^\circ \end{aligned}$$

$$P.f = \cos \phi$$

$$= \cos 44.37$$

$$P.f = 0.71 \text{ (Lagging)}$$

- **Ex.-3.** Two coils having resistance $5\ \Omega$ and $10\ \Omega$ and inductances $0.04\ \text{H}$ and $0.05\ \text{H}$ respectively are connected in parallel across a $200\ \text{V}$, $50\ \text{Hz}$ supply. Calculate: i. Conductance, susceptance and admittance of each coil. ii. Total current drawn by the circuit and its power factor. iii. Power absorbed by the circuit. **AKTU 2021-22**



$$X_{L1} = 2\pi f L_1 = 100\pi \times 0.04 = 12.56\ \Omega$$

$$X_{L2} = 2\pi f L_2 = 100\pi \times 0.05 = 15.70\ \Omega$$

$$Z_1 = 5 + 12.56j$$

$$Z_2 = 10 + 15.70j$$

$$Z^{-1} = (Z_1^{-1} + Z_2^{-1}) = Y$$

$$Y = 0.056 - 0.114j$$

$$\begin{array}{ccc} \downarrow & \downarrow & \downarrow \\ \text{Admitt} & \text{Cond} & \text{Suscep} \\ & (G) & (B) \end{array}$$

$$Z = Y^{-1} = \underbrace{3.47}_R + \underbrace{7.05j}_X$$

$$(ii) I = \frac{V}{Z} = \frac{200}{Z}$$

$$I = 25.42 \angle -63.70^\circ$$

$\hookrightarrow \phi$

$$(iii) \text{ p.f. } = \cos \phi = 0.44 (\text{lagg})$$

$$(iv) P = I^2 R$$

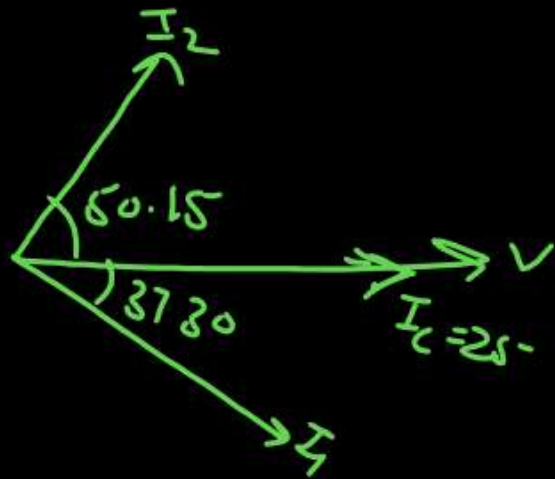
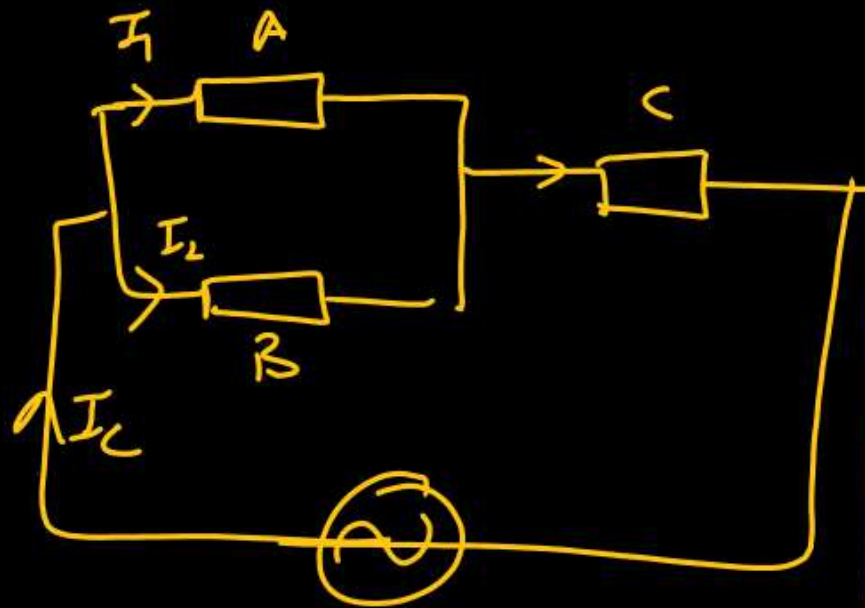
$$= (25.42)^2 \times 3.47$$

$$= 2242.23 \text{ Watt}$$

Ans

- **Ex.-4.** In series parallel circuit A and B are in parallel and in series with C. The impedances are $Z_A = 4+3j$ ohm, $Z_B = 4-j5$ ohm and $Z_C = 2+j8$ ohm. If the current $I_C = 25+j0$ amp Calculate (i) Branch voltage (ii) Branch Currents (iii) Total Power (iv) Phasor Diagram

AKTU 2015-16



$$I_1 = I_C \times \frac{B}{A+B}$$

$$= 25 \times \frac{(4-j5)}{(4+3j) + (4-j5)}$$

$$= \frac{25 \times (4-j5)}{8-2j}$$

$$I_1 = 19.41 \angle -37.30^\circ$$

$$I_2 = I_C \times \frac{A}{A+B}$$

$$= 25 \times \frac{(4+3j)}{8-2j} = 15.15 \angle 50.90^\circ$$

$$V_1 = I_1 Z_1 = ?$$

$$V_2 = I_2 Z_2 = ?$$

$$(ii) Z_{eq} = (Z_A || Z_B) + Z_C$$

$$Z_{eq} = 5.88 + 7.97j$$

$$P = I_C^2 R_{eq} = 25^2 \times 5.88$$

$$P = 3675 \text{ watt}$$

Ex.-5. In the circuit shown in figure find (i) Current in each branch (ii) Voltage V_{ab} and V_{bc} (iii) Power Loss in the circuit.

AKTU- 2001-02

$$Z_{eq} = (Z_1 || Z_2) + Z_3 = \left[(4 + 8j)^{-1} + (-8j)^{-1} \right]^{-1} + (-4j)$$

$$Z_{eq} = \frac{16}{-12j}$$

$$I = \frac{V}{Z} = \frac{120}{-12j} = 10 \angle 90^\circ$$

$$I_1 = I \times \frac{(4 + 8j)}{(4 + 8j) - 8j} =$$

$$I_1 = 13.4 \angle 100.30^\circ$$

$$I_2 = I \times \frac{(-8j)}{4} = I \times (-2j) = 20 \angle -90^\circ$$

$$V_{ab} = I \times (-4j)$$

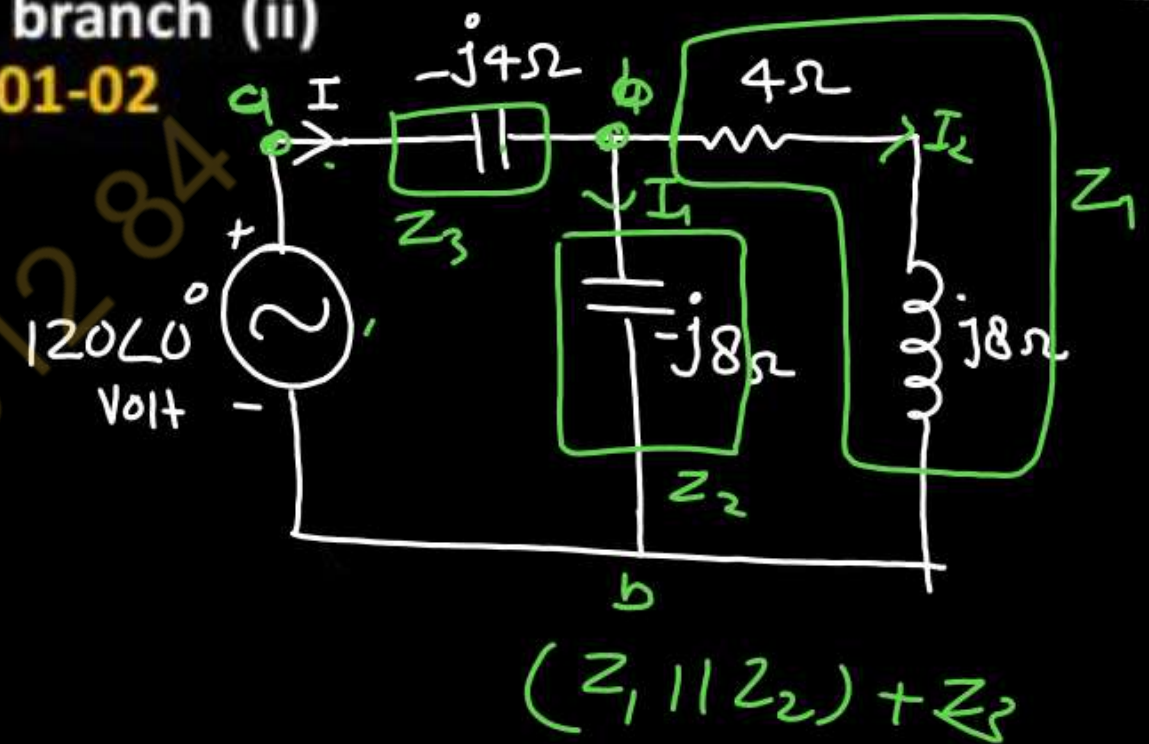
$$V_{bc} = (-8j) I_1$$

$$(iii) P = I^2 R$$

$$= 6^2 \times 16$$

$$P = 576 \text{ watt}$$

$$VI \cos \phi$$



Q.1 Obtain the power factor of a two branch parallel circuit where the first branch has $Z_1 = (2+4j)$ ohm and second $Z_2 = 6+j0$ ohm . To what value must the 6 ohm resistor be changed to result in the overall power factor 0.9 lagging. **AKTU- 2012-13**

$$\Delta R = 3.19 \Omega$$

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FUNDAMENTALS OF ELECTRICAL ENGINEERING

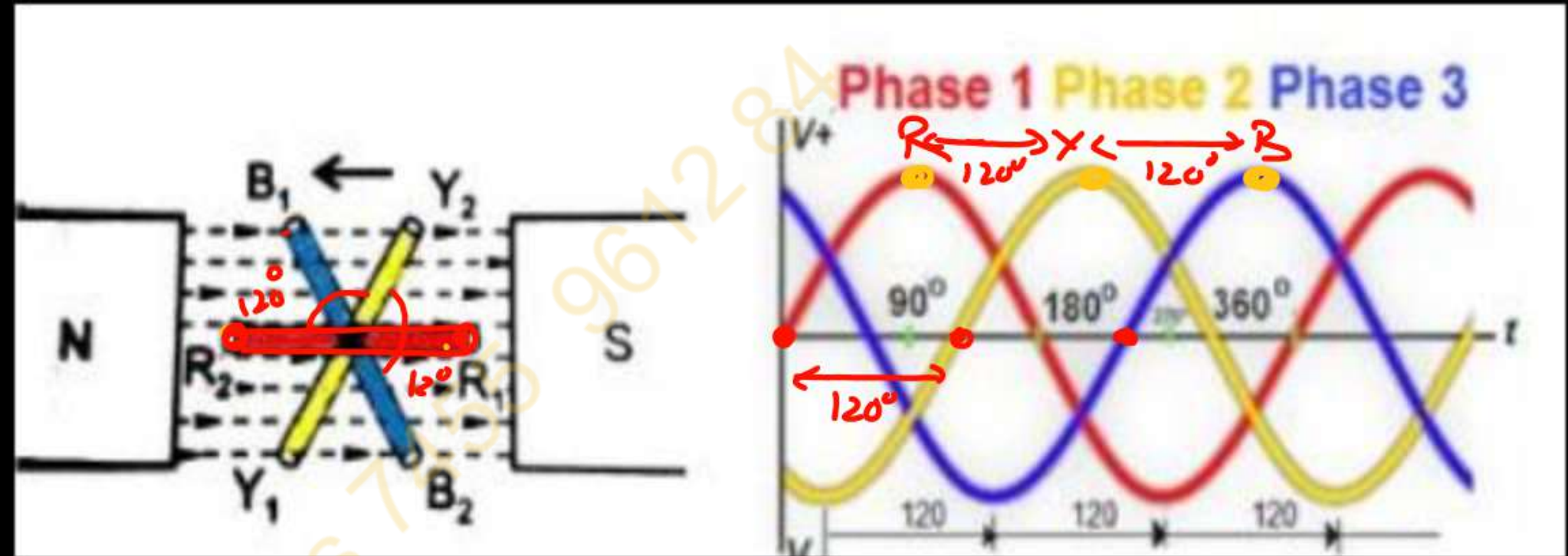
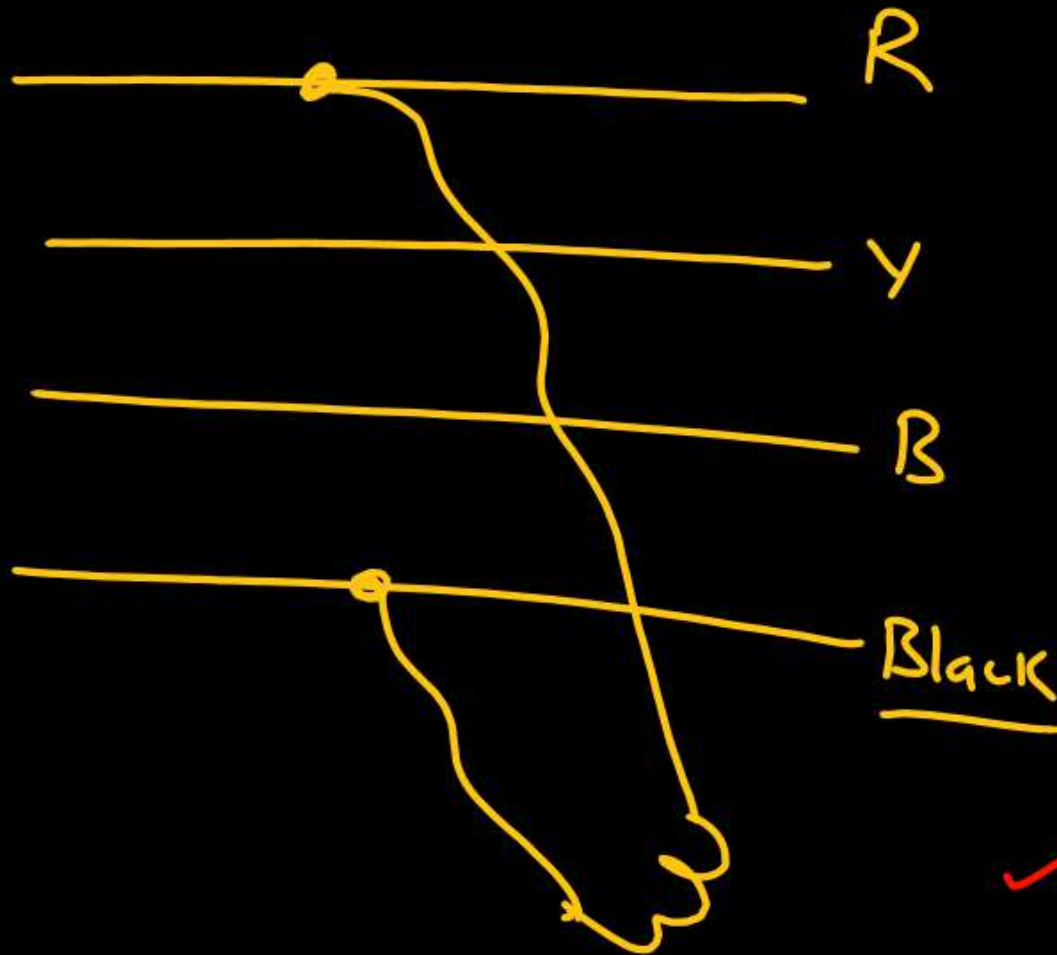
UNIT -2 : Steady State Analysis of 1- Φ Circuits

Lecture- 7

Today's Target-

- ✓ Three Phase Supply System ✓
- ✓ Connections in 3-phase system ✓
- ✓ PYQ Parallel Circuit, Series Parallel Circuit ✓
- ✓ DPPs

Generation of Three Phase Supply

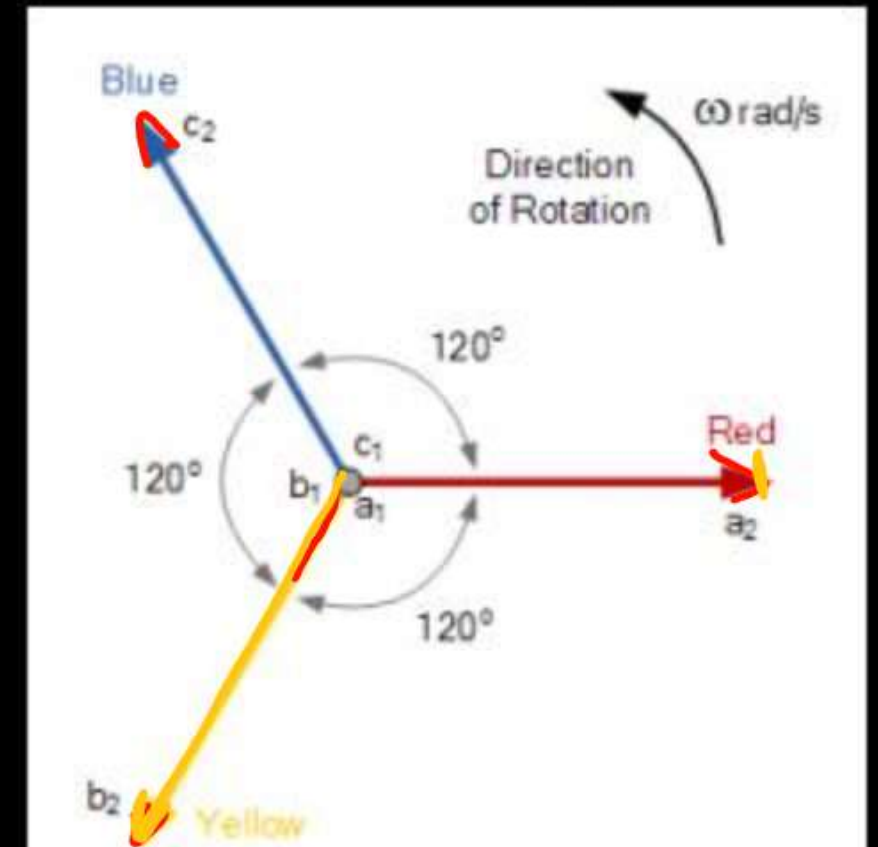


$$E_R = E_m \sin \omega t$$

$$E_Y = E_m \sin(\omega t - 120^\circ)$$

$$E_B = E_m \sin(\omega t - 240^\circ)$$

Sum of all three phasors at any instant is always zero



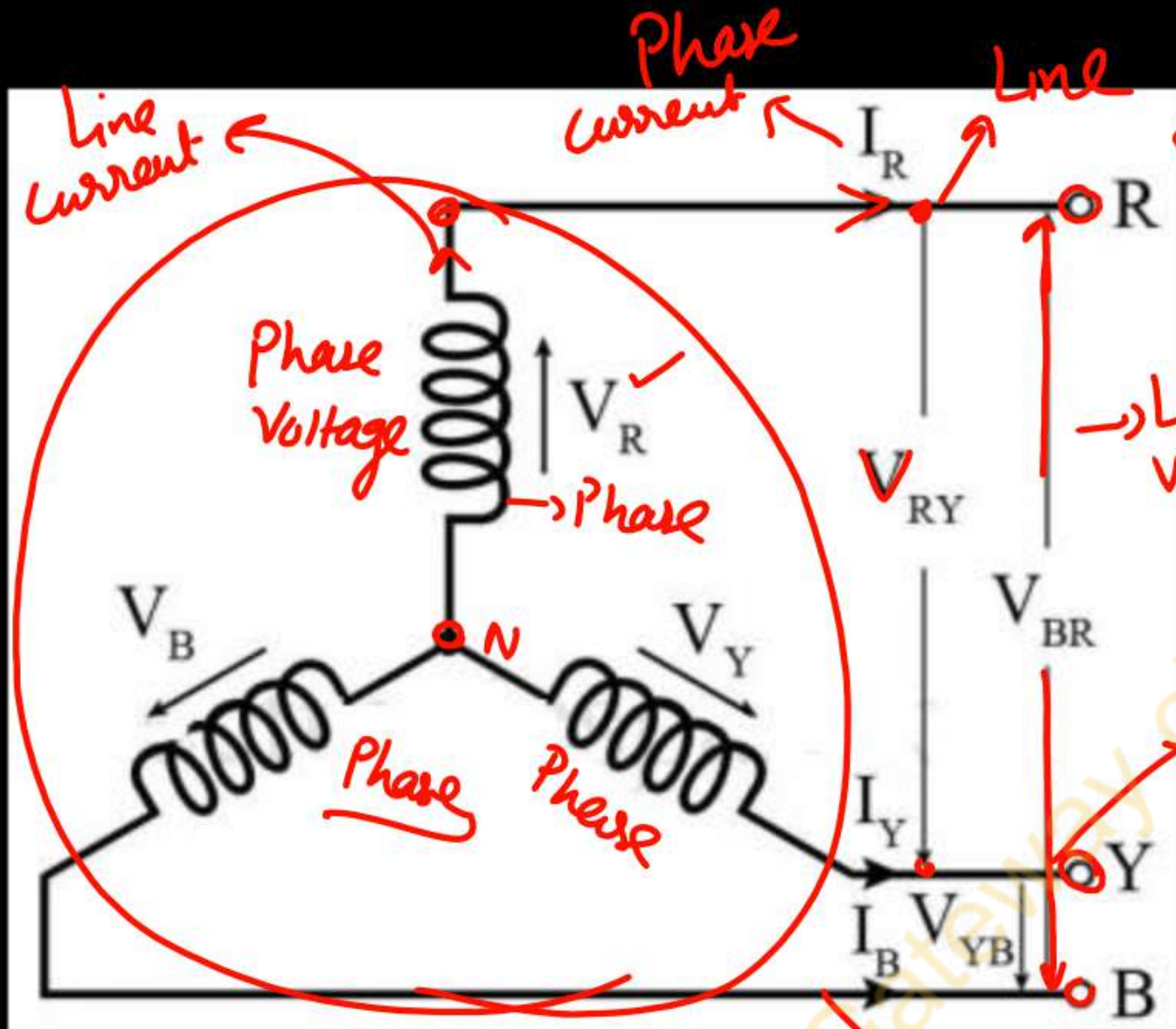
- **Phase Sequence** – The *phase sequence* is defined as the order in which the emf in three phase or coils of an alternator attains the positive maximum value. It is determined by the direction of rotation of alternator.
- ✓ **Naming the Phases** – The name of the three phases are given by the natural colors viz. Red (R), Yellow (Y) and Blue (B). Thus, the phase sequence being RYB.
- ✓ **Balanced Three-Phase Supply System** – In a balanced three-phase supply system, the three phase voltages are equal in magnitude and frequency. and displaced at 120° from each other

Advantages of Three Phase Supply

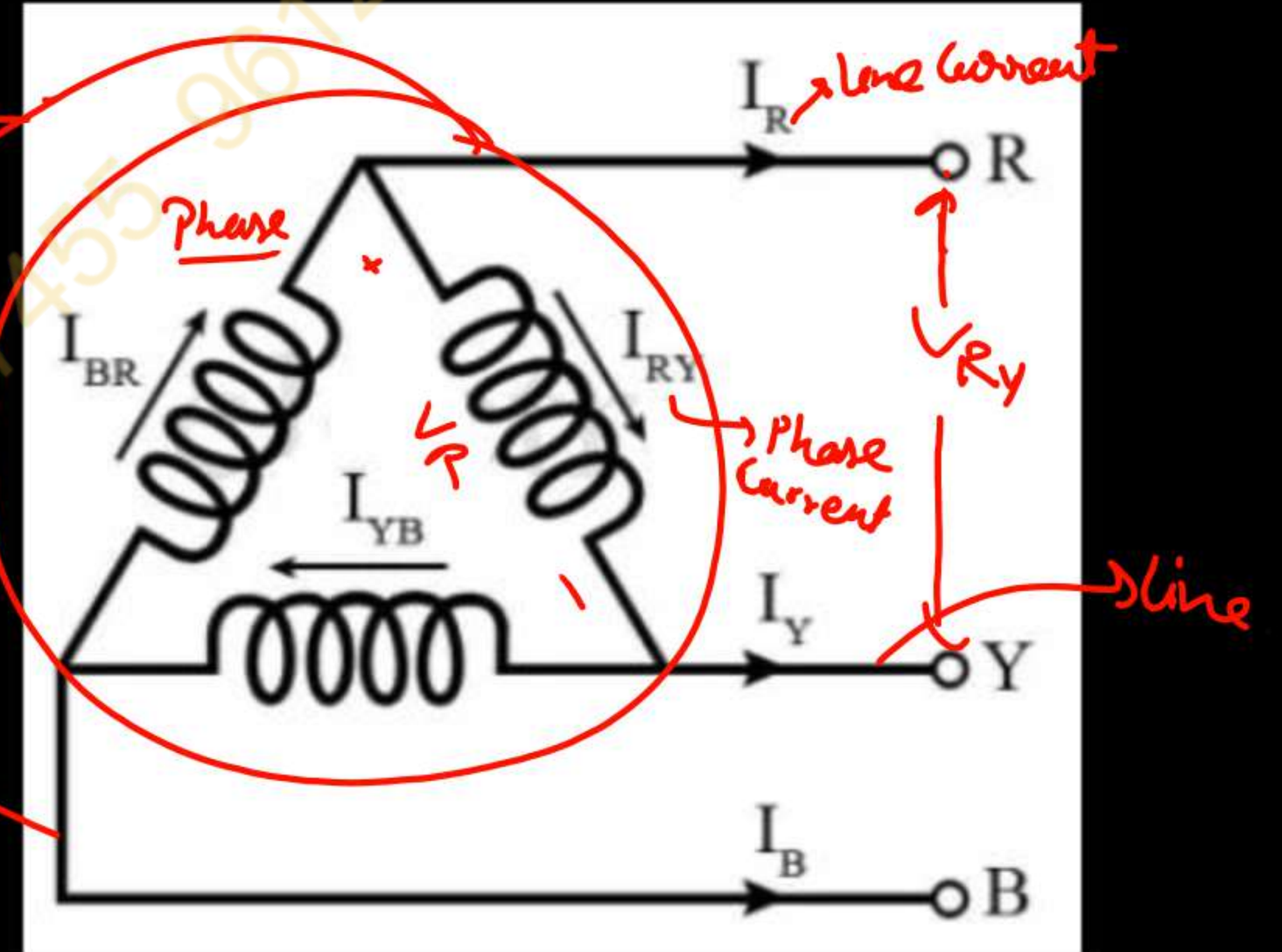
- The three phase power has a constant magnitude while the power in single phase system is the function of supply frequency i.e. pulsates from zero to maximum value at twice the supply frequency.
- ✓ In the 3-phase system, a rotating magnetic field can be created in stationary windings.
- For the same rating, three phase machines are smaller and simpler in construction
- To transmit the same amount of power over certain distance at a given voltage, the 3- ϕ system requires less (or $3/4$) of the weight of copper than that required by the 1- ϕ system.
- ✓ The voltage regulation of three phase transmission lines is better than that of 1-phase line.
- A single phase load can be supplied by a three phase system but, the converse is not true.

Types of Connections (Line and Phase Values)

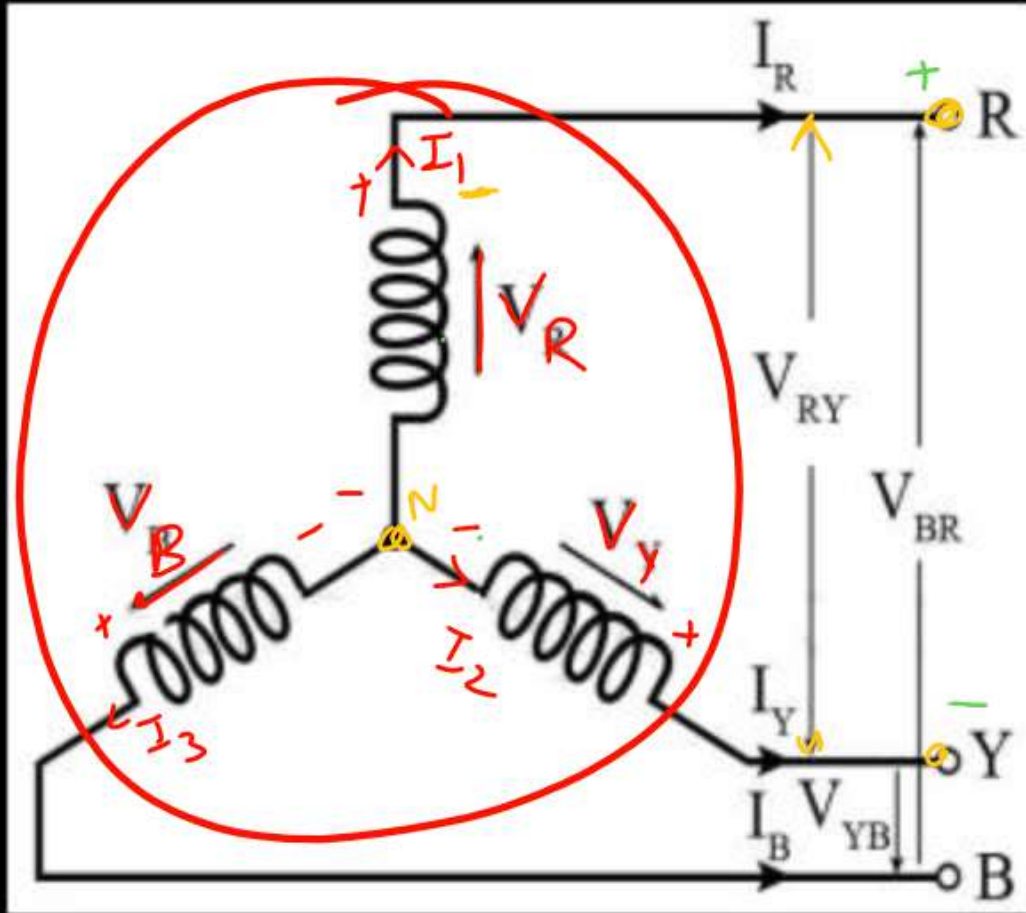
Star (Y)



Delta (Δ)



Relation Between Line and phase Values in Star Connection



from diagram

$$I_1 = I_R$$

$$I_2 = I_Y$$

$$I_3 = I_B$$

$$I_{ph} = I_L$$

from circuit

$$V_L = V_{RY}, V_{YB}, V_{BR}$$

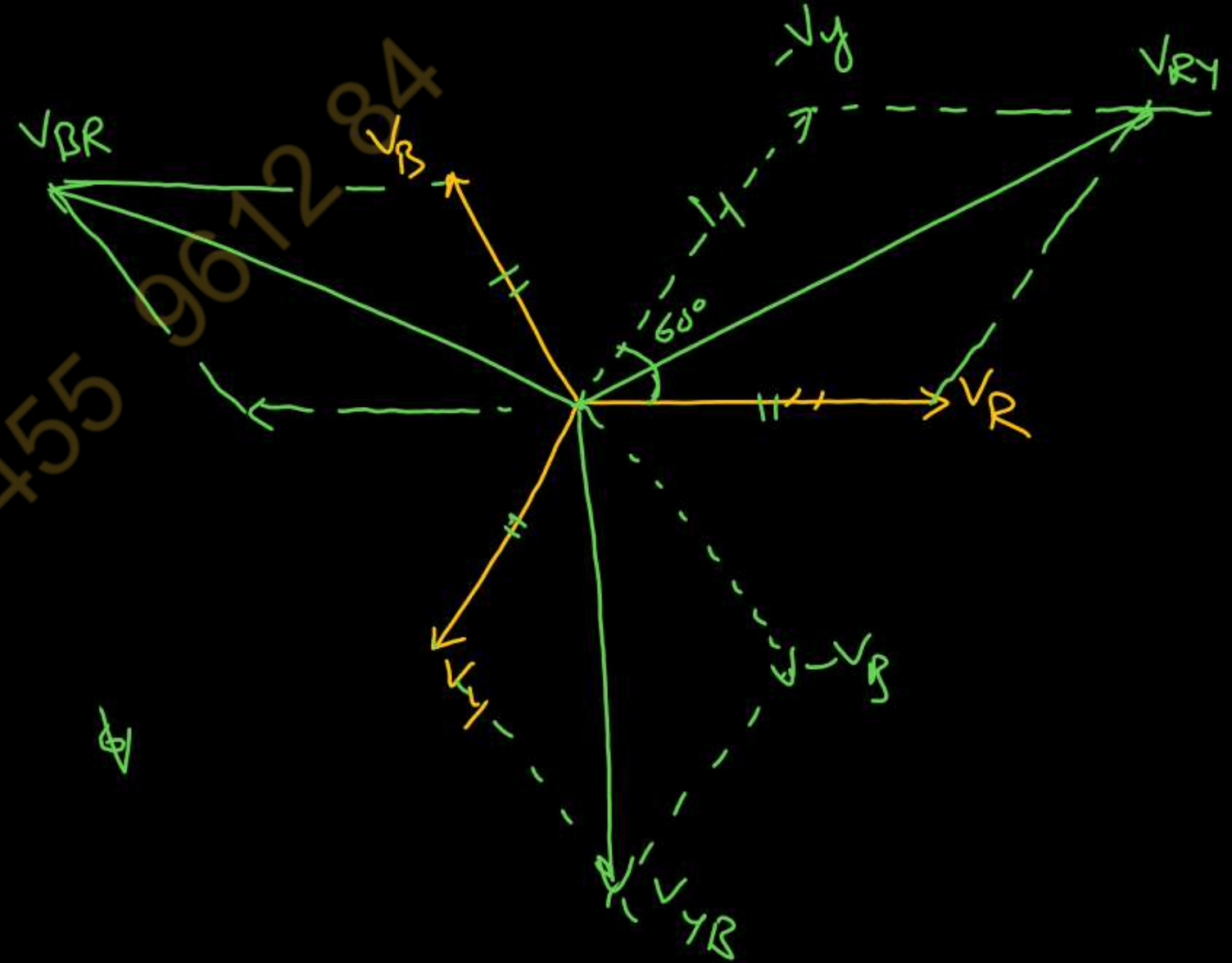
$$V_{ph} = V_R, V_Y, V_B$$

and

$$V_{RY} = V_R - V_Y$$

$$V_{YB} = V_Y - V_B$$

$$V_{BR} = V_B - V_R$$

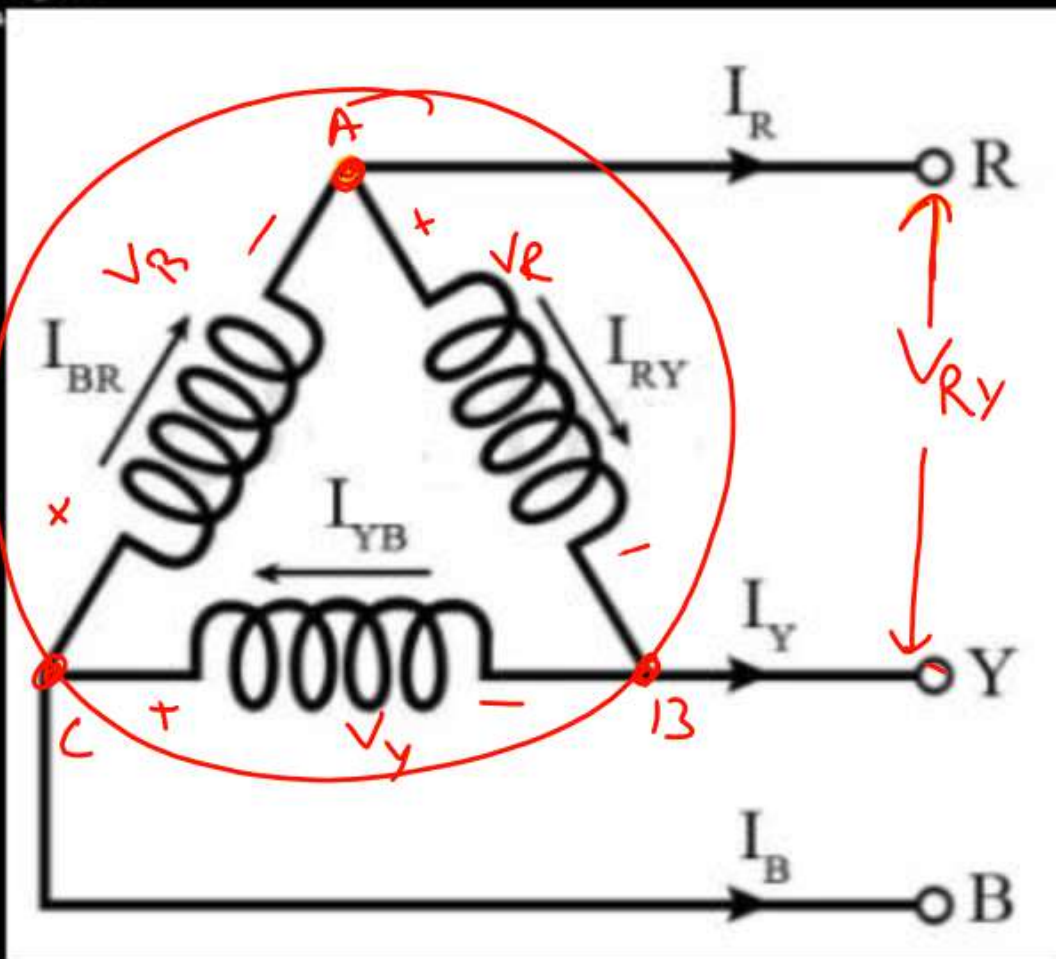


$$V_{RY} = \sqrt{|V_R|^2 + |V_Y|^2 + 2|V_R||V_Y|\cos 60}$$

$$V_L = \sqrt{V_{ph}^2 + V_{ph}^2 + 2V_{ph}^2 \times \frac{1}{2}}$$

$$V_L = \sqrt{3V_{ph}^2}$$

$$V_L = \sqrt{3}V_{ph}$$



$$I_R = I_Y = I_B = I_L$$

$$I_{RY} = I_{YB} = I_{BR} = I_{ph}$$

from circuit

$$V_{RY} = V_R$$

$$V_{YB} = V_Y$$

$$V_{BR} = V_B$$

$$V_L = V_{ph}$$

for current

At node A

$$I_R + I_{RY} = I_{BR}$$

$$I_R = I_{BR} - I_{RY} \quad (1)$$

Similarly

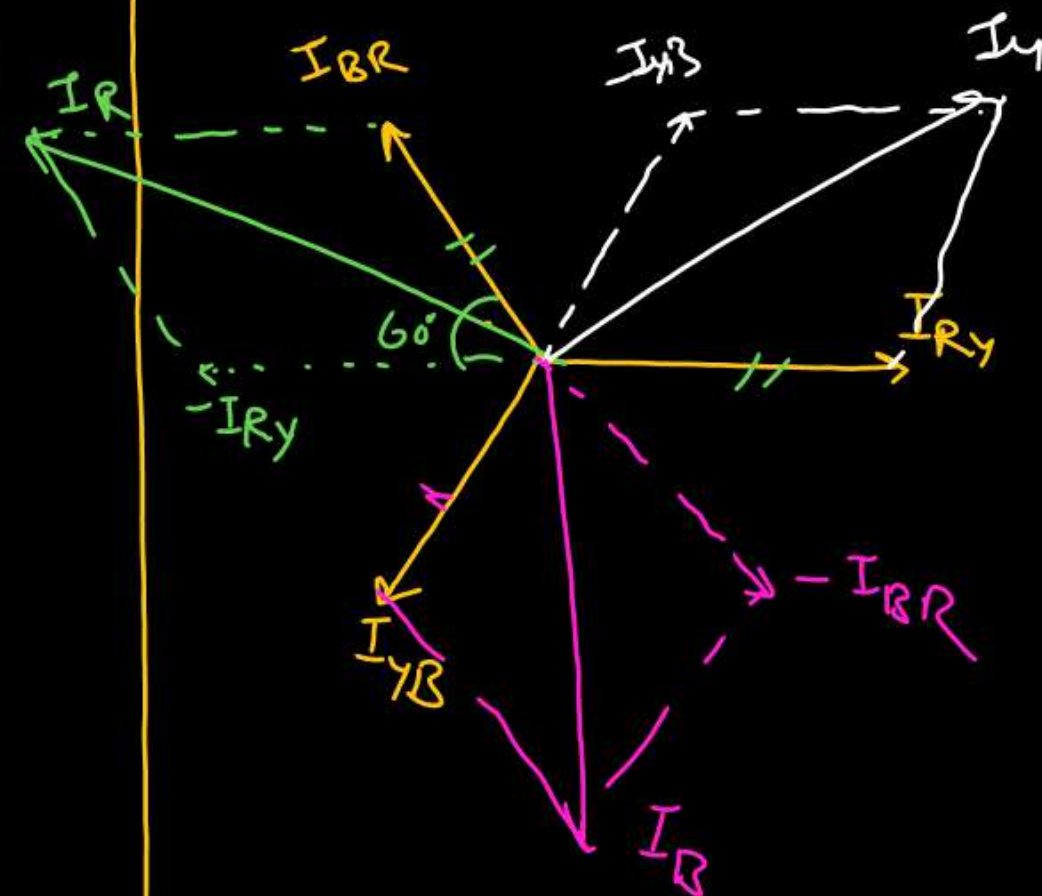
$$I_Y = I_{RY} - I_{YB} \quad (2)$$

$$I_B = I_{YB} - I_{BR} \quad (3)$$

$$I_R = \sqrt{I_{BR}^2 + I_{RY}^2 + 2I_{BR}I_{RY}\cos 60}$$

$$I_L = \sqrt{I_{ph}^2 + I_{ph}^2 + 2I_{ph}^2 \frac{1}{2}}$$

$$I_L = \sqrt{3} I_{ph}$$



Power Relations in Star and Delta connection

$$I_L = I_{ph}$$

$$V_L = \sqrt{3} V_{ph}$$

$$P_{3-\phi} = 3 V_{ph} I_{ph} \cos \phi$$

$$P_{3-\phi} = 3 \times \frac{V_L}{\sqrt{3}} \cdot I_L \cos \phi$$

$$P_{3-\phi} = \sqrt{3} V_L I_L \cos \phi$$

$$V_{ph} = V_L$$

$$I_L = \sqrt{3} I_{ph}$$

$$P_{3-\phi} = 3 V_{ph} I_{ph} \cos \phi$$

$$P_{3-\phi} = 3 V_L \times \frac{I_L}{\sqrt{3}} \cos \phi$$

$$P_{3-\phi} = \sqrt{3} V_L I_L \cos \phi$$

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- Ex.-1** A balanced star connected load of $(8 + j6)$ ohm per phase is connected to a balanced 3- phase, 400 V supply. Find the line current, phase current, power factor, power and total volt-amperes. **AKTU- 2021-22**

$$Z_{ph} = 8 + j6$$

$$V_L = 400V$$

$$V_L = \sqrt{3} V_{ph}$$

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.9 \approx 231$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{231}{(8 + j6)} = 23.1 \angle -36.86^\circ = I_L$$

$$P.f = \cos 36.86^\circ = 0.8 (\text{lagg})$$

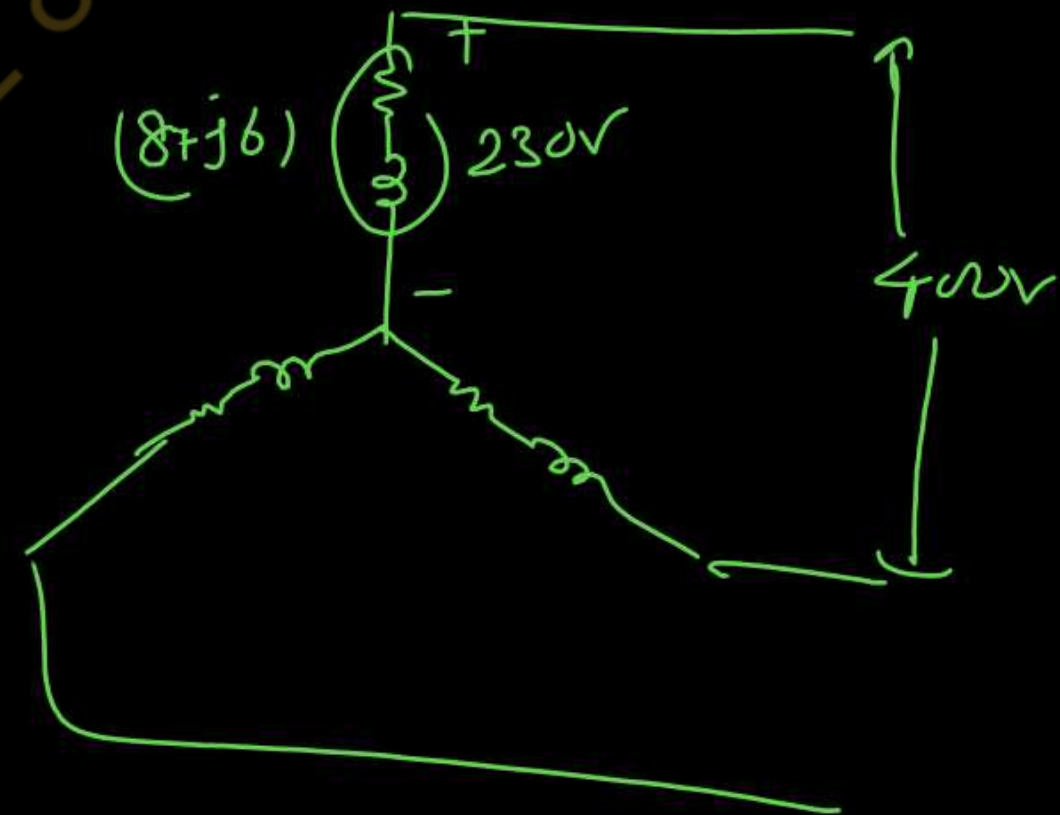
$$P = \sqrt{3} 400 \times 23.1 \times 0.8$$

$$P = 12804.97 \text{ watt}$$

$$S = 3VI$$

$$= 3 \times 231 \times 23.1$$

$$S = 16008.3 \text{ VA}$$



- Ex.-2.** A balanced 3-Phase star connected load takes 30KW at a leading current of 48A from a 3- ϕ source of 500V, 50Hz. Find the circuit parameters per phase. **AKTU- 2021-22**

$$P = 30 \text{ KW}$$

$$I_L = 48 \text{ A} = I_{ph}$$

$$V_L = 500 \text{ V}$$

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{500}{\sqrt{3}} = 288.67 \text{ volt}$$

$$Z_{ph} = \frac{V_{ph}}{I_{ph}} = 6.01 \Omega$$

$$P = 3 V_{ph} I_{ph} \cos \phi$$

$$\cos \phi = \frac{P}{3 V_{ph} I_{ph}}$$

$$\cos \phi = \frac{30 \times 10^3}{3 \times 288.67 \times 48}$$

$$\cos \phi = 0.67$$

$$\phi = 43.80^\circ$$

$$Z_{ph} = 6.01 \angle -43.80^\circ$$

$$= 4.33 - j4.15$$

$\downarrow$ $\downarrow$
 R X_C

Numerical Examples

Self

- **Ex.-3.** A balanced star-connected load of impedance, $Z=10\angle 60^\circ\Omega$ is connected to line voltage of 230V. Obtain the phase current, phase voltage and power supplied to load **AKTU- 2021-22**

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- Ex.-4.** A balanced delta connected load of $(12 + j 9) \Omega$ / phase is connected to 3- phase 400 V supply. Calculate line current, power factor and power drawn by it. **AKTU-2014-15**

$$Z_{ph} = 12 + j9$$

$$V_L = 400V = V_{ph}$$

$$I_{ph} = \frac{V_{ph}}{Z_{ph}} = \frac{400}{(12 + j9)}$$

$$= 21.33 - j16$$

$$I_{ph} = 26.7 \angle -36.86^\circ$$

$$I_L = \sqrt{3} \times I_{ph} = 46.18 \angle -36.86^\circ$$

$$p.f. = \cos 36.86^\circ = 0.8 (\text{lagg})$$

$$P = \sqrt{3} V_L I_L \cos \phi$$

$$= \sqrt{3} \times 400 \times 46.18 \times 0.8$$

$$P = 25.59 \text{ kW}$$

- **Ex.-5.** A 3- ϕ , 500 V motor has a power factor of 0.4 (lagging). Two wattmeters are connected to measure the input and show the total input power to be 30 kW. Find out the readings of each wattmeter **AKTU- 2021-22**

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